Boundary Selection and Sharp Asymptotics for Exact Burg Mirror Descent

Technical proof supplement

Abstract

This supplement supplies the active technical dependencies of the paper. The complete general-codimension finite-length proof, normal-power variation and exact scalar lift appear in the main text. Here we prove planar analytic selection and its local germ lemma; stationary, cluster and moving cancellation; oriented flag capture and an exact split-clock obstruction; nonlinear two-clock tracking; coupled and tilted sharp endpoints; and the mixed polynomial rate criterion. The critical quadratic forcing proof retains its separate uniform analytic assumptions and two finite-horizon absorptions. Historical qualitative alternatives and profile catalogues are preserved in a noncompiled source archive rather than repeated here. The supplement and main text form one paper, with the original ambient restriction $\eta L < 1$ throughout.

Keywords: Burg mirror descent; boundary selection; finite physical length; normal-power variation; scalar cancellation; sharp asymptotics

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S1 Planar analytic selection: complete proof

We prove Main Theorem 2. In this section the objective, denoted by p, is analytic on a neighborhood of the whole closed triangle; convexity is not assumed. The exact map is Main (M.1), with the admitted ambient restriction $\eta L < 1$.

S1.1 Exact reciprocal formula and descent

In the following single-step notation, k is suppressed: $x^+ = x + d$, $g = \nabla p(x)$, and $w_i = x_i x_i^+$. The first-order conditions and $\sum_i d_i = 0$ give

$$\frac{1}{x_i^+} - \frac{1}{x_i} = \eta r_i, \qquad d_i = -\eta w_i r_i, \qquad r_i = g_i - \mu, \quad \mu = \frac{\sum_i w_i g_i}{\sum_i w_i}. \tag{S.1}$$

Let $B = \max_\Delta \|\nabla p\|_\infty$ and $Q = 1 + 2\eta B$. Since μ is an average with positive weights, $|r_i| \le 2B$. Using $x_i/x_i^+ = 1 + \eta x_i r_i$ and $x_i^+/x_i = 1 - \eta x_i^+ r_i$, respectively, yields

$$Q^{-1} \le x_i^+/x_i \le Q. \tag{S.2}$$

Define $V = \sum_i w_i r_i^2$ and $S = \sum_i d_i^2/w_i = \eta^2 V$. Since $w_i \le 1$, we have $\|d\|^2 \le S$; also, $\langle g, d\rangle = -S/\eta$. The ambient descent lemma therefore gives

$$p(x) - p(x^+) \ge \frac{1 - \eta L/2}{\eta} S = \eta(1 - \eta L/2)V \ge c\|d\|^2. \tag{S.3}$$

The constant $c > 0$ is globally fixed. Thus $p(x_k) \downarrow \ell$, $\sum_k V_k < \infty$, and $\sum_k \|d_k\|^2 < \infty$. If $p(x_k) = \ell$ at any step, then (S.3) forces $d_k = 0$, and uniqueness of the exact update makes the trajectory constant thereafter. Hence every nonconstant trajectory with a coordinate tending to zero has a strictly positive objective gap.

Lemma S1 (Cluster sets and relative-face criticality)**.** *The cluster set ω of every trajectory is nonempty, compact, and connected, with* $\operatorname{dist}(x_k, \omega) \to 0$ *and $p \equiv \ell$ on ω. For any $x_* \in \omega$, all indices $i \in \operatorname{supp} x_*$ have equal values of $\partial_i p(x_*)$.*

Proof. Compactness gives a nonempty compact cluster set, the function-value conclusion, and the distance conclusion; if the latter failed, one could take a subsequence staying away from ω and then extract a cluster point, a contradiction. If ω were disconnected, it could be partitioned into two nonempty disjoint compact sets at positive distance from each other. Take disjoint small neighborhoods that are still separated by a positive distance. The entire trajectory eventually lies in their union, while $d_k \to 0$ prevents it from passing from one neighborhood to the other thereafter; this contradicts the fact that both parts contain cluster points.

For $x_{k_j} \to x_*$, the fact that $d_k \to 0$ gives $x_{k_j+1} \to x_*$. Use the identity

$$V_k = \frac{1}{\sum_i w_{i,k}} \sum_{i<j} w_{i,k} w_{j,k} (g_{i,k} - g_{j,k})^2.$$

If $x_{*,i}, x_{*,j} > 0$, the corresponding product of weights tends to a positive number; since $\sum_i w_{i,k} \le 1$ and $V_k \to 0$, we have $g_{i,k_j} - g_{j,k_j} \to 0$. Continuity gives the required equality. Only relative criticality within the support is proved here; the KKT inequalities for the missing coordinates have not been assumed. ∎

Lemma S2 (KL capture from a single interior cluster contact)**.** *If ω contains an interior point, then the entire trajectory has finite Euclidean length and converges to that point.*

Proof. Let $\bar{x} \in \omega \cap \operatorname{ri} \Delta$. By Lemma S1, $\bar{x}$ is a critical point on the affine hyperplane, and $p(\bar{x}) = \ell$. Choose a relative ball centered at $\bar{x}$, denoted by U, whose closure lies in the interior and in an ordinary KL neighborhood. For a source point $x_k \in U$, (S.2) gives a uniform positive lower bound for the target coordinates, so projecting (S.1) onto the tangent space $T = \{d : \sum d_i = 0\}$ yields

$$\|\nabla_T p(x_k)\| = \eta^{-1} \|\Pi_T \operatorname{diag}(w_{i,k}^{-1}) d_k\| \le C \|d_k\|.$$

If the trajectory becomes constant in finitely many steps, the conclusion is immediate. Otherwise, $e_k = p(x_k) - \ell > 0$; at every sufficiently late step whose source point lies in U, the ordinary KL inequality and (S.3) give

$$\varphi(e_k) - \varphi(e_{k+1}) \ge \varphi'(e_k)(e_k - e_{k+1}) \ge c' \|d_k\|.$$

Take a late visiting point x_j whose distance to $\bar{x}$ is less than one quarter of the radius of U, and such that $\varphi(e_j)/c'$ is less than one quarter of that radius. If there were a first step leaving U, the preceding estimate would still apply at the source point of the exit step; summing over the entire block would place the exit point at distance less than one quarter of the radius from x_j, a contradiction. The trajectory therefore never leaves U again, and the same summation gives finite remaining length. Its limit can only be the known subsequential limit $\bar{x}$. ∎

S1.2 Finite length on the closed triangle

The planar case admits finite Euclidean length and point convergence without prescribing a limiting face or a protected tail. The next theorem has its own regularity hypothesis: real analyticity on a neighborhood of the entire closed triangle suffices, and semialgebraicity is not required. Its point-selection argument uses exact source–target identities, boundary topology, and analytic division at the edge endpoints as well as in the edge interiors. A single analytic germ lemma then gives finite length at every boundary limit.

Theorem S1 (Planar analytic finite length for the exact Burg map)**.** *Let*

$$\Delta = \{x \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 = 1\}, \qquad h(x) = -\sum_{i=1}^{3} \log x_i.$$

Let p be real analytic on an open neighborhood of Δ, and let $L > 0$ be a Euclidean Lipschitz constant for its ambient gradient on Δ. For every fixed $\eta > 0$ satisfying $\eta L < 1$ and every $x_0 \in \operatorname{ri}\Delta$, the original exact iteration (M.1) *converges to one point $x_* \in \Delta$ and satisfies*

$$\sum_{k=0}^{\infty} \|x_{k+1} - x_k\| < \infty. \tag{S.4}$$

There is a scalar γ_* such that

$$\partial_i p(x_*) = \gamma_* \quad (x_{*,i} > 0), \qquad \partial_i p(x_*) \geq \gamma_* \quad (x_{*,i} = 0). \tag{S.5}$$

Thus the limit is a KKT point for the full simplex; it need not be a global minimizer. In particular, the conclusion applies to every polynomial objective on the closed 2-simplex.

Lemma S3 (Eventual signs for an analytic half-plane germ)**.** *Let $T = (F, U)$ be a real analytic map germ at $(0,0)$ satisfying*

$$\begin{aligned} T(0,0) &= (0,0), \qquad & U(x,u) &= uB(x,u), \\ B(0,0) &= \beta > 0, \qquad & \partial_x F(0,0) &= \lambda > 0. \end{aligned} \tag{S.6}$$

Let $q_k = (x_k, u_k)$ be an orbit with $u_k > 0$ and $q_k \to (0,0)$. For each real analytic scalar germ G, there is an index N_G after which $G(q_k)$ is strictly positive, strictly negative, or

identically zero. The index may depend on G and the orbit. Moreover, for some N, u_k is strictly decreasing and x_k is strictly monotone toward zero or identically zero, and

$$\sum_{k\geq N}\big(|x_{k+1}-x_k|+|u_{k+1}-u_k|\big)=|x_N|+u_N. \tag{S.7}$$

No sign or relative-scale hypothesis on x_k is imposed. The sign conclusion also has the following parameterized form. Let $\mathcal{T}=(\mathcal{R},\mathcal{F},\mathcal{U})$ be an analytic germ at $(\rho,x,u)=0$, fixing zero, with

$$\mathcal{U}(\rho,x,u)=u\mathcal{B}(\rho,x,u),\qquad \mathcal{B}(0)>0,\qquad \partial_x\mathcal{F}(0)>0. \tag{S.8}$$

Suppose an actual orbit tends to zero, has $u_k>0$, and either $\rho_k=0$ on a tail or

$$\rho_k>0,\qquad \rho_k=o(u_k^M)\quad \textit{for every fixed } M>0. \tag{S.9}$$

Then every fixed analytic $G(x,u)$ has one eventual strict sign or is zero. This last assertion concerns observables independent of ρ and does not assert monotonicity of the actual parameterized increments.

Proof. We prove the two sign statements by one exact discrete barrier. The only plane-curve input is finite convergent Puiseux decomposition of a real analytic zero-set germ; see [8, Chapter 2]. The unparameterized case is obtained by adjoining the constant parameter $\rho=0$.

Write $F=\mathcal{F}$, $U=\mathcal{U}$, and $B=\mathcal{B}$ temporarily. Consider one convergent Puiseux graph $x=h(u)$, $u>0$, with $h(u)\to 0$. If it is nonzero, its first term is cu^r, where $c\neq 0$ and $r>0$ is rational, so $|h'(v)|\leq Cv^{r-1}$; for $h=0$ the derivative is zero. On a small full outer box choose $0<m\leq B\leq M$, $|B_x|\leq K$, and $|F_x-\lambda|<\lambda/4$, where $\lambda=F_x(0)>0$. Thus $mu\leq U\leq Mu$ and $|U_x|\leq Ku$, uniformly in both transverse variables. For $\psi(x,u)=x-h(u)$,

$$\partial_x(\psi\circ\mathcal{T})=F_x-h'(U)U_x=\lambda+o(1)+O(u^r)>\lambda/2>0. \tag{S.10}$$

The factor u in $U_x=uB_x$ compensates the possibly unbounded Puiseux slope. Fix the outer transverse interval first, and shorten the positive u interval until $h(u)$ lies in its middle half and every argument in $[mu,Mu]$ is in the positive domain of h. Late actual sources and their entire horizontal segments to $(\rho,h(u),u)$ then lie in this box. Convergence supplies these domain conditions; no forward invariance or comparison orbit is assumed.

The exact boundary displacement is $\beta(\rho,u)=F(\rho,h(u),u)-h(U(\rho,h(u),u))$. Choose $D\geq 1$ so that $H_b(t)=h(t^D)$ is analytic. The positive unit $A(t,\rho)=B(\rho,H_b(t),t^D)$ has an analytic positive Dth root, and

$$\widehat{\beta}(t,\rho)=F(\rho,H_b(t),t^D)-H_b\big(tA(t,\rho)^{1/D}\big)=\beta(\rho,t^D) \tag{S.11}$$

is jointly analytic near zero. This is a convergent composition, rather than a formal truncation. On a zero-parameter tail, $\widehat{\beta}(t,0)$ is either zero or has a fixed strict sign for small $t>0$. On a positive-parameter tail, if $\widehat{\beta}$ is not identically zero, take its least nonzero ρ coefficient and then that coefficient's least t order. Analytic division gives

$$\widehat{\beta}(t,\rho)=\rho^{j}\big[t^{m}c(t)+\rho R_b(t,\rho)\big],\qquad c(0)\neq 0, \tag{S.12}$$

with finite integers $j,m\geq 0$ and bounded analytic R_b on a fixed box. At $t_k=u_k^{1/D}$, (S.9) gives $\rho_k=o(t_k^m)$ if $m>0$; if $m=0$, $\rho_k\to 0$ suffices. Consequently the boundary displacement again has an eventual fixed strict sign. If $\widehat{\beta}=0$, it is exactly zero.

Integration on the full source comparison segment gives

$$\psi(\mathcal{T}(\rho,x,u))-\beta(\rho,u)=\int_{h(u)}^{x}\partial_x(\psi\circ\mathcal{T})(\rho,v,u)\,dv. \tag{S.13}$$

The right side has exactly the sign of $x-h(u)$. A positive boundary displacement makes the nonnegative side map strictly positive, allowing at most one entry into that side and no return. A negative displacement gives the reverse rule. When it is zero, both strict sides and equality are preserved. The parameter horn is used only at the actual source pair (ρ_k,u_k), which stays fixed on this comparison segment; no property of the comparison endpoint's $\mathcal{R}$ image is required. Thus every graph has an eventual fixed side, including possible equality.

For a nonzero analytic $G(x,u)$ vanishing at zero, its local zero set has finitely many geometric analytic half-branches and no isolated zeros accumulating at zero. Omit branches in $u\leq 0$. A remaining branch has an analytic parametrization with $u(t)=dt^n+\cdots$, $d>0$; its convergent Puiseux inverse expresses it as $x=h_j(u)$, including the zero graph. Distinct Puiseux differences have a fixed sign after a common shrinking, so these graphs can be ordered. Their complementary strips in a small half-box are connected: interpolation between each pair of continuous bounding graphs identifies its strip with $(0,\sigma)\times(0,1)$. Apply the one-way rule to all finitely many graphs and take a common late index. The projected orbit is then on a zero branch or stays in one complementary strip, where G has one strict sign. The cases $G=0$, $G(0)\neq 0$, or no positive half-branch are immediate. No analyticity or definability of the orbit is used.

For the original two-variable map, apply this result to its analytic observables $U-u$ and $F-x$. Positive or zero normal displacement contradicts $u_k>0$ and $u_k\to 0$, so u_k is eventually strictly decreasing. The tangential displacement has a fixed strict sign or is zero; convergence forces monotonicity toward zero or eventual equality to zero. Telescoping proves (S.7). This last step uses observables of the two-variable map and is not claimed for general three-variable parameterized displacements. ∎

Proof of Theorem S1. We retain the exact source–target weights $w_i=x_ix_i^{+}$ throughout. The reciprocal identities (S.1), the uniform ratios (S.2), and descent (S.3) apply under the

present hypotheses with the original L and η. No asymptotic replacement of the mirror map is used.

1. Cluster sets and interior capture. Write $\Delta x_k = x_{k+1} - x_k$ and $p(x_k) \downarrow \ell$. Descent gives $\Delta x_k \to 0$ and $V_k \to 0$, where $V_k = \sum_i w_{i,k}(g_{i,k} - \mu_k)^2$ is defined above. Compactness supplies a nonempty compact cluster set ω, with

$$\operatorname{dist}(x_k, \omega) \to 0, \qquad p \equiv \ell \text{ on } \omega. \tag{S.14}$$

The distance assertion follows by extracting a cluster point from any subsequence on which it failed. If ω were disconnected, it would be the union of two disjoint nonempty compact sets at positive distance. Choose disjoint neighborhoods of these sets that are still separated by a positive distance. By (S.14) the entire late tail lies in their union, while $\Delta x_k \to 0$ prevents passages between them. Both sets therefore cannot contain cluster points, a contradiction. Hence ω is connected. Moreover,

$$V_k = \frac{1}{\sum_i w_{i,k}} \sum_{i<j} w_{i,k} w_{j,k} (g_{i,k} - g_{j,k})^2$$

shows that the supported gradient components agree at every cluster point: along a subsequence tending to that point, the target has the same limit and every corresponding supported weight tends to a positive square. These are the conclusions of Lemma S1; they require no semialgebraicity.

If the trajectory is eventually stationary, its interior stationary point already satisfies (S.5). Otherwise descent and uniqueness of the exact step give

$$P_k := p(x_k) - \ell > 0 \quad \text{for every } k. \tag{S.15}$$

Indeed, a zero gap would force that step, and all subsequent steps, to vanish.

For clarity, the interior capture argument of Lemma S2 also applies here using the ordinary analytic gradient KL inequality. If $\bar{x} \in \omega \cap \operatorname{ri} \Delta$, choose a relative ball U about $\bar{x}$ whose closure is interior and contained in an analytic KL neighborhood. At each source $x_k \in U$, the ratio bound gives a uniform positive lower bound for the target coordinates, whether or not the target lies in U. For $T = \{v : \sum_i v_i = 0\}$, projection of (S.1) therefore yields

$$\|\nabla_T p(x_k)\| = \eta^{-1} \|\Pi_T \operatorname{diag}(w_{i,k}^{-1}) \Delta x_k\| \le C \|\Delta x_k\|.$$

Let φ be a concave KL desingularizer with $\varphi(0) = 0$. For every sufficiently late source in U, the positive-gap KL inequality and (S.3) imply

$$\varphi(P_k) - \varphi(P_{k+1}) \ge \varphi'(P_k)(P_k - P_{k+1}) \ge c_0 \|\Delta x_k\|.$$

Choose a late visit within one quarter of the radius of U, with $\varphi(P_k)/c_0$ smaller than one quarter of that radius. Summing through a hypothetical first exit contradicts the distance

required to leave U; the source-local estimate includes the exit step. The remaining length is finite and the orbit converges to $\bar{x}$. Consequently, any nonsingleton cluster set must satisfy $\omega \subset \partial\Delta$.

2. An edge interval and a global finite-order factorization. Assume for contradiction that ω is not a singleton. A compact connected nonsingleton subset of the boundary circle contains an arc. Indeed, if it is a proper subset, remove a boundary point outside it and identify the punctured circle with $\mathbb{R}$; its image is a nondegenerate compact interval. The whole-circle case is immediate. Since the triangle has only three vertices, a nonempty open interval I in the relative interior of one edge lies in ω. Relabel that edge as $F = \{x \in \Delta : x_3 = 0\}$, set $u = x_3$, and use affine coordinates $(s, u) = (x_2, x_3)$, so $x_1 = 1-s-u$. The function

$$P(s, u) = p(1 - s - u, s, u) - \ell$$

is analytic on a connected convex open neighborhood $\mathcal{U}$ of the closed affine triangle, after shrinking the domain if necessary. Such a neighborhood exists because the triangle is compact and convex. The intersection of $\mathcal{U}$ with $u = 0$ is a connected interval extending beyond both endpoints of F. By (S.14), $P(s, 0) = 0$ on I; the one-variable identity theorem makes it zero on that entire line interval.

Let $a \geq 1$ be the least integer for which $c_a(s) = (a!)^{-1}\partial_u^a P(s, 0)$ is not identically zero on the line interval. If no such integer existed, analyticity would make P vanish on a nonempty affine open set and then throughout $\mathcal{U}$ by the identity theorem. The objective would be constant on Δ, and the exact Burg map would be the identity, contrary to the assumed nonsingleton cluster set. Every lower normal coefficient vanishes identically. Local analytic division near $u = 0$, patched with P/u^a off that line, now gives

$$P = u^a H, \qquad H|_F \not\equiv 0, \tag{S.16}$$

with H analytic on a neighborhood of the whole closed triangle. The local quotients agree on overlaps because they agree wherever $u \neq 0$. In particular, H and its first two derivatives are bounded on the triangle, including its vertices.

To use ambient derivatives in the exact identities, extend $H(s, u)$ independently of x_1 and replace p there by the equivalent representative $\widetilde{p}(x) = \ell + x_3^a H(x_2, x_3)$. The two representatives agree on an affine neighborhood of Δ; their gradients differ there only by a common multiple of $(1, 1, 1)$. This common shift cancels both in the constrained mirror subproblem and in $g_i - \mu$ in (S.1). The orbit is unchanged, and the original L is still used only for descent of the original objective. No Lipschitz bound for $\widetilde{p}$ is imposed on the step size. Below, derivatives of H refer to this ambient extension.

By (S.15), $H(x_k) = P_k/u_k^a > 0$. Thus $H \geq 0$ on I. The analytic function $H|_F$ cannot vanish on all of I, so there is an actual cluster point $\bar{x} \in I$ with

$$A := H(\bar{x}) > 0. \tag{S.17}$$

No positivity is assumed at any other point of the edge.

3. Exact active and normal identities, uniformly up to vertices. Suppress k and write $h_i = \partial_i H$ and

$$S_A = w_1 + w_2, \quad w = w_3, \quad \alpha_i = \frac{w_i}{S_A}\ (i = 1, 2), \quad \theta = \frac{S_A}{S_A + w}, \quad \Lambda = w\theta.$$

Also set

$$\bar{h} = \alpha_1 h_1 + \alpha_2 h_2, \qquad V_A = \sum_{i=1}^{2} w_i (h_i - \bar{h})^2.$$

For the representative in (S.16), the active gradient average is $u^a \bar{h}$, while the normal component is $a u^{a-1} H + u^a h_3$. Substitution into (S.1) gives the exact decomposition

$$\begin{aligned} \mathcal{D} &= u^{a-1}\big[aH + u(h_3 - \bar{h})\big], \qquad \delta := u - u^+ = \eta\Lambda\mathcal{D}, \\ \Delta x_i &= z_i + \alpha_i \delta \quad (i = 1, 2), \qquad z_i = -\eta u^a w_i (h_i - \bar{h}), \quad z_1 + z_2 = 0. \end{aligned} \tag{S.18}$$

When $u \le 1/2$, (S.2) yields

$$S_A \ge Q^{-1}(x_1^2 + x_2^2) \ge \frac{(1-u)^2}{2Q} \ge \frac{1}{8Q}, \qquad Q^{-1}u^2 \le w \le Qu^2.$$

It follows that fixed positive constants satisfy

$$c_\Lambda u^2 \le \Lambda \le C_\Lambda u^2 \qquad (0 < u \le 1/2). \tag{S.19}$$

Neither active coordinate is required to have its own positive lower bound. These estimates therefore include sources arbitrarily close to either endpoint of F.

Let

$$B_H = \max_{\Delta, i} |\partial_i H|, \qquad M_H = \max_{\Delta} \|\nabla^2 H\|, \qquad H_{\max} = \max_{\Delta} |H|.$$

For every fixed $b > 0$, sufficiently small sources with $H \ge b$ satisfy $aH + u(h_3 - \bar{h}) \ge ab/2$. Together with (S.19) and boundedness of H and its derivatives, this proves

$$0 < c_b \eta u^{a+1} \le \delta \le C_b \eta u^{a+1}. \tag{S.20}$$

The target remains interior, so $0 < \delta < u$ at those sources. All source-smallness thresholds may depend on the already fixed η; they do not strengthen $\eta L < 1$.

4. A corrected height decreases at every sufficiently small positive-height source. Embed $z = (z_1, z_2, 0)$ and set $e = (\alpha_1, \alpha_2, -1)$. Then (S.18) gives

$$\Delta x = z + \delta e, \quad \|e\|^2 \le 2, \quad \|z\|^2 \le \eta^2 u^{2a} V_A, \quad \|\Delta x\|^2 \le 2\eta^2 u^{2a} V_A + 4\delta^2,$$

and

$$\langle \nabla H, \Delta x \rangle = -\eta u^a V_A + \delta(\bar{h} - h_3).$$

Taylor's theorem on the actual feasible source–target segment gives

$$H(x^+) - H(x) \le -\eta u^a (1 - M_H \eta u^a) V_A + 2B_H \delta + 2M_H \delta^2.$$

Choose once and for all

$$K = 2B_H + 2M_H + 1, \qquad W = H + Ku.$$

For every fixed $b > 0$, at each sufficiently small source with $H \geq b$, (S.20) holds and $M_H \eta u^a \leq 1/2$. Using $0 < \delta < u \leq 1$, we obtain

$$W(x) - W(x^+) \geq \frac{\eta}{2} u^a V_A + \delta. \tag{S.21}$$

This inequality requires only the source conditions. It assumes neither a target-height lower bound, an individual active-coordinate lower bound, nor a future stay in a collar of F.

5. Positive upcrossings cannot be supplied by excursions. The global gap $P_k = u_k^a H_k \to 0$ and $H_k > 0$ give, for each fixed $b > 0$,

$$H_k \geq b \quad \Longrightarrow \quad u_k \leq (P_k/b)^{1/a} \longrightarrow 0 \tag{S.22}$$

uniformly over such late indices. Hence every sufficiently late source with $H_k \geq b$ satisfies (S.21), regardless of where the active coordinates lie. Bounded derivatives of H and $\Delta x_k \to 0$ also give $|H_{k+1} - H_k| \to 0$.

Fix $0 < b < c$. There cannot be arbitrarily late passages from a value $H \leq b$ to a subsequent value $H \geq c$. Suppose otherwise, and in such a late passage let m be the first index with $H_m \geq c$ and let $j - 1$ be the last preceding index with $H_{j-1} \leq b$. Then

$$H_j \leq b + o(1), \qquad H_k > b \quad (j \leq k \leq m).$$

If $j = m$, vanishing height increments contradict $c > b$ directly. Otherwise (S.21) applies at every source $j, \ldots, m-1$. Since (S.22) gives $u_j \to 0$, telescoping yields

$$c \leq H_m \leq W_m \leq W_j = H_j + Ku_j \leq b + o(1),$$

again a contradiction. This argument allows excursions of arbitrary duration and location before the last low index.

The contact (S.17) supplies arbitrarily late values near A. If values at most $A/2$ also occurred arbitrarily late, subsequent returns above $3A/4$ would be forbidden upcrossings. Thus $H_k > A/2$ eventually. Now (S.22) forces $u_k \to 0$ on the whole tail, and (S.21) makes W_k eventually decreasing and bounded below. Its limit is identified by the subsequence tending to $\bar{x}$. We have therefore proved, before assuming any protected tail,

$$u_k \to 0, \qquad H_k \to A > 0, \qquad W_k \downarrow A \quad \text{eventually.} \tag{S.23}$$

6. A second identity theorem and summable tangential drift. Every point of the original interval I remains a cluster point. Equation (S.23) implies $H = A$ on I. The

one-variable identity theorem now makes $H|_F$ identically equal to A, including an interval beyond both endpoints. The same analytic division argument used in step 2 gives

$$H = A + uJ \tag{S.24}$$

on a neighborhood of the whole closed triangle, with J analytic there and its first derivatives bounded on Δ. For the active directions this gives $h_1 - h_2 = O(u)$ uniformly, including near the vertices: for the chosen ambient extension, $h_1 = 0$ and $h_2 = u\partial_s J$. Since each $h_i - \bar{h}$ is bounded by $|h_1 - h_2|$, (S.18) yields

$$|z_i| \le C\eta u^{a+1} \quad (i = 1, 2).$$

On the late positive-height tail, (S.20) bounds these terms by a fixed multiple of δ. The mass-transfer terms obey $|\alpha_i \delta| \le \delta$, and $|\Delta u| = \delta$. Consequently,

$$\|x_{k+1} - x_k\| \le C'(u_k - u_{k+1}) \quad \text{for every sufficiently large } k. \tag{S.25}$$

The right-hand side telescopes, so the tail has finite Euclidean length and one limit. This contradicts the assumed nonsingleton ω. Thus ω is a singleton in every case, and compactness gives $x_k \to x_*$.

7. The missing-coordinate KKT inequalities. The relative-face criticality established in step 1 gives a common supported gradient value γ_* at x_*. Since $w_{i,k} \to x_{*,i}^2$, the weighted multiplier in (S.1) satisfies $\mu_k \to \gamma_*$. If a missing coordinate j had $\partial_j p(x_*) < \gamma_*$, continuity would give an $\varepsilon > 0$ such that

$$\frac{1}{x_{j,k+1}} - \frac{1}{x_{j,k}} = \eta\big(\partial_j p(x_k) - \mu_k\big) \le -\eta\varepsilon$$

for all sufficiently large k. Summation would eventually make a positive reciprocal negative. This contradiction proves (S.5).

8. Finite length at every boundary limit. An interior limit already has finite remaining length by step 1. For a boundary limit, relabel the coordinates so that it is $X_* = (1 - a, a, 0)$ with $0 \le a < 1$; $a = 0$ is the vertex case. Write $X = (z, s, u) = (1 - s - u, s, u)$ and $f(s, u) = p(1 - s - u, s, u)$, which is analytic on a full neighborhood of $(a, 0)$. Subtracting the common gradient component gives the representative $(0, f_s, f_u)$. The exact update and its mass constraint are

$$\begin{aligned} z^+ = \frac{z}{1 - \eta z\mu}, \qquad s^+ = \frac{s}{1 + \eta s(f_s - \mu)}, \qquad u^+ = \frac{u}{1 + \eta u(f_u - \mu)}, \\ C(s, u, \mu) := \frac{z}{1 - \eta z\mu} + \frac{s}{1 + \eta s(f_s - \mu)} + \frac{u}{1 + \eta u(f_u - \mu)} - 1 = 0. \end{aligned} \tag{S.26}$$

At $(s, u, \mu) = (a, 0, 0)$ this constraint vanishes. When $a > 0$, step 1 gives $f_s(a, 0) = 0$; when $a = 0$, the corresponding coordinate terms vanish without requiring that derivative to be zero. In either case

$$C_\mu(a, 0, 0) = \eta\big[a^2 + (1 - a)^2\big] > 0.$$

The analytic implicit-function theorem supplies an analytic multiplier $\mu(s,u)$ with $\mu(a,0) = 0$. The displayed formulas therefore define an analytic germ $T(x,u) = (s^+(a+x,u) - a, u^+(a+x,u))$ fixing zero. Its normal component is $U = uB(x,u)$ with $B(0,0) = 1$ and $B > 0$ on a full small neighborhood. All denominators are positive after shrinking. At feasible interior sources the three image coordinates are positive, sum to one, and satisfy the original mirror optimality equations, so uniqueness of the strictly convex subproblem identifies this germ with the actual exact update.

At a vertex, $s^+ = sA(s,u)$ with $A(0,0) = 1$, so $\partial_x F(0,0) = 1$. At an edge-interior point put $b = 1 - a > 0$. Differentiating the constraint and the supported update at $(a,0,0)$ gives

$$\begin{aligned} C_s &= -\eta a^2 f_{ss}(a,0), \qquad \mu_s = \frac{a^2 f_{ss}(a,0)}{a^2+b^2}, \\ \lambda &:= \partial_x F(0,0) = 1 - \eta \frac{a^2 b^2}{a^2+b^2} f_{ss}(a,0). \end{aligned} \tag{S.27}$$

The affine edge direction is $(-1,1,0)$, of squared ambient norm two; the original gradient-Lipschitz bound on the simplex gives $|f_{ss}(a,0)| \le 2L$ by tangent secants along that edge. Since $a + b = 1$, one has $a^2b^2/(a^2+b^2) \le 1/8$, and hence

$$\lambda \ge 1 - \eta L/4 > 3/4. \tag{S.28}$$

The bound uses the original objective's L and imposes no additional step restriction.

Lemma S3 now applies to the actual boundary tail. Each original-coordinate displacement is an analytic germ, so each physical coordinate is eventually monotone or constant. The missing coordinate u_k eventually strictly decreases, and $s_k - a$ is eventually strictly monotone toward zero or identically zero. At a vertex, $s_k > 0$ forces strict decrease of that coordinate as well. For a sufficiently late N, the affine physical embedding yields

$$\begin{aligned} \|X_{k+1} - X_k\| &\le \sqrt{2}\big(|s_{k+1} - s_k| + |u_{k+1} - u_k|\big), \\ \sum_{k \ge N} \|X_{k+1} - X_k\| &\le \sqrt{2}\big(|s_N - a| + u_N\big) < \infty. \end{aligned} \tag{S.29}$$

Adding the finite prefix proves (S.4) in every case. The local germ argument uses convergence already established in steps 1–7; it is not an input into that global point-selection proof. ∎

Remark (Scope of the planar conclusion). Theorem S1 gives finite Euclidean length for every interior orbit, including all vertex and edge limits, without a sign restriction on the objective or its leading boundary coefficients. Near a boundary limit, each fixed analytic germ has an eventual strict sign or vanishes along the entire tail; the onset index may depend on that germ and the orbit. No orbit scale comparison is assumed. The general analytic finite-length theorem adds no rate or uniform onset bound and makes no claim of global minimization. No higher-dimensional arbitrary-objective convergence or finite Burg-metric boundary length follows. Analyticity across the whole closed simplex remains essential to the proof: it supplies the edge divisions and the boundary map germs.

S2 Structured scalar cancellation

The normal-power budget and exact scalar lift are proved in Main Sections 4.1–4.2. The following applications establish their center-variation hypothesis from the actual reciprocal normal dynamics.

S2.1 A stationary normal contrast

Keep exactly two supported coordinates. Select two literal normals $u = x_3$, $v = x_4$, and write w for any further normals. In the fixed physical chart let

$$r = u - v, \qquad s = u + v, \qquad y = (x_1 - x_2)/2.$$

Suppose locally the normalized objective is

$$F(u, v, w, y) = C(r, y) + B(s, w) - c_*, \qquad C_r(0, y) = 0. \tag{S.30}$$

The total objective is subject to the standing convex analytic and unique-minimum hypotheses. Separate convexity of C and B, with C even in r, is sufficient but not required. The supported Hessian may vanish. The selected pair has equal limiting KKT gaps, which may be positive; further normal gaps and orders are unrestricted subject to the stated total hypotheses.

The ambient realization

$$f(x) = C(x_3 - x_4, (x_1 - x_2)/2) + B(x_3 + x_4, x_5, \ldots) - c_*$$

makes the derivative differences explicit: $\partial_3 f - \partial_4 f = 2C_r$ and $\partial_1 f - \partial_2 f = C_y$. The same differences hold for any ambient representative agreeing on the simplex affine hull. Its gradient difference is normal to that hull, hence a common-coordinate scalar. No normalization contribution from B is discarded.

Proposition S2.1 (Stationary-contrast scalar length)**.** Every interior exact Burg orbit satisfying (S.30) has finite Euclidean physical length. On a late tail, the sign of $u - v$ is fixed, and every positive power of its absolute value has finite total variation. No monotonicity of $u - v$ or of the literal normals is asserted.

Proof. Convexity along r at fixed s, w, y and $C_r(0, y) = 0$ give

$$C_r(r, y) = rb(r, y), \qquad b(r, y) = \int_0^1 C_{rr}(tr, y)\, dt \geq 0.$$

The function b is analytic and bounded on a protected neighborhood. Define the literal reciprocal contrast

$$z = \frac{1}{u} - \frac{1}{v} = -\frac{r}{uv}.$$

Subtracting the actual normal reciprocal equations cancels the multiplier exactly:

$$z^+ = z + 2\eta C_r(r, y) = z\big(1 - 2\eta uvb(r, y)\big). \tag{S.31}$$

Because $u, v \to 0$ and b is bounded and nonnegative, the factor lies in $(0, 1]$ on a late tail. Thus $z = 0$ remains zero, or its sign is fixed and $|z|$ is nonincreasing. Every $|z|^\alpha$, $\alpha > 0$, is bounded and has finite variation. The exact identity

$$|r|^\alpha = |z|^\alpha u^\alpha v^\alpha \tag{S.32}$$

and Proposition 4.1 show that every $|r|^\alpha$ has finite variation. Here the product rule for bounded finite-variation sequences follows, for example, from $|a^+b^+ - ab| \le |a^+||b^+ - b| + |b||a^+ - a|$.

The supported center solves $C_y(r, \psi(r)) = 0$ and is independent of s, w. Its local existence and uniqueness follow from Section 4.2. On either sign of r, the one-variable Puiseux lemma [11, Proposition 1.8.4] gives a positive α such that

$$|\psi(r_2) - \psi(r_1)| \le C\big||r_2|^\alpha - |r_1|^\alpha\big|$$

for points of that sign sufficiently near zero, extending to zero by continuity. A constant branch is immediate. The actual contrast has an eventual fixed sign by (S.31); if it is zero, the center is constant. Equation (S.32) therefore proves actual center variation is finite. Theorem 3 supplies full physical length. ∎

This argument uses the objective's exact reciprocal contrast, not an arbitrary bounded-variation normal path. It does not extend as stated to a general $C(r, s, w, y)$ or to a contrast whose zero-restoration sheet moves with s, w. Section S2.3 treats a particular moving, one-signed forcing class by a different argument.

S2.2 Exchange-symmetric normal clusters

Still assume exactly two supported coordinates. Let $v_1, \dots, v_h$ be a cluster of literal missing coordinates, with $h \ge 2$, and let w denote any remaining normals. Define

$$S = \sum_{i=1}^{h} v_i, \qquad W = \sum_{i<j} (v_i - v_j)^2.$$

Suppose the local affine objective has the form

$$F(v, w, y) = \Phi(W, y) + G(S, w) - c_*. \tag{S.33}$$

Here Φ is analytic near $(0, y_*)$, G is independent of y, and the standing total convexity, analyticity and unique-minimum conditions remain in force.

Proposition S2.2 (Cluster cancellation and scalar length)**.** Every interior exact Burg orbit in (S.33) has finite Euclidean physical length. Every positive power of each pairwise contrast $|v_i - v_j|$ and of W has finite total variation. The cluster size and the number of further normals are arbitrary finite numbers.

Proof. Total convexity forces $\Phi_W \geq 0$ for small $W \geq 0$ and protected y. To see this, fix a centered vector a with $\sum_i a_i = 0$, hold S, w, y fixed, and write $v = S\mathbf{1}/h + ta$. The resulting scalar function

$$t \longmapsto \Phi\left(t^2 \sum_{i<j} (a_i - a_j)^2, y\right)$$

is convex and even. Its derivative has the sign of t, proving $\Phi_W \geq 0$ for positive W; continuity covers $W = 0$. This argument can be made on the segment from the equal-mass cluster point to any sufficiently small feasible cluster point, inside the protected neighborhood.

Differentiation gives $\partial_{v_i} W - \partial_{v_j} W = 2h(v_i - v_j)$. Thus the actual reciprocal contrast $z_{ij} = 1/v_i - 1/v_j$ satisfies

$$z_{ij}^+ = z_{ij}\big(1 - 2h\eta v_i v_j \Phi_W(W, y)\big). \tag{S.34}$$

The normalizing multiplier and common G_S term cancel. Since Φ_W is bounded and nonnegative, the factor is eventually in $(0, 1]$. Each pairwise ordering is then fixed and each $|z_{ij}|^\alpha$ is bounded and nonincreasing. As

$$|v_i - v_j|^\alpha = |z_{ij}|^\alpha v_i^\alpha v_j^\alpha,$$

Proposition 4.1 and the bounded-variation product rule prove all pairwise power assertions.

For $0 < \alpha \leq 1$ and nonnegative finite vectors a, b, concavity gives

$$\left| \left(\sum_j a_j\right)^\alpha - \left(\sum_j b_j\right)^\alpha \right| \leq \sum_j |a_j^\alpha - b_j^\alpha|. \tag{S.35}$$

One may prove this by replacing the coordinates successively and using $|(c + a)^\alpha - (c + b)^\alpha| \leq |a^\alpha - b^\alpha|$ for $c \geq 0$. Apply it to $a_{ij} = (v_i - v_j)^2$ at consecutive iterates. The already proved variation of $|v_i - v_j|^{2\alpha}$ gives finite variation of W^α. For $\alpha > 1$, ordinary variation of W and local Lipschitzness of the power suffice.

The supported center now solves $\Phi_y(W, \Psi(W)) = 0$. It is unique and continuous by Section 4.2 and depends on the single nonnegative variable W. Its bounded analytic zero graph has a convergent one-variable Puiseux parametrization [11, Proposition 1.8.4], giving

$$|\Psi(W_2) - \Psi(W_1)| \leq C|W_2^\alpha - W_1^\alpha|$$

for some $\alpha > 0$, or a constant center. Thus actual center variation is finite, and the scalar lift proves physical length. ∎

Both stationary contrast and cluster cancellation work because the supported center sees one scalar parameter whose positive powers have an exact reciprocal variation control. They do not authorize a general bounded-variation composition rule for a center depending on all normal parameters.

S2.3 A moving one-signed normal contrast

We prove Main Theorem 5. Keep exactly two supported coordinates and write the literal normal vector as $n = (u, v, \xi)$, with any finite number of extra normals ξ. Put $r = u - v$, $s = u + v$, and use the supported coordinate y of Section 4.2. Suppose locally

$$F(n, y) = C(r, y) + B(n) - c_*, \qquad C_r(r, y) = rb(r, y), \qquad 0 < b_0 \le b(r, y) \le b_1. \tag{S.36}$$

The functions are analytic on fixed protected neighborhoods and the total objective has the standing convex analytic and unique-minimum properties. Separate convexity of C, B is not required. After exchanging u, v if necessary, assume that

$$h(n) = B_v(n) - B_u(n), \qquad cs^\nu \le h(n) \le Cs^\nu, \qquad \nu > 2, \tag{S.37}$$

$$|n_j \partial_j h(n)| \le Cs^\nu \quad \text{for every literal normal } n_j. \tag{S.38}$$

These bounds hold uniformly throughout a sufficiently small positive normal box, including all extra-normal parameters. They are objective-level assumptions, not assertions along a presumed asymptotic path.

S2.3.1 Exact forcing and a barrier before comparability

Let

$$z = \frac{1}{u} - \frac{1}{v} = -\frac{r}{uv}, \qquad a = 2\eta uv b(r, y), \qquad g = \eta h(n) > 0.$$

The affine normal derivatives each include the common supported normalization term; it cancels in their difference. Thus the physical derivative contrast is exactly $2C_r - h$. Subtract the actual reciprocal equations to obtain

$$z^+ = (1 - a)z - g. \tag{S.39}$$

On a late tail $0 < a < 1$. While $z \ge 0$, this recurrence decreases z until a possible crossing; once $z < 0$, it stays negative. Hence at most one sign crossing occurs on that tail. If z stays nonnegative forever, it is bounded and every z^γ is nonincreasing. The identity $|r|^\gamma = z^\gamma u^\gamma v^\gamma$ already gives all contrast power variation in that branch. The center argument below then proves length directly.

For the other branch, discard the finite prefix through the crossing and put $t = -z \geq 0$. Then $r = uvt \geq 0$ and the exact recurrence is

$$t^+ = (1-a)t + aq, \qquad q = \frac{h(n)}{2uvb(r,y)} > 0. \tag{S.40}$$

No comparability of u, v is yet known. The symbol estimates give $q \leq Ds^\nu/(uv)$ for a fixed D. Use the exact identity $s^2 = r^2 + 4uv$ and $r = uvt$:

$$\frac{s^\nu}{uv} = s^{\nu-2}(4 + rt) \leq 4s^{\nu-2} + s^{\nu-1}t. \tag{S.41}$$

Choose K so late that $Ds_k^{\nu-1} \leq 1/2$ for every $k \geq K$, and set $\delta = \sup_{k \geq K} s_k$. Equations (S.40)–(S.41) give

$$t_{k+1} \leq (1 - a_k/2)t_k + 4Da_k\delta^{\nu-2} \leq \max\{t_k, 8D\delta^{\nu-2}\}. \tag{S.42}$$

Consequently $t_k \leq T_0 := \max\{t_K, 8D\delta^{\nu-2}\}$ on the tail. In the nonnegative-z branch, z was already bounded. In either case,

$$\frac{|r|}{s} = \frac{uv|z|}{s} \leq \frac{s|z|}{4} \to 0, \qquad \frac{u}{v} = \frac{s+r}{s-r} \to 1. \tag{S.43}$$

Only now may we use, after enlarging K, $1/2 \leq u/v \leq 2$ and $c_0 s^2 \leq uv \leq s^2/4$. This order excludes a circular estimate of q by $Cs^{\nu-2}$ before the barrier. No monotonicity of t, q, r or the normals is claimed.

S2.3.2 Variation of the moving reciprocal equilibrium

Continue on the forced-sign branch. Fix $\alpha > 0$, and set

$$V = t^\alpha, \qquad E = q^\alpha, \qquad \beta = \alpha(\nu - 2) > 0.$$

Here E is an equilibrium power, unrelated to the earlier supported-error norm. In the comparable positive-normal region $q \leq Cs^{\nu-2}$. Logarithmic differentiation gives

$$\begin{aligned} n_j\partial_j \log q &= n_j\partial_j \log h - \mathbf{1}_{j=u} - \mathbf{1}_{j=v} - n_j\partial_j \log b, \\ \partial_y \log q &= -b_y/b. \end{aligned}$$

The lower bound on h and (S.38) bound its logarithmic normal derivatives. Analyticity and $b \geq b_0$ bound the remaining terms; b depends only on r, y. Therefore

$$|n_j\partial_j E(n,y)| \leq C_\alpha s^\beta, \qquad |\partial_y E(n,y)| \leq C_\alpha s^\beta. \tag{S.44}$$

These constants are fixed on a protected comparable box, independently of a later choice of tail index.

To compare consecutive values, first interpolate all normals geometrically with $y = y_k$ fixed, then interpolate y linearly with target normals fixed. The normal interpolation

remains in $1/2 \le u/v \le 2$, because $\log(u/v)$ interpolates linearly. Each intermediate normal is at most the maximum of its endpoints. With $S_k = \sum_j \max(n_{j,k}, n_{j,k+1})$, integration of (S.44) gives

$$|\Delta E_k| \le C_\alpha S_k^\beta \sum_j |\log(n_{j,k+1}/n_{j,k})| + C_\alpha s_{k+1}^\beta |\Delta y_k|. \tag{S.45}$$

The target mass in the second term is retained. Lemma 8 proves, for every finite $T \ge K$,

$$\sum_{k=K}^{T} |\Delta E_k| \le H_K + C_\alpha \delta^\beta Y_{K,T}, \qquad Y_{K,T} = \sum_{k=K}^{T} |\Delta y_k|, \tag{S.46}$$

where H_K is a known finite normal budget independent of T. We do not yet assert finite variation of E; its supported-motion term remains explicit.

S2.3.3 Ordered powers and the product telescope

Equation (S.40) places t_{k+1} between t_k and q_k. A positive power is increasing, so V_{k+1} lies between V_k and E_k. Hence

$$|\Delta V_k| = |V_k - E_k| - |V_{k+1} - E_k|.$$

With $e_k = |V_k - E_k|$, the triangle inequality at the next equilibrium yields $|\Delta V_k| \le e_k - e_{k+1} + |\Delta E_k|$. Sum only over a finite interval and use (S.46):

$$\sum_{k=K}^{T} |\Delta V_k| \le e_K + H_K + C_\alpha \delta^\beta Y_{K,T}. \tag{S.47}$$

No uniform positive lower bound on a or divergence of $\sum a_k$ is needed.

Put $P = (uv)^\alpha$. Proposition 4.1 makes it bounded and of finite variation. The identities

$$r^\alpha = PV, \qquad \Delta(PV)_k = P_{k+1}\Delta V_k + V_k \Delta P_k$$

retain the target factor. Since $\sup P \le \delta^{2\alpha}/4^\alpha$ and $\sup V \le T_0^\alpha$ by the barrier,

$$\sum_{k=K}^{T} |\Delta(r^\alpha)_k| \le K_K + C_\alpha \delta^{2\alpha+\beta} Y_{K,T} = K_K + C_\alpha \delta^{\alpha\nu} Y_{K,T}. \tag{S.48}$$

One finite choice is

$$K_K = (\sup P)(e_K + H_K) + (\sup V) \sum_{k \ge K} |\Delta P_k|.$$

It depends only on known finite quantities, not on T or unknown supported variation. In particular, the possibly large barrier constant T_0 occurs only in this additive term. The feedback coefficient has a fixed local constant times $\delta^{\alpha\nu}$.

S2.3.4 One Puiseux power closes the finite horizon

The actual supported center solves $C_y(r, \psi(r)) = 0$ because B is independent of y. Section 4.2 gives a unique continuous local center, with $\psi(0) = y_*$. Total convexity suffices: any small signed r can be realized by small positive u, v, and $F_{yy} = C_{yy}$ on the resulting slice. Its bounded analytic zero graph is subanalytic. On each fixed sign, the one-variable Puiseux lemma [11, Proposition 1.8.4] gives $\alpha > 0$ and K_ψ with

$$|\psi(r_2) - \psi(r_1)| \le K_\psi \big||r_2|^\alpha - |r_1|^\alpha\big|. \tag{S.49}$$

The inequality extends to zero by continuity. For a constant branch any positive exponent works. No multivariate Puiseux parametrization is asserted.

If z stays nonnegative, its already established contrast power variation and (M.34) prove supported variation directly. In the forced-sign branch, combine (S.48) for the exponent in (S.49) with the exact finite-horizon scalar lift:

$$Y_{K,T} \le A_K + K_\psi K_K + K_\psi C_\alpha \delta^{\alpha\nu} Y_{K,T}. \tag{S.50}$$

Increase K until the last coefficient is at most $1/2$. This is legitimate because $s_k \to 0$ and the local derivative, Puiseux and scalar-lift constants were fixed independently of K. The additive A_K, K_K may change but are finite for the chosen index. Thus every finite horizon obeys

$$Y_{K,T} \le 2(A_K + K_\psi K_K).$$

Only now let $T \to \infty$. Supported variation is finite without having been assumed earlier. Ordinary normal variation and the affine reconstruction prove physical finite length from every interior start.

After this closure, repeat the equilibrium and ordered-power estimates for any $\gamma > 0$. The supported-motion term in (S.46) is now known finite, so t^γ is of finite variation, and so is $|r|^\gamma = (uv)^\gamma t^\gamma$. The other sign branch was already monotone in reciprocal powers. This proves all claimed reciprocal and physical contrast power conclusions in the correct order.

The assumption $\nu > 2$ is used in the strictly positive exponent $\beta = \alpha(\nu - 2)$ of the weighted-logarithmic budget. At $\nu = 2$ this argument does not close as stated. The positive lower bound on b is used in both the barrier and the logarithmic derivatives. Arbitrary sign-changing forcing, forcing with an interior zero, and vanishing transverse restoration are not covered. These are limitations of this theorem, not finite-length counterexamples. No summability of h_k and no monotonicity of the individual normals is used.

S2.4 Globally convex cusp benchmarks

The following polynomial verifies that the scalar results include genuinely degenerate supported curvature and a center that is not locally Lipschitz at an interior normal

diagonal. Set

$$\varepsilon = \frac{1}{100}, \quad A(y) = 1+2\varepsilon y+5\varepsilon^2 y^2, \quad K_0 = \frac{25}{3}\varepsilon^4, \quad C_0(r,y) = r^2 A(y)+y^4+K_0 r^4. \tag{S.51}$$

The quadratic satisfies $A(y) = 5(\varepsilon y+1/5)^2+4/5 > 0$ globally. Direct differentiation gives

$$(C_0)_{rr} = 2A + 12K_0 r^2 > 0, \quad (C_0)_{ry} = 2rA', \quad (C_0)_{yy} = 10\varepsilon^2 r^2 + 12y^2.$$

Expanding their determinant with the displayed value of K_0 yields the exact identity

$$\det \nabla^2 C_0 = 24A(y)y^2 + 4\varepsilon^2 r^2(1-15\varepsilon y)^2 + 1000\varepsilon^6 r^4 \geq 0. \tag{S.52}$$

Thus C_0 is convex on all of $\mathbb{R}^2$. It is nonnegative and vanishes only at $(0,0)$.

The center equation is

$$4\psi^3 + 2\varepsilon r^2 + 10\varepsilon^2 r^2 \psi = 0. \tag{S.53}$$

The $K_0 r^4$ term has no supported derivative. For $r \neq 0$ write $\psi = |r|^{2/3} w$. The equation becomes $4w^3 + 2\varepsilon + 10\varepsilon^2 |r|^{2/3} w = 0$. Its limiting real root has nonzero derivative in w, so the implicit-function theorem gives

$$\psi(r) \sim -(\varepsilon/2)^{1/3}|r|^{2/3}. \tag{S.54}$$

This is a supported-center equivalent, not an asserted orbit clock or physical-tail equivalent.

S2.4.1 Stationary pairs and arbitrary clusters

For every integer $M \geq 2$, the ambient polynomial

$$f(x) = C_0(x_3 - x_4, (x_1 - x_2)/2) + (x_3 + x_4)^{2M} \tag{S.55}$$

is globally convex, being an affine composition of C_0 plus an even power of a linear form. On the simplex its unique minimizer is $(1/2, 1/2, 0, 0)$; its supported Hessian and both normal KKT gaps vanish. Since $(C_0)_r = r(2A + 4K_0 r^2)$, Proposition S2.1 proves finite physical length for every interior start. The center has the cusp (S.54) on $u = v$, including interior points with $u = v > 0$. Componentwise fractional-power changes are smooth local coordinate changes at such positive points, so they cannot turn this center into a locally Lipschitz function there. The exact contrast argument supplies its variation instead.

For a cluster of any size $h \geq 2$, take

$$\Phi(W,y) = WA(y) + y^4 + K_0 W^2, \qquad G(S,w) = S^{2M} + \sum_j w_j^{2M_j}, \quad M, M_j \geq 2. \tag{S.56}$$

To check global convexity, use orthonormal coordinates r on the centered cluster subspace. Then $W = h\|r\|^2$, and the additional fixed rescaling $t = \sqrt{h}\, r$ gives $W = \|t\|^2$. The

radial/y Hessian is the two-variable Hessian of C_0, with radial coordinate $\|t\|$; every angular eigenvalue is $2(A + 2K_0W) > 0$. Continuity handles $W = 0$. Convexity transfers through the fixed linear change of coordinates, and G is convex. The resulting original-coordinate polynomial is therefore globally convex.

Its only feasible zero has every normal zero and $y = 0$, so the simplex minimizer is unique. The supported Hessian vanishes. Its center satisfies

$$4\Psi^3 + 10\varepsilon^2 W\Psi + 2\varepsilon W = 0, \qquad \Psi(W) \sim -(\varepsilon/2)^{1/3}W^{1/3}.$$

Proposition S2.2 applies with arbitrary additional normals. Nonnegative linear terms in S or the extra normals can add common cluster or extra-normal strict gaps without changing the center or contrast identities.

S2.4.2 Asymmetric forcing for every degree

The asymmetric ambient polynomial

$$f(x) = C_0(x_3 - x_4, (x_1 - x_2)/2) + (x_3 + 2x_4)^{2M}, \qquad M \geq 2, \tag{S.57}$$

is likewise globally convex and analytic. On the simplex it has the same unique minimizer, vanishing supported Hessian and zero pair gaps. In the notation of Section S2.3,

$$b(r, y) = 2A(y) + 4K_0r^2 \geq 8/5, \qquad h(u, v) = 2M(u + 2v)^{2M-1}, \qquad \nu = 2M - 1 \geq 3.$$

Since $s \leq u + 2v \leq 2s$,

$$2Ms^\nu \leq h \leq 2M2^\nu s^\nu, \qquad \frac{uh_u}{h} = \frac{\nu u}{u + 2v} \leq \nu, \qquad \frac{vh_v}{h} = \frac{2\nu v}{u + 2v} \leq \nu.$$

These formulas verify all hypotheses for every $M \geq 2$. The theorem therefore proves finite physical length and every positive contrast-power variation for every such degree. Constants and admitted ambient bounds may depend on M; no finite sample of degrees is used as proof. Global convexity of a polynomial is not an assertion of a finite gradient-Lipschitz constant on all of ambient space: the admitted bound is on the simplex.

The center equation remains (S.53), so $\alpha = 2/3$ is available in the finite-horizon absorption. The actual normal stationary sheet is nevertheless moving. Hold s, y fixed and substitute $u = (s + r)/2$, $v = (s - r)/2$. The actual derivative is

$$F_r = rb(r, y) - M\left(\frac{3s - r}{2}\right)^{2M-1}. \tag{S.58}$$

At $r = 0$, $s > 0$, this is negative. Its r derivative at $(s, r) = (0, 0)$ is $b(0, y) > 0$, uniformly for protected y. The implicit-function theorem gives a local analytic zero sheet $r = \rho(s, y)$. To find its first term, put $\nu = 2M - 1$ and test $r_0 = M(3/2)^\nu s^\nu/b(0, y)$. Its leading residual cancels. The first forcing correction is $O(s^{\nu-1}r_0) = O(s^{2\nu-1})$, while

$b(r_0, y) - b(0, y) = O(r_0^2)$ contributes only $O(r_0^3)$. A uniformly nonzero r derivative converts the residual to the same root error. Thus

$$\rho(s, y) = \frac{M(3/2)^{2M-1}}{b(0, y)} s^{2M-1} + O(s^{4M-3}), \tag{S.59}$$

uniformly on that y interval. Its positive leading coefficient gives $0 < \rho < s$ for small positive s. This is a genuinely moving cancellation sheet. The recurrence (S.39) retains its nonzero forcing throughout. The one-crossing claim concerns the fixed diagonal $r = 0$, not the moving sheet ρ; no finite-crossing claim for ρ is used.

More generally, $C_0(u - v, y) + (c_1 u + c_2 v)^{2M} + G(u + v, \xi)$ with $0 < c_1 < c_2$ has force $2M(c_2 - c_1)(c_1 u + c_2 v)^{2M-1}$. Analytic convex G, or further same-oriented terms and higher-order perturbations preserving the uniform symbol assumptions, fit whenever the total constrained minimizer remains unique. Common pair gaps and extra-normal gaps can be included through G.

S2.5 Scope and established inputs

All degenerate-supported length classes here have exactly two supported coordinates. The normal-power estimate allows arbitrary support, but by itself supplies no variation of a multidimensional degenerate supported block. The scalar lift is conditional on variation of the actual center; arbitrary multivariate center dependence, arbitrary sign-changing contrast forcing and vanishing restoration in the forced theorem remain outside the proved structural alternatives. No general degenerate-face or nonunique-minimizer theorem is asserted.

The one-normal Puiseux argument is an alternative proof within earlier single-face coverage. The stationary benchmark (S.55) has transverse minimizer $r = 0$, so its reduced value is exactly $s^{2M} + y^4$. The coupled full-jet hypothesis of Main Theorem 9 requires pure supported order equal to radial order plus two. Here that would require $4 = 2M + 2$, which fails for every $M \geq 2$. The exact homogeneous formulation also excludes the cubic mixed term $2\varepsilon r^2 y$.

The external analytic input is the classical one-variable convergent Puiseux lemma [11, Proposition 1.8.4]; no multiparameter analogue is invoked. Existing continuous Hessian-barrier flow results [9] do not automatically sum the literal increments of the nonlinear exact fixed-step map. Similarly, finite-continuous-kernel boundary results [12, Theorem 1 and Section 4.3] do not directly apply to the Burg kernel, which diverges at a zero coordinate. Every infinite-tail conclusion above follows from the displayed analytic estimates and exact recurrences; finite diagnostics are not used as proofs.

S2.6 Critical quadratic forcing

We prove Main Theorem 6. The standing convex analytic and unique-minimum assumptions of Main Section 4 remain in force, with exactly two supported coordinates and

the original ambient $\eta L < 1$. Use the literal normal vector $n = (u, v, \xi)$, scalar supported coordinate y, and $r = u - v$, $s = u + v$. Suppose locally

$$F = C(r, y) + B(u, v, \xi) - c_*, \qquad C_r = rb(r, y), \qquad 0 < b_0 \le b \le b_1, \tag{S.60}$$

with analytic functions on protected neighborhoods. The total objective is convex; separate convexity of the summands is not required. After exchanging the selected normals if necessary, assume

$$h(u, v, \xi) := B_v - B_u, \qquad c(u + v)^2 \le h(u, v, \xi) \le C(u + v)^2 \tag{S.61}$$

uniformly in a sufficiently small positive normal box, with every extra normal independently variable. An estimate only along a presumed orbit would not suffice. There is no additional derivative-symbol hypothesis in this quadratic theorem; its needed derivative cancellation will follow from analyticity and this full-box bound.

Proposition 4.1 supplies entry and finite variation of every positive literal-normal power. Theorem 3 supplies, for every finite $T \ge N$,

$$Y_{N,T} := \sum_{k=N}^{T} |\Delta y_k| \le A_N^y + \sum_{k=N}^{T} |\Delta \psi(r_k)|, \qquad A_N^y < \infty. \tag{S.62}$$

The constant is known independently of T. The actual supported center solves $C_y(r, \psi(r)) = 0$. Its unique continuous bounded analytic-zero graph depends on r alone. On either fixed sign, the one-variable Puiseux argument in Section 4.2 gives some $\alpha > 0$ and a fixed K_ψ with

$$|\psi(r') - \psi(r)| \le K_\psi \big| |r'|^\alpha - |r|^\alpha \big|. \tag{S.63}$$

A constant branch permits any positive exponent. No supported or center variation has yet been assumed.

S2.6.1 A bounded positive reciprocal interval, fixed before shrinking

As in Section S2.3, the common supported normalization contribution cancels in the derivative contrast, and the actual multiplier cancels in the reciprocal difference. Hence

$$z^+ = (1 - a)z - \eta h, \qquad z = \frac{1}{u} - \frac{1}{v} = -\frac{r}{uv}, \qquad a = 2\eta uvb. \tag{S.64}$$

Eventually $0 < a < 1$. While $z \ge 0$, it decreases; once negative, it stays negative. A zero is followed by a negative value. Thus there is at most one eventual crossing of the fixed diagonal. If z remains nonnegative forever, every z^γ, $\gamma > 0$, is bounded and nonincreasing. The exact product $|r|^\gamma = z^\gamma u^\gamma v^\gamma$ and normal-power variation give every contrast power finite variation. Equations (S.62)–(S.63) then prove supported variation and physical length in this branch.

Otherwise discard the finite prefix through the crossing and put $t = -z > 0$. Then

$$r = uvt, \qquad t^+ = (1-a)t + aq, \qquad q = \frac{h}{2uvb} > 0. \tag{S.65}$$

Before assuming comparability, set $D = C/(2b_0)$. The identity $s^2 = 4uv + r^2$ gives

$$q \le D\frac{s^2}{uv} = D(4 + rt) \le D(4 + st).$$

Choose a late index N_b such that $Ds_k \le 1/2$ for every $k \ge N_b$. Then

$$t_{k+1} \le (1 - a_k/2)t_k + 4Da_k \le \max\{t_k, 8D\}. \tag{S.66}$$

Thus $t_k \le M_b := \max\{t_{N_b}, 8D\}$, with no prior lower bound on uv/s^2. This proves

$$\frac{|r_k|}{s_k} = \frac{u_k v_k t_k}{s_k} \le \frac{s_k M_b}{4} \to 0, \qquad u_k/v_k \to 1.$$

The forever-nonnegative branch also has bounded z and the same ratio conclusion.

There is a fixed lower bound in the positive-t branch. Indeed,

$$q \ge \frac{cs^2}{2uvb_1} \ge \frac{2c}{b_1}.$$

Fix $N_0 \ge N_b$ with $t_{N_0} > 0$ and put

$$m_0 = \min\{t_{N_0}, 2c/b_1\} > 0, \qquad M_0 = M_b. \tag{S.67}$$

The convex-combination recurrence preserves $m_0 \le t_k \le M_0$. These constants are fixed now and are never recomputed when the tail mass cutoff is subsequently decreased. They may depend on the orbit. No uniform bound over all interior starts is claimed.

S2.6.2 Uniform analytic divisibility and the contrast chart

Expand the analytic force in u, v, with analytic coefficient functions of all extra normals:

$$h(u, v, \xi) = h_0(\xi) + h_{10}(\xi)u + h_{01}(\xi)v + \sum_{i+j\ge 2} h_{ij}(\xi)u^i v^j.$$

Fix any small positive ξ and any $a_1, a_2 > 0$. Substitute $(u, v) = \lambda(a_1, a_2)$ into (S.61). Letting $\lambda \to 0$ gives $h_0(\xi) = 0$; dividing the remaining identity by λ and taking the limit gives $a_1 h_{10}(\xi) + a_2 h_{01}(\xi) = 0$. Varying (a_1, a_2) in the positive cone makes both linear coefficients zero. Their analytic dependence on ξ extends these identities from the open positive box to a connected full neighborhood. With no extra normals the same argument is immediate. Grouping the convergent series, or using analytic Taylor remainder, gives the exact analytic ideal identity

$$h = u^2 A(u, v, \xi) + uvE(u, v, \xi) + v^2 G(u, v, \xi) \tag{S.68}$$

on a smaller neighborhood. It is not a finite-order truncation. In particular no extra normal was required to scale proportionally to s.

On the forced-sign branch introduce

$$d(s,t) = 2 + \sqrt{4 + s^2t^2}, \qquad r(s,t) = \frac{ts^2}{d(s,t)}, \qquad uv = \frac{s^2}{d(s,t)}, \tag{S.69}$$
$$u(s,t) = \frac{s + r(s,t)}{2}, \qquad v(s,t) = \frac{s - r(s,t)}{2}.$$

These solve $r = uvt$ and $u+v = s$ exactly. The square root has an analytic branch near $s = 0$ for t in a fixed compact interval. Choose once a slightly larger positive interval containing $[m_0, M_0]$, a small s interval, a small extra-normal box and a protected supported interval. For small positive s, the chart has positive u, v; the supported coordinates are positive throughout the required interpolation box. All bounds below are fixed on this compact box before any further tail shrinkage.

Both $u/s = (1 + ts/d)/2$ and $v/s = (1 - ts/d)/2$ extend analytically to zero. Equation (S.68) consequently yields

$$h(u(s,t), v(s,t), \xi) = s^2 H(s,t,\xi), \qquad H(0,t,\xi) = h_2(1/2, 1/2, \xi), \tag{S.70}$$

where h_2 is the homogeneous degree-two part in u, v. This boundary value is independent of t. Evaluating the force bounds on $u = v = s/2$ and taking $s \to 0$ proves $c \le H(0,t,\xi) \le C$ for small positive extra normals, with the boundary values included by continuity. Analyticity permits a smaller protected box with the needed positive lower bound.

The exact equilibrium in the new chart is therefore

$$q(s,t,\xi,y) = \frac{d(s,t)H(s,t,\xi)}{2b(r(s,t),y)}, \qquad q(0,t,\xi,y) = \frac{2h_2(1/2,1/2,\xi)}{b(0,y)}. \tag{S.71}$$

It extends analytically through $s = 0$, is positive on the protected box, and its boundary value is independent of t. Compact analytic derivative bounds and integration of q_{st} from $s = 0$ give

$$|q_s| + \sum_j |q_{\xi_j}| + |q_y| \le C_0, \qquad |q_t| \le C_1|s|. \tag{S.72}$$

This full analytic cancellation replaces the strictly positive weighted-logarithmic exponent used in Section S2.3. Its zero-exponent version is not invoked.

S2.6.3 First finite-horizon absorption

Write $\mathrm{TV}_N^T(f) = \sum_{k=N}^T |f_{k+1} - f_k|$, with the componentwise sum for a vector, and set $X_{N,T} = \mathrm{TV}_N^T(t)$. The ordered update (S.65) implies

$$|\Delta t_k| = |t_k - q_k| - |t_{k+1} - q_k|.$$

Use the triangle inequality at q_{k+1} and telescope over the finite horizon:

$$X_{N,T} \le |t_N - q_N| + \mathrm{TV}_N^T(q). \tag{S.73}$$

No monotonicity of q, t or lower bound on a_k is used. In particular no assertion about $\sum a_k$ is needed.

Let $s_k \le \delta$ for every $k \ge N$, including all targets. Interpolate consecutive chart coordinates one at a time within the fixed product box. On the t segment the fixed mass is at most δ. Equation (S.72) gives

$$|\Delta q_k| \le C_n\big(|\Delta s_k| + \|\Delta \xi_k\|_1\big) + C_t\delta|\Delta t_k| + C_y|\Delta y_k|.$$

Ordinary normal variation makes $B_N = \mathrm{TV}_N^\infty(s) + \mathrm{TV}_N^\infty(\xi)$ already finite. Thus

$$X_{N,T} \le |t_N - q_N| + C_n B_N + C_t\delta X_{N,T} + C_y Y_{N,T}.$$

Choose the tail so that $C_t\delta \le 1/2$. Then, for every finite T,

$$X_{N,T} \le A_N^t + BY_{N,T}, \qquad A_N^t = 2(|t_N - q_N| + C_n B_N) < \infty, \quad B = 2C_y. \tag{S.74}$$

The coefficient B is fixed independently of subsequent shrinkage, while A_N^t may depend on the chosen index but not on T or any unknown supported variation. This is not yet a claim that t has finite variation.

S2.6.4 Physical contrast powers and the second absorption

Fix the one Puiseux exponent α from (S.63). Since $t \in [m_0, M_0]$,

$$r^\alpha = s^{2\alpha}V(s,t), \qquad V(s,t) = \left(\frac{t}{d(s,t)}\right)^\alpha \tag{S.75}$$

has uniformly bounded V, V_s, V_t on the fixed box. The earlier positive lower bound on t prevents a fractional-power derivative singularity. At fixed t,

$$\partial_s(s^{2\alpha}V) = 2\alpha s^{2\alpha-1}\left(V + \frac{sV_s}{2\alpha}\right).$$

The bracket is uniformly bounded. Integrating between the two positive endpoint masses controls this segment by $C_R|S^{2\alpha} - s^{2\alpha}|$. At the target mass $S \le \delta$, the remaining t segment costs at most $C_R\delta^{2\alpha}|\Delta t|$. Therefore

$$\mathrm{TV}_N^T(r^\alpha) \le C_R\,\mathrm{TV}_N^T(s^{2\alpha}) + C_R\delta^{2\alpha}X_{N,T}. \tag{S.76}$$

Every positive power of $s = u+v$ is already of finite variation. For $0 < \theta \le 1$, the concave-power sum inequality (S.35) bounds its increment by the sum of the u^θ, v^θ increments. For $\theta > 1$, ordinary variation of s and local Lipschitzness of its power suffice. In particular $B_N^\alpha = \mathrm{TV}_N^\infty(s^{2\alpha}) < \infty$ without using supported or reciprocal variation.

Combining (S.63), (S.62) and (S.76), put $D_\alpha = K_\psi C_R$ and $E_N = A_N^y + D_\alpha B_N^\alpha$. All finite horizons obey

$$Y_{N,T} \le E_N + D_\alpha\delta^{2\alpha}X_{N,T}.$$

Substitute (S.74):

$$Y_{N,T} \le E_N + D_\alpha \delta^{2\alpha} A_N^t + D_\alpha B \delta^{2\alpha} Y_{N,T}. \tag{S.77}$$

Keep the original protected box and the original m_0, M_0. Increase N until both $C_t\delta \le 1/2$ and $D_\alpha B\delta^{2\alpha} \le 1/2$. This is possible since $s_k \to 0$ and $\alpha > 0$, with all multiplying constants fixed. The additive constants may change with N but remain finite. Thus

$$Y_{N,T} \le 2(E_N + D_\alpha \delta^{2\alpha} A_N^t)$$

uniformly over finite T. Only now let $T \to \infty$. Supported variation is finite, and (S.74) then proves reciprocal variation. Smallness of C_y was not assumed: the return to the supported center supplies the factor $\delta^{2\alpha}$.

For every $\gamma > 0$, the function $t \mapsto t^\gamma$ is Lipschitz on the fixed interval $[m_0, M_0]$. Hence every t^γ has finite variation. The product $|r|^\gamma = t^\gamma u^\gamma v^\gamma$ proves every physical contrast-power assertion. The other sign branch was already treated. Finally $\|\Delta x\|_2 \le 2\|\Delta n\|_1 + 2|\Delta y|$ proves physical length; every discarded finite prefix is harmless. No rate, specified reciprocal limit, attraction rate to a stationary sheet, or finite number of crossings of that moving sheet has been proved or used.

S2.7 A globally convex quadratic-forcing cusp

For arbitrary fixed parameters $e, d > 0$, set

$$A_e(y) = 1 + 2ey + 5e^2y^2, \qquad K_e = 25e^4/3, \tag{S.78}$$
$$C_e(r,y) = r^2 A_e(y) + y^4 + K_e r^4, \qquad D_d(s,r) = s^2(1 - 2dr + 5d^2r^2) + r^4 + K_d s^4.$$

The global certificate from Section S2.4 holds for every positive e, not just the earlier fixed value:

$$A_e(y) = 5(ey + 1/5)^2 + 4/5, \qquad (C_e)_{rr} = 2A_e + 12K_e r^2 > 0,$$
$$\det \nabla^2 C_e = 24A_e(y)y^2 + 4e^2r^2(1 - 15ey)^2 + 1000e^6r^4 \ge 0.$$

Thus C_e is globally convex and nonnegative, with zero only at $(0,0)$. The identity $D_d(s,r) = C_d(s,-r)$ proves its global convexity by reflection. Equivalently its determinant is

$$24(1 - 2dr + 5d^2r^2)r^2 + 4d^2s^2(1 + 15dr)^2 + 1000d^6s^4 \ge 0,$$

and its ss Hessian entry is positive. Its only zero is also the origin.

Consequently the original-coordinate polynomial

$$f(x) = C_e(x_3 - x_4, (x_1 - x_2)/2) + D_d(x_3 + x_4, x_3 - x_4) \tag{S.79}$$

is globally convex and analytic. Its only feasible zero has $r = s = y = 0$, hence its simplex minimizer is uniquely $(1/2, 1/2, 0, 0)$. The supported face objective is y^4, so the supported

Hessian vanishes. All first derivatives vanish at the minimum, so both normal KKT gaps vanish. A finite original ambient smoothness bound exists on the compact simplex; no globally finite gradient-Lipschitz bound on all of ambient space is asserted.

For $B(u,v) = D_d(u+v, u-v)$, the operator $\partial_v - \partial_u$ equals $-2\partial_r$ at fixed s. The exact force is

$$h = 4ds^2 - 20d^2rs^2 - 8r^3. \tag{S.80}$$

Both remainder terms are retained. On the positive normal cone $|r| \le s$, so

$$|h - 4ds^2| \le (20d^2 + 8)s^3, \qquad 2ds^2 \le h \le 6ds^2 \quad \left(0 < s \le \frac{2d}{20d^2+8}\right).$$

Also $b = 2A_e(y) + 4K_e r^2 \ge 8/5$. These are uniform neighborhood bounds for every fixed $e, d > 0$, not finite parameter tests. Theorem 6 applies. At $r = 0$, $h = 4ds^2$, so this example lies outside every uniform upper bound $h \le Cs^\nu$ with $\nu > 2$.

Since D_d is independent of y, the center equation is unchanged:

$$4\psi^3 + 2er^2 + 10e^2r^2\psi = 0, \qquad \psi(r) \sim -(e/2)^{1/3}|r|^{2/3}.$$

For completeness, setting $\psi = |r|^{2/3}w$ gives $4w^3 + 2e + 10e^2|r|^{2/3}w = 0$; its limiting real root has nonzero derivative, proving the equivalent by the implicit-function theorem. This is a supported-center equivalent, not an orbit asymptotic.

The actual fixed-s contrast derivative is

$$F_r = rb(r,y) - 2ds^2 + 10d^2rs^2 + 4r^3.$$

Its derivative in r at $(s,r) = (0,0)$ is $b(0,y) > 0$, uniformly on a protected supported interval. The implicit-function theorem gives the genuine moving stationary sheet

$$\rho(s,y) = \frac{2d}{b(0,y)}s^2 + O(s^4). \tag{S.81}$$

Indeed substituting the leading term leaves $O(s^4)$ in the exact equation, and the uniformly nonzero r derivative gives the same root error. The leading coefficient is positive, so $0 < \rho < s$ for small positive s; at $y = 0$ it gives $\rho(s,0) \sim ds^2$. No dynamic tracking equivalent or moving-sheet crossing bound is inferred.

The critical theorem removes the forcing-order exclusion at two under its uniform analytic independent-box assumptions. The earlier $\nu > 2$ theorem and its weighted-logarithmic proof remain intact. Both results still require exactly two supported coordinates, positive transverse restoration, the separated scalar center and one-signed force. General degenerate supported blocks, arbitrary multivariate centers, sign-changing force, vanishing restoration and nonunique minimizers remain outside these theorems.

S3 Oriented capture and split clocks

S3.1 Finite-codimension entry and exact coordinates

This section proves the additional capture geometry in Main Theorem 7. Qualitative length for the broader analytic class follows from Main Corollary 2.1. Let f be convex and analytic on a two-sided open neighborhood of the closed standard simplex, let x_* be its unique constrained minimizer, and let N be the set of its $q \geq 1$ missing coordinates. The entire supported-face Hessian is positive definite, with the empty-tangent convention. All normal KKT gaps are zero. The admitted ambient bound and fixed step remain $0 < \eta L < 1$.

Write $V = \{v : \mathbf{1}^T v = 0\}$ and $F(v) = f(x_* + v) - f(x_*)$. Zero gaps and the supported KKT equality imply $D_V F(0) = 0$. Hence $F \geq 0$ on a sufficiently small full convex neighborhood of zero. This full-neighborhood nonnegativity is used in the flag construction; convexity only on the feasible side would not provide the same argument. The local feasible cone is

$$C = \{v \in V : v_i \geq 0 \ (i \in N)\}. \tag{S.82}$$

Let X be the actual target of the Burg map. With $p_i = \partial_i f(x)$, the exact equations are

$$\frac{1}{X_i} - \frac{1}{x_i} = \eta(p_i - \mu), \qquad \Delta x_i = -\eta\omega_i(p_i - \mu), \tag{S.83}$$

$$\omega_i = x_i X_i, \qquad S_\omega = \sum_i \omega_i, \qquad \mu = \frac{\sum_i \omega_i p_i}{S_\omega}. \tag{S.84}$$

These identities retain every source–target weight. A static rotation used below never turns a rotated normal into a literal Burg coordinate.

Lemma S4 (Entry for any fixed finite number of normals)**.** *Every interior-start orbit satisfies*

$$F(v_k) = O(k^{-1}), \qquad x_k \longrightarrow x_*. \tag{S.85}$$

Its supported coordinates are eventually bounded away from zero, the exact map has a local analytic extension at x_*, and $U_i/u_i \to 1$ for every literal normal $u_i = x_i$, $i \in N$.

Proof. Main Lemmas 2 and 3 give convergence, protected support and the inverse-linear value bound. No normal comparability is used for this entry.

At x_*, the mass equation for $X_i = x_i/[1 + \eta x_i(p_i - \mu)]$ has derivative $\eta \sum_{i \notin N} x_{*,i}^2 > 0$ with respect to μ. The analytic implicit-function theorem therefore gives analytic μ and X near x_*. The denominators are nonzero there. For each missing coordinate, its denominator tends to one, proving $U_i/u_i \to 1$ without any normal comparability assumption. ∎

S3.2 A finite convex analytic flag

The next two lemmas are static statements on a full neighborhood. They allow mixed principal monomials and a nonzero analytically flat terminal space.

Lemma S5 (Nested zero spaces)**.** *Let A be convex and analytic near zero in a finite-dimensional Euclidean space V, with $A(0) = 0$ and $\nabla A(0) = 0$. Put $V_0 = V$. Whenever $A|_{V_j}$ is not the zero germ, let P_j be its first nonzero homogeneous term, of degree m_j, and define*

$$V_{j+1} = \{z \in V_j : P_j(z) = 0\}, \qquad E_j = V_j \cap V_{j+1}^{\perp}.$$

Then P_j is globally convex and nonnegative, $m_j \geq 2$ is even, and V_{j+1} is a proper linear subspace of V_j. The degrees strictly increase. After at most $\dim V$ nonzero stages the construction terminates in a subspace W on which A is identically zero as an analytic germ. The spaces and degrees are independent of the choice of complements.

Proof. Stationarity and convexity imply local nonnegativity. For each restriction, positive dilations divided by the first nonzero power converge uniformly on fixed compact sets to P_j. Convexity and nonnegativity pass to this limit on all of V_j. A nonzero nonnegative homogeneous polynomial has even degree, and stationarity excludes degree one.

If $P_j(z) = 0$, convexity and nonnegativity make $P_j(tz) = 0$ for $0 \leq t \leq 1$. The polynomial identity in t extends this equality to all real t. The zero set is convex and closed under every scalar, so it is linear. Since $P_j \not\equiv 0$, this space is proper. The degree-m_j term vanishes identically on it. Therefore any next nonzero restricted term has larger degree. Every nonzero stage drops dimension, so the procedure ends in at most $\dim V$ stages. If the remaining restriction has no nonzero Taylor coefficient, analyticity makes it identically zero near zero. The case $W = \{0\}$ is included. Only the complements, not the nested zero spaces or their orders, depend on the Euclidean splitting. ∎

Write $V = E_0 \oplus \cdots \oplus E_s \oplus W$, and $v = (z_0, \ldots, z_s, w)$ accordingly. Define

$$T(z) = \sum_{j=0}^{s} \|z_j\|^{m_j}. \tag{S.86}$$

Lemma S6 (Uniform weighted coercivity and derivatives)**.** *On a sufficiently small full neighborhood one has*

$$cT(z) \leq A(z, w) \leq CT(z), \tag{S.87}$$

$$\|D_{E_j} A(z, w)\| \leq CT(z)^{1-1/m_j}, \qquad \|D_W A(z, w)\| \leq CT(z). \tag{S.88}$$

The constants are uniform as z approaches any coordinate subspace and w ranges over a smaller flat-variable box.

Proof. Choose orthonormal bases in the whole groups E_j and in W. Every nonzero E_j axis has first positive order m_j: its leading term is $P_j|_{E_j}$, which is positive away from

zero. Every W axis is identically zero. Express a small vector as the average of suitably enlarged signed axis points and apply Jensen. Equivalence of the finite coordinate power sums and the block powers gives

$$0 \le A(z,w) \le CT(z)$$

uniformly on a sufficiently small box. Also $A(0,w) = 0$ throughout the full small W box.

Assign weight $1/m_j$ to every coordinate in E_j and weight zero to W. For each fixed small w, the bound above forces every z-Taylor coefficient of weighted degree below one to vanish. Indeed, under $z_j \mapsto t^{1/m_j} z_j$, a nonzero least-weight polynomial of weight less than one would violate the $O(t)$ bound for some fixed z. Its coefficients cannot cancel for every z. The constant term is already zero. These coefficient functions are analytic in w and vanish throughout its box.

Let $H_w(z)$ be the polynomial of weighted degree one in z. There are finitely many such z monomials; their coefficients are analytic functions of w. All positive weights are rational, so there is a discrete positive gap δ from weight one to the next possible weight. Uniform convergence of the analytic series and its differentiated series on a smaller polydisc gives

$$\begin{aligned} A(z,w) &= H_w(z) + O(T(z)^{1+\delta}), \\ \|D_{E_j} A(z,w)\| \le CT(z)^{1-1/m_j}, &\qquad \|D_W A(z,w)\| \le CT(z). \end{aligned}$$

For uniformity, when $T > 0$ write $z_j = T^{1/m_j} v_j$ with $\sum_j \|v_j\|^{m_j} = 1$ and use this compact weighted unit section. Differentiation in w does not lower positive weighted degree. The estimates at $T = 0$ follow by continuity.

At $w = 0$, H_0 is a globally convex nonnegative polynomial: it is the compact-uniform limit of $A(t^{1/m_0} z_0, \dots, t^{1/m_s} z_s, 0)/t$. Its zero set is linear. To see this even without ordinary homogeneity, convexity and nonnegativity imply vanishing along the segment from zero to any zero, and the polynomial identity extends it to the full line; convexity then gives addition closure.

That zero space is invariant under every weighted dilation. Fix a dilation parameter $t \ne 1$. The distinct weights have distinct eigenvalues; each projection onto a whole group E_j is a polynomial in this fixed dilation. Consequently the zero space splits across the E_j. On each such group $H_0 = P_j|_{E_j}$, positive away from zero. Thus H_0 has no nonzero zero. Compactness of the weighted unit section and weighted homogeneity give $H_0 \ge cT$.

There are only finitely many coefficient functions in H_w. Their continuity preserves this positive lower bound, uniformly on the weighted unit section, after shrinking the w box. Finally shrink the z box to absorb the $O(T^{1+\delta})$ remainder. This proves (S.87) and (S.88). ∎

The whole equal-weight groups are essential in the positivity argument. Positivity merely on selected axes would not suffice. Neither additive separability nor translation

invariance in W is asserted. For instance, $A(z,w) = z^2/(1-w)$ is convex near zero and vanishes identically on $z = 0$, yet its principal coefficient depends on w.

Lemma S7 (Directional control and cone coercivity)**.** *For a fixed $a \in V_j$,*

$$|D_a A(v)| \le C_a A(v)^{1-1/m_j}. \tag{S.89}$$

If a fixed closed cone C satisfies $V_{j+1} \cap C = \{0\}$, then, near zero,

$$A(v) \ge c\|v\|^{m_j} \qquad (v \in C). \tag{S.90}$$

Proof. A vector in V_j has no component in any earlier E_l, $l < j$. Later orders satisfy $m_l \ge m_j$. For small A, their derivative bounds A^{1-1/m_l} are no larger than A^{1-1/m_j}; the possible W component contributes $O(A)$. Lemma S6 proves (S.89).

On the compact unit section of C, projection to $E_0 \oplus \cdots \oplus E_j$ is bounded away from zero, since its kernel is V_{j+1}. Write $v = r\xi$ with $r \le 1$ and $\|\xi\| = 1$. For every $l \le j$, $r^{m_l} \ge r^{m_j}$. The sum of the earlier block powers of ξ has a positive minimum on this unit section. Apply (S.87) to obtain (S.90). This proof permits supported lineality in C. ∎

S3.3 Exact reciprocal capture from the last visible space

Apply the flag to the full affine germ F of Section S3.1. Its terminal flat space obeys $W \cap C = \{0\}$. Otherwise a nonzero vector v in that intersection would yield small feasible points $x_* + tv$ with $F(tv) = 0$, contrary to uniqueness. Since $V_0 \cap C \ne \{0\}$, there is a last index j_* with

$$V_{j_*} \cap C \ne \{0\}, \qquad V_{j_*+1} \cap C = \{0\}.$$

Main Theorem 7 assumes a fixed $a \in V_{j_*}$ whose literal missing-coordinate entries are all positive. The other entries may have either sign and satisfy $\sum_i a_i = 0$. Set $m = m_{j_*}$. Lemma S7 gives

$$F(v) \ge c\|v\|^m \quad (v \in C), \qquad |D_a F(v)| \le C F(v)^{1-1/m}. \tag{S.91}$$

The second estimate holds on the full neighborhood; the first is needed only on the feasible cone.

Lemma S8 (Uniform normal comparability)**.** *Under the hypotheses above, the bounds* (M.45) *and* (M.46) *hold for every interior orbit. In particular, with $R = \sum_{i \in N} u_i$,*

$$R_k^2 \kappa_k^2 = O(k^{-2/m}) \longrightarrow 0. \tag{S.92}$$

Proof. Pair (S.83) with the fixed physical tangent vector a. The multiplier cancels exactly:

$$Q(x) = \sum_i \frac{a_i}{x_i}, \qquad Q(X) - Q(x) = \eta D_a F(v). \tag{S.93}$$

Protected support makes its supported reciprocal contribution uniformly bounded, possibly signed. Therefore

$$Q(x) = \sum_{i \in N} \frac{a_i}{u_i} + O(1), \qquad a_i > 0.$$

The value bound (S.85) and derivative estimate (S.91) yield

$$|Q_{k+1} - Q_k| \le Ck^{-1+1/m}.$$

Summing the positive upper bound, or comparing with its integral, gives

$$Q_k \le Ck^{1/m}.$$

The finite initial segment changes only the constant. Add a fixed bound for the signed supported contribution. The resulting positive normal reciprocal sum bounds every $a_i/u_{i,k}$ separately, so

$$u_{i,k} \ge ck^{-1/m}. \tag{S.94}$$

Independently, cone coercivity and value entry give

$$\|v_k\| \le Ck^{-1/m}, \qquad u_{i,k} \le Ck^{-1/m}. \tag{S.95}$$

Dividing these bounds gives $\kappa_k \le C$. Summing the upper normal bounds proves (S.92). No limiting ray or tracking estimate has been assumed. The resulting comparability is a conclusion. Cone coercivity combined with any lower normal bound gives $F_k \ge c/k$, hence $F_k = \Theta(k^{-1})$. The main physical-tail theorem gives $\mathcal{L}_k = O(k^{-1/m})$ under the retained supported-Hessian hypothesis. ∎

S3.4 Covered planes and nested orders

The examples use the four-coordinate simplex with vertex minimizer $x_* = (1, 0, 0, 0)$ and $u = (x_1, x_2, x_3)$. Every displayed objective is a sum of even powers of linear forms, hence is globally convex and analytic. Each has a unique feasible minimizer and zero normal gaps. Any fixed step admitted by its actual ambient Hessian bound is allowed.

For

$$f(u) = (u_1 + u_2 - u_3)^4 + u_1^6 + u_2^6 + u_3^6,$$

the first degree is four, its zero space is the plane $u_1 + u_2 = u_3$, and the sixth-degree restriction is positive definite there. The terminal space is zero. The last visible plane contains $(1, 1, 2)$ in normal coordinates, which is the physical tangent vector $(-4, 1, 1, 2)$. Main Theorem 7 gives uniform normal comparability without selecting a ray in this plane. Finite physical length follows from Main Corollary 2.1.

For

$$f(u) = (u_1 - u_2)^2 + (u_1 + u_2 - 2u_3)^4 + (u_1 + u_2 + u_3)^6,$$

the successive zero spaces are the plane $u_1 = u_2$, the line $u_1 = u_2 = u_3$, and zero, with orders $2, 4, 6$. The last visible line is strictly positive in all normal coordinates, so the theorem applies with $m = 6$. These adapted linear forms are static coordinates only; their Burg mobility is not claimed diagonal. Adding a positive quadratic form in independent supported affine coordinates produces a nonempty supported tangent without changing the conclusion. Neither example supplies or claims a sharp rate or tail-length/distance equivalent.

S3.5 An exact polynomial obstruction to universal balance

Proposition S3.1 (Polynomially diverging normal ratio)**.** On the four-coordinate simplex, let

$$f(u) = (u_1 - u_2)^2 + (u_1 + u_2)^6 + u_3^4.$$

For an interior start with $u_1 = u_2 = a > 0$, $u_3 = b > 0$, every exact fixed-step orbit satisfying the ambient step condition retains this symmetry and obeys

$$a_k \sim (1152\eta k)^{-1/6}, \qquad b_k \sim (16\eta k)^{-1/4}, \tag{S.96}$$

$$\frac{a_k}{b_k} \sim \frac{(16\eta)^{1/4}}{(1152\eta)^{1/6}} k^{1/12}. \tag{S.97}$$

The orbit has finite physical length. Thus it disproves universal fixed-power polylogarithmic normal comparability, not finite length.

Proof. The objective is globally convex. Zero value on the nonnegative cone forces every normal to vanish, so the supported vertex is its unique minimizer. All normal gaps are zero, and the supported tangent is empty. The exact mirror subproblem is strictly convex and invariant under swapping u_1, u_2 whenever the source has that symmetry. Uniqueness of its solution proves invariance of $u_1 = u_2$.

On this plane the source derivatives are

$$p_1 = p_2 = 192a^5, \qquad p_3 = 4b^3, \qquad p_0 = 0.$$

Protected x_0 and the exact target ratios give

$$0 \le \mu_k \le C(a_k^7 + b_k^5). \tag{S.98}$$

Indeed the normal numerator in (S.84) has these orders and its supported denominator stays positive. Entry and $F = (2a)^6 + b^4$ give

$$a_k \le C(\log k/k)^{1/6}, \qquad b_k \le C(\log k/k)^{1/4}.$$

Both terms on the right of (S.98) are summable. Define

$$c_k = \eta \sum_{j<k} \mu_j, \qquad A_k = \frac{1}{a_k} + c_k, \qquad B_k = \frac{1}{b_k} + c_k.$$

The correction c_k is bounded and the exact reciprocal equations give

$$A_{k+1} - A_k = 192\eta a_k^5, \qquad B_{k+1} - B_k = 4\eta b_k^3. \tag{S.99}$$

Since $a_k, b_k \to 0$, one has $A_k a_k \to 1$, $B_k b_k \to 1$, and both corrected reciprocals diverge. Their relative increments tend to zero. The mean-value theorem or binomial expansion in (S.99) yields

$$A_{k+1}^6 - A_k^6 \longrightarrow 1152\eta, \qquad B_{k+1}^4 - B_k^4 \longrightarrow 16\eta.$$

Cesaro summation proves (S.96), and their quotient gives (S.97). Since $k^{1/12}/(\log k)^b \to \infty$ for every fixed b, no such universal polylogarithmic bound is possible.

The flag has orders $2, 4, 6$ and zero spaces

$$V_1 = \{u_1 = u_2\}, \qquad V_2 = \{u_1 = u_2,\ u_3 = 0\}, \qquad V_3 = \{0\}.$$

The last visible space lies in the coordinate face $u_3 = 0$, precisely excluding the positivity hypothesis of Main Theorem 7.

Finally the proved equivalents imply

$$\frac{a_k^7 + b_k^5}{a_k^5} \longrightarrow 0, \qquad \frac{a_k^7 + b_k^5}{b_k^3} \longrightarrow 0.$$

Thus μ_k is eventually negligible relative to both positive source derivatives. The exact reciprocal equations make a_k and b_k eventually decreasing; mass conservation makes $x_{0,k}$ increasing. Their coordinatewise variations are finite, hence so is the Euclidean physical length. ∎

S4 Nonlinear two-clock tracking

S4.1 Entry and exact boundary coordinates

This section proves Main Theorem 8. Throughout, the simplex has arbitrary finite dimension, the unique constrained minimizer x_* has exactly two zero coordinates, and its full supported-face Hessian is positive. Convexity and analyticity hold on a two-sided open neighborhood of the closed simplex. No optional one-sided or germ-only weakening is used. The admitted ambient bound L and fixed $0 < \eta L < 1$ remain unchanged.

Write $X = \mathcal{T}(x)$ for the exact target and $g = \nabla f(x)$. Its reciprocal equations and mass constraint are

$$\frac{1}{X_i} - \frac{1}{x_i} = \eta(g_i - \mu), \qquad X_i = \frac{x_i}{1 + \eta x_i (g_i - \mu)}, \qquad \sum_i X_i = 1. \tag{S.100}$$

Multiplying the increment equation by the source–target weights $\omega_i = x_i X_i$ and summing gives

$$\mu = \frac{\sum_i \omega_i g_i}{\sum_i \omega_i}. \tag{S.101}$$

Every use of this identity below retains the target weights. Equality on the simplex hyperplane changes an ambient gradient only by a common scalar and hence leaves this map unchanged; it does not license a different step restriction.

Lemma S9 (Value entry and protected support)**.** *Every interior-start orbit satisfies*

$$f(x_k) - f(x_*) = O(\log k/k), \qquad x_k \longrightarrow x_*. \tag{S.102}$$

Each supported coordinate is bounded below by a positive constant on a tail.

Proof. This is the comparator and compactness argument of Main Lemma 2; its hypotheses hold here. Positive limiting supported coordinates are protected. No rate or finite-length conclusion is used for entry. ∎

Let N be the two missing coordinates and $A = N^c$. Choose a fixed affine chart

$$x = x_* + Bu + Jy, \qquad B_N = I_2, \quad J_N = 0, \quad \mathbf{1}^T B = 0, \quad \mathbf{1}^T J = 0,$$

where J has full column rank and spans the entire supported tangent. Normalize the local objective by $F(u, y) = f(x_* + Bu + Jy) - f(x_*)$, so $F(0, 0) = 0$. The implicit-function theorem gives the analytic active center $y = \psi(u)$ because $F_{yy}(0, 0) \succ 0$. All these center points with small $u \geq 0$ are feasible: their supported coordinates remain positive. Its reduced objective

$$\phi(u) = F(u, \psi(u)), \qquad \phi(0) = 0,$$

has Hessian equal to the Schur complement

$$\nabla^2\phi = F_{uu} - F_{uy}F_{yy}^{-1}F_{yu} \succeq 0$$

on a two-sided neighborhood. Hence ϕ is locally convex, its gradient at zero is the normal KKT-gap vector $\gamma \geq 0$, and $S = \nabla^2\phi(0) \succeq 0$. Uniform strong active convexity gives

$$c\|y - \psi(u)\|^2 \leq F(u, y) - \phi(u) \leq C\|y - \psi(u)\|^2. \tag{S.103}$$

An empty supported tangent needs no elimination: $\phi = F$ and all active terms are absent.

The exact map extends analytically to a neighborhood of x_* in these affine coordinates. At x_* all supported gradients agree, and the derivative of the mass equation in (S.100) with respect to μ is $\eta \sum_{a \in A} x_{*,a}^2 > 0$. Its implicit solution gives analytic μ and X with nonzero denominators. This extension is used to estimate the original feasible map, not to define a different iteration.

S4.2 The automatic convex center estimate

Lemma S10 (Flatness of a minimizing center)**.** *Let $F(z, r)$ be real analytic and convex on a two-sided neighborhood of zero, with $F(0) = 0$, $\nabla F(0) = 0$ and $\nabla^2 F(0) = \mathrm{diag}(A, 0)$,*

$A \succ 0$. *Let* $z = \Phi(r)$ *be its analytic stationary center and* $f_0(r) = F(\Phi(r), r)$. *If* f_0 *is nonconstant, there are an even integer* $m \geq 4$ *and* $c > 0$ *such that*

$$f_0(r) = cr^m + O(r^{m+1}), \qquad \Phi'(r) = O(|r|^{m/2}), \qquad \Phi(r) = O(|r|^{m/2+1}). \tag{S.104}$$

With $A(r) = F_{zz}(\Phi(r), r)$ *and* $\zeta = z - \Phi(r)$, *one also has uniformly*

$$F_r(z, r) = f_0'(r) - \Phi'(r)^T A(r)\zeta + O(\|\zeta\|^2), \tag{S.105}$$

$$|F_r(z, r)| \leq C\big(|r|^{m-1} + |r|^{m/2}\|\zeta\| + \|\zeta\|^2\big). \tag{S.106}$$

Proof. Local partial minimization preserves convexity, as also follows from the Schur complement. Its scalar value has zero first and second derivatives at zero. The first nonzero Taylor coefficient of a nonconstant analytic convex function at its minimum is positive and has even order, yielding $m \geq 4$.

For fixed small r, the function $q_r(z) = F_{rr}(z, r)$ is nonnegative on a common z-ball by two-sided convexity. Its gradient and Hessian in z are uniformly bounded on a larger compact box. Choose M large enough both to bound that Hessian and to ensure that $z - \nabla q_r(z)/M$ remains in the larger ball whenever z is in the smaller one. Taylor's upper bound and nonnegativity give

$$0 \leq q_r\big(z - \nabla q_r(z)/M\big) \leq q_r(z) - \frac{\|\nabla q_r(z)\|^2}{2M}.$$

Consequently $\|F_{zrr}\| \leq C\sqrt{F_{rr}}$ on this box.

Differentiate $F_z(\Phi(r), r) = 0$ once. It gives $F_{zr} = -A(r)\Phi'(r)$ at the center, and differentiation of the scalar minimum gives

$$F_{rr}(\Phi(r), r) = f_0''(r) + \Phi'(r)^T A(r)\Phi'(r).$$

A second differentiation of stationarity gives

$$A(r)\Phi'' + F_{zrr} + 2F_{zzr}\Phi' + F_{zzz}[\Phi', \Phi'] = 0.$$

Uniform positivity of $A(r)$, bounded analytic derivatives and smallness of Φ' therefore imply

$$\|\Phi''(r)\| \leq C\big(|r|^{(m-2)/2} + \|\Phi'(r)\|\big).$$

The Hessian hypothesis gives $\Phi'(0) = 0$. Integration followed by Gronwall's inequality, separately on either side of zero, yields $\Phi' = O(|r|^{m/2})$. Integrating again gives (S.104). Finally, Taylor expansion of F_r in z about $\Phi(r)$, using $F_{rz} = -\Phi'^T A(r)$, proves (S.105) and (S.106). All estimates concern analytic functions on actual fixed neighborhoods. ∎

Now assume $\gamma = 0$, $\operatorname{rank} S = 1$ and $\ker S = \operatorname{span}\{a\}$ for $a > 0$. Put

$$\ell_i = 1/a_i, \quad r = \ell^T u/2, \quad P = I - a\ell^T/2, \quad w = Pu \in E_0 := \ker \ell^T. \tag{S.107}$$

A fixed affine shear in the supported variable completes the quadratic square. In coordinates (r, w, e) the quadratic objective is

$$\tfrac{1}{2}w^T S w + \tfrac{1}{2}e^T H e, \qquad H \succ 0.$$

Its entire Hessian kernel is the r-axis. The stable vector $z = (w, e)$ may have arbitrary finite dimension, so Lemma S10 applies to its full analytic center $\Phi(r) = (W(r), E(r))$.

The minimized germ is nonconstant. If it were constant, every nearby center point would have minimum value; for small $r > 0$, its literal normals would be $ra + O(r^2) > 0$, and its supported coordinates would remain positive. These would be distinct feasible minimizers, contradicting uniqueness. Thus the automatic positive even order in (S.104) exists.

For the dynamics, use the exact active center $e = \psi(r, w)$ in the sheared chart and define

$$\delta = w - W(r), \qquad \epsilon = e - \psi(r, w). \tag{S.108}$$

The reused notation ψ denotes active minimization after the shear; in particular $\psi(r, W(r)) = E(r)$. The difference $e - E(r)$ would be the wrong fast error for the following estimates: its active-normal cross forcing can be of order $r\|\delta\|$, rather than $r^2\|\delta\|$.

S4.3 Reciprocal capture before tracking

Strong convexity in the full transverse variable gives $F \geq f_0(r) + c\|z - \Phi(r)\|^2$. Since $\|\Phi(r)\|^2 = O(r^{m+2})$, shrinking the neighborhood yields

$$F(r, w, e) \geq c_0\big(r^m + \|w\|^2 + \|e\|^2\big) \qquad (r \geq 0). \tag{S.109}$$

By Lemma S9, along the actual orbit

$$r_k \leq C(\log k/k)^{1/m}, \tag{S.110}$$

$$\|w_k\| + \|e_k\| + \|\delta_k\| + \|\epsilon_k\| \leq C\sqrt{\log k/k}. \tag{S.111}$$

For the last term, use $\psi(r, w) - E(r) = \psi(r, w) - \psi(r, W(r)) = O(\|\delta\|)$. A crude estimate of ψ about the origin would not suffice at high radial orders. The center itself is smaller than the square-root scale by (S.104) and (S.110).

Let d be the constant physical derivative of the affine chart in the r direction with (w, e) fixed. Its normal block is $a > 0$ and $\mathbf{1}^T d = 0$. Pairing (S.100) with d eliminates the multiplier exactly:

$$q_k = d^T(1/x_k), \qquad q_{k+1} - q_k = \eta F_r(x_k), \qquad q_k = \frac{a_1}{u_{1,k}} + \frac{a_2}{u_{2,k}} + O(1). \tag{S.112}$$

The bounded term comes only from the protected supported coordinates and need not have a fixed sign. Equations (S.106) and (S.110)–(S.111) imply

$$|q_{k+1} - q_k| \le C\left[(\log k/k)^{(m-1)/m} + \log k/k\right].$$

Summation, using $\sum_{j\le k}(\log j/j)^{(m-1)/m} = O(k^{1/m}(\log k)^{(m-1)/m})$, gives

$$q_k \le Ck^{1/m}(\log k)^{(m-1)/m}.$$

The accumulated $O((\log k)^2)$ term is smaller. Each positive normal reciprocal in (S.112) is separately bounded by $q_k + C$, so

$$u_{i,k} \ge ck^{-1/m}(\log k)^{-(m-1)/m}, \qquad i = 1, 2. \tag{S.113}$$

Because $m > 2$, dividing the square-root bound by this lower radial scale proves

$$\frac{w_k}{r_k},\ \frac{e_k}{r_k},\ \frac{\delta_k}{r_k},\ \frac{\epsilon_k}{r_k} \longrightarrow 0, \qquad \frac{u_{i,k}}{r_k} \longrightarrow a_i. \tag{S.114}$$

Moreover $\sum_k r_k^2 = \infty$ by (S.113). No sharp rate, normal monotonicity, prescribed coordinate ratio, or length assertion has been used to establish this captured tube.

S4.4 Two contraction clocks for the exact map

Equip E_0 with the fixed metric $\langle v, w\rangle_* = v^T D_a^{-2} w$, where $D_a = \operatorname{diag}(a_1, a_2)$. For the supported tangent use the fixed positive metric

$$G = J_A^T \operatorname{diag}(x_{*,A}^{-2}) J_A.$$

All finite-dimensional norms below are uniformly equivalent. Constants can depend on the admitted fixed η.

Lemma S11 (Exact slow and fast inequalities)**.** *On a late captured tail there are $\rho < 1$, $c_1 > 0$ and $C < \infty$ such that*

$$\|\epsilon^+\|_G \le \rho\|\epsilon\|_G + Cr^2\|\delta\|_* + Cr^{m+1}, \tag{S.115}$$

$$\|\delta^+\|_* \le (1 - c_1 r^2)\|\delta\|_* + Cr^2\|\epsilon\|_G + Cr^{m+1}, \tag{S.116}$$

$$\frac{|\Delta r|}{r} \le C\big[r(\|\delta\|_* + \|\epsilon\|_G) + r^m\big] = o(r^2). \tag{S.117}$$

If the supported tangent is empty, the fast inequality and every ϵ term are omitted.

Proof. At an active-sheet source all supported gradients equal a common scalar λ. In the literal normal variables, put $d_i = \partial_i\phi(u)$; thus the normal gradients equal $\lambda + d_i$. With the actual weights $\omega_i = x_i X_i$ and $S_\omega = \sum_j \omega_j$, equations (S.100)–(S.101) give

$$D_0 := \mu - \lambda = \frac{\sum_{i\in N}\omega_i d_i}{S_\omega},$$

$$\Delta u_i = -\eta\omega_i(d_i - D_0), \qquad \Delta x_a = \eta\omega_a D_0 \quad (a \in A). \tag{S.118}$$

The denominator is bounded away from zero. At the full center $w = W(r)$ one has $\phi_w = 0$, and hence

$$d_i = \frac{\ell_i}{2} f_0'(r) = O(r^{m-1}).$$

Since $u = ra + o(r)$ there, the full physical defect at that source is $O(r^{m+1})$. The bounded chart, $D\psi$ and W' give the same order for both centered target defects. For any sheet point on the captured tube, bounded Hessian of ϕ gives

$$\|d(r, w)\| \le C(\|\delta\| + r^{m-1}) = o(r). \tag{S.119}$$

We next verify the slow derivative with its exact denominator errors. Fix r and interpolate w between $W(r)$ and $W(r) + \delta$ on the active sheet. These segments still satisfy $u/r \to a$, uniformly, by (S.114) and $W = o(r)$. Along them,

$$\omega_i/r^2 \to a_i^2, \qquad D_w d = (S + o(1))|_{E_0}, \qquad D_w\omega_i = O(r), \qquad d = o(r).$$

Indeed the analytic rational map has $U_i/u_i \to 1$; differentiating its numerator and denominator gives the $O(r)$ derivative of $u_i U_i$. Differentiation of the exact quotient in (S.118) yields

$$D_w D_0 = O(r^2), \qquad D_0 = O(r^2\|d\|).$$

The numerator derivative consists of $O(r)\|d\| + O(r^2)\|D_w d\|$, and the denominator and its derivatives are bounded. Differentiating the normal increment now gives

$$D_w\Delta u = -\eta r^2 D_a^2 S|_{E_0} + o(r^2). \tag{S.120}$$

The term $(D_w\omega_i)(d_i - D_0)$ is $o(r^2)$, while $\omega_i D_w D_0 = O(r^4)$. Thus neither target weights nor the multiplier are being silently discarded.

Since $\delta^+ = w + P\Delta u - W(r + \Delta r)$, differentiation at fixed r gives

$$D_w\delta^+ = I + PD_w\Delta u - W'(r^+)D_w\Delta r = I - \eta r^2 A_0 + o(r^2), \quad A_0 = (PD_a^2 S)|_{E_0}.$$

Here $D_w\Delta r = O(r^2)$ and $W' = O(r^{m/2})$, so the moving-center correction is $o(r^2)$. The projection P is orthogonal in the fixed $*$ metric: $D_a^{-2}a = \ell$ and $\ell^T w = 0$ on E_0. Consequently

$$\langle w, A_0 w\rangle_* = w^T S w > 0 \qquad (0 \ne w \in E_0).$$

The kernel of S is $\mathrm{span}\{a\}$, disjoint from E_0. This makes A_0 a positive self-adjoint operator on the one-dimensional E_0; its exact discrete derivative has norm at most $1 - c_1 r^2$ on a sufficiently late tube. Integrating from the full center, whose defect is $O(r^{m+1})$, proves (S.116) on the sheet.

Off the sheet, differentiate the exact rational normal map in the active error ϵ at fixed u. The derivatives of g and μ are bounded, and the derivative of X_i has a factor u_i^2. Thus

the difference of normal targets is $O(r^2\|\epsilon\|)$. The same bound holds after applying P and $W(r^+)$. This gives the off-sheet term in (S.116).

For the fast error, differentiate $\epsilon^+ = e^+ - \psi(u^+)$ in ϵ at the boundary. Its linearization is

$$A_f = I - \eta G^{-1} H, \quad H = J^T \nabla^2 f(x_*) J \succ 0. \tag{S.121}$$

This follows directly by differentiating the supported reciprocal equations along J, since the normal target derivative in an active direction vanishes at zero normals. The admitted ambient bound gives

$$0 \prec H \preceq L J^T J \preceq L G.$$

The last inequality uses $0 < x_{*,a} \le 1$. Hence A_f is self-adjoint in the G metric with all eigenvalues in $(0, 1)$; no replacement of L is needed. Analyticity yields a fixed contraction factor $\rho < 1$ in a neighborhood.

On the active sheet the physical defect in (S.118) is at most $C \sum_i u_i^2 |d_i|$. Bounded coordinate and center derivatives give

$$\|\epsilon^+(u, 0)\|_G \le C r^2 \|\delta\| + C r^{m+1}.$$

Integrating the ϵ derivative proves (S.115). Measuring the fast error from $\psi(u)$, rather than from $E(r)$, is essential to this estimate.

Finally (S.119), the exact sheet update, and the off-sheet $O(r^2\|\epsilon\|)$ bound prove

$$\|\Delta u\| \le C r^2 \big(\|\delta\| + \|\epsilon\| + r^{m-1}\big). \tag{S.122}$$

Since r is a fixed linear functional of u, this proves (S.117). The last equality follows from $\delta, \epsilon = o(r)$ and $m \ge 4$. All source–target segments stay in the protected analytic box; throughout this proof only the eventual neighborhood has been reduced. ∎

Choose a sufficiently small fixed $\kappa > 0$. Combining (S.115)–(S.116) gives

$$Z^+ \le (1 - c_2 r^2) Z + C r^{m+1}, \qquad Z = \|\delta\|_* + \kappa \|\epsilon\|_G. \tag{S.123}$$

Indeed choose $\kappa C < c_1/2$ to absorb the fast forcing from δ; the reverse coupling is absorbed by the fixed positive gap $1 - \rho$ on a tail. Equation (S.117) implies $(r^+/r)^{m-1} = 1 + o(r^2)$. Dividing (S.123) by $(r^+)^{m-1}$ gives, for $z = Z/r^{m-1}$,

$$z^+ \le (1 - c_3 r^2) z + C_3 r^2.$$

A constant supersolution larger than C_3/c_3 and the starting value proves boundedness by induction. The divergent slow clock is already available from (S.113), although this boundedness alone does not require it. Thus $\delta, \epsilon = O(r^{m-1})$. Substitute this into (S.115) and divide by $(r^+)^{m+1}$. Since $r^+/r \to 1$, the resulting fixed contraction with bounded forcing proves

$$\delta = O(r^{m-1}), \qquad \epsilon = O(r^{m+1}). \tag{S.124}$$

When the supported tangent is empty, the same argument starts directly from (S.116). The vector active case changes only the fixed G norm, not the slow normal space or any exponent.

S4.5 The radial equivalent and physical length

The full-center error in Lemma S10 is

$$z - \Phi(r) = \big(\delta,\, \psi(r,w) - \psi(r,W(r)) + \epsilon\big) = O(\|\delta\| + \|\epsilon\|).$$

By (S.124), it is $O(r^{m-1})$. Formula (S.105) therefore gives

$$F_r = mcr^{m-1}(1 + o(1)).$$

The error terms are $O(r^{3m/2-1})$ and $O(r^{2m-2})$, both smaller than r^{m-1}. By capture and supported protection, the exact quantity in (S.112) satisfies $q \sim 2/r > 0$. Hence

$$\Delta q = \eta mcr^{m-1}(1 + o(1)), \qquad \frac{\Delta q}{q} = O(r^m) \longrightarrow 0.$$

The binomial expansion at the positive q gives

$$\Delta(q^m) = mq^{m-1}\Delta q\,(1 + o(1)) \longrightarrow \eta m^2 c\, 2^{m-1}.$$

Cesaro summation yields $q_k^m/k \to \eta m^2 c\, 2^{m-1}$. Using $q_k r_k \to 2$ proves

$$r_k \sim \left(\frac{2}{\eta m^2 ck}\right)^{1/m}. \tag{S.125}$$

This coefficient is consistent with any fixed positive rescaling of the kernel generator a and its induced radial coefficient c.

The exact normal estimate (S.122) and (S.124) give $\|\Delta u\| = O(r^{m+1})$. Also

$$\|\Delta\epsilon\| \le \|\epsilon^+\| + \|\epsilon\| = O(r^{m+1}),$$

where consecutive radial ratios tend to one. The active center has bounded derivative, so $e = \psi(u) + \epsilon$ gives $\|\Delta e\| = O(r^{m+1})$. Reversing the fixed shear and the affine physical chart proves

$$\|x_{k+1} - x_k\| = O(r_k^{m+1}) = O(k^{-1-1/m}).$$

Its sum is finite; the omitted finite prefix makes no difference. This completes the quantitative clause of Main Theorem 8. A position equivalent has not been differentiated, and no sharp literal increment vector, eventual individual-normal monotonicity, or tail/distance equivalent has been inferred from these upper estimates.

S4.6 Nonlinear witness and scope comparison

For Main (M.50), each summand is a convex even power of an affine form, so the objective is globally convex. A zero must have $w = y = 0$ and then $r = 0$, yielding the unique simplex minimizer $(1/2, 1/2, 0, 0)$. The supported Hessian is positive. Its normal quadratic form is w^2 and hence has the positive kernel $u = v$.

Both full transverse center equations are $z + 2(r + z)^3 = 0$. Analytic implicit solution gives $z = -2r^3 + O(r^5)$. Substitution yields $f_0(r) = 2r^4 + O(r^6)$. The squared terms $w^2 + y^2$ contribute at most 2 to an ambient operator-norm Hessian bound. Since $r + w = x_3$, the term $(r + w)^4$ contributes at most 12 on the simplex. The affine form $r + y = (x_3 + x_4 + x_1 - x_2)/2$ has gradient norm one and absolute value at most $1/2$ on the simplex, so its fourth power contributes at most 3. Thus $L = 17$ is an admitted, not necessarily optimal, bound. Formula (S.125) with $c = 2, m = 4$ gives $kr_k^4 \to 1/(16\eta)$ whenever $\eta < 1/17$.

Static partial minimization is not itself an exact dynamic reduction: the original projected normal sequence retains target weights and supported tracking errors. The two-clock estimates provide that bridge. The present positive-supported-Hessian result differs from the degenerate supported block in the tilted endpoint theorem.

S5 Coupled jets and sharp supported endpoints

S5.1 Analytic reduced class and fixed-clock comparisons

This part proves Main Theorem 9, then derives its exact homogeneous corollary with the stronger explicit bias. All hypotheses of the main statement remain in force: $N \geq 2$ literal missing coordinates, a strictly positive one-dimensional normal kernel, the entire supported tangent of dimension $d \geq 1$, no neutral block, and open-neighborhood convexity and analyticity. In the fixed physical chart $x = x_* + Bu + Ey$, use

$$\ell_i = a_i^{-1}, \quad r = \ell^T u/N, \quad P = I - a\ell^T/N, \quad w = Pu \in E_0 = \ker \ell^T, \quad D_a = \operatorname{diag}(a_i).$$

The normal Hessian is $K \succeq 0$ with $\ker K = \operatorname{span}\{a\}$. Fixed linear coordinates on E_0 define transverse derivatives; embedded vectors define physical displacements. The stationary center and reduced jet are

$$\begin{gathered} F_w(\phi(r,y), r, y) = 0, \quad f_{\mathrm{red}}(r,y) = F(\phi(r,y), r, y), \\ f_{\mathrm{red}}(r,y) = cr^m + c_1 r^{m+1} + \mathcal{J}_p(r,y) + O(s^{p+1}), \\ p = m+2, \quad q = p/2, \quad s = |r| + \|y\|, \quad c > 0, \quad \mathcal{J}_p(0,y) > 0 \quad (y \neq 0). \end{gathered} \tag{S.126}$$

Here $m \geq 4$ is even and the expansion is a full mixed Taylor jet, with differentiated analytic remainders. No joint convexity of $\mathcal{J}_p$ is assumed. Put

$$\delta = w - \phi(r,y), \qquad \lambda = mc/N, \qquad \lambda_1 = (m+1)c_1/N, \qquad C_r = (\eta m \lambda)^{-1/m}. \tag{S.127}$$

The original ambient bound and fixed step $0 < \eta L < 1$ are retained. Bounds may depend on this fixed η; there is no additional small-step limit.

Fix a tail origin k_0 and set $T_k = 1 + \eta(k - k_0)$. If a nonnegative sequence satisfies

$$D_{k+1} \le (1 - c_0 \eta T_k^{-\gamma}) D_k + C\eta T_k^{-\gamma-\beta}, \qquad 0 \le \gamma < 1, \quad \beta > 0,$$

with nonnegative coefficient, then $D_k = O(T_k^{-\beta})$. Indeed

$$A(T + \eta)^{-\beta} \ge AT^{-\beta} - \beta A \eta T^{-\beta-1}.$$

Take $A \ge 2C/c_0$ and then T sufficiently large that $\beta T^{\gamma-1} \le c_0/2$; increasing A to dominate the starting value gives a supersolution and induction. This also applies to a norm contracted by a positive linear map.

For the nonlinear comparison, suppose $R_k \to 0$ and

$$R_{k+1} \le R_k - c_0 \eta R_k^{m+1} + C\eta T_k^{-1-1/m}. \tag{S.128}$$

On a sufficiently small fixed interval the map $R \mapsto R - c_0 \eta R^{m+1}$ is increasing. Choose a late starting index and A so that $c_0 A^{m+1} - C \ge A/m$, $AT^{-1/m}$ dominates the starting value, and this supersolution stays in the interval. The same convex tangent estimate proves $R_k = O(T_k^{-1/m})$. The choice is possible because $R_k \to 0$ and $T_k^{-1/m} \to 0$: one may choose the initial supersolution value as the larger of R_k and a fixed multiple of $T_k^{-1/m}$. Neither comparison assumes radial monotonicity.

S5.2 The analytic transverse center

Throughout this subsection, $t = (r, y) \in \mathbb{R} \times \mathbb{R}^d$, $s = |r| + \|y\|$, and $w \in \mathbb{R}^h$ denotes fixed linear transverse coordinates. All estimates are local and uniform in these coordinates.

Lemma S12 (Radial first jet of the static center)**.** *Let $F(w, t)$ be real analytic and convex on an open neighborhood of $(0, 0)$, with*

$$F_w(0) = 0, \qquad H = F_{ww}(0) > 0, \qquad F_{wt}(0) = 0.$$

Let ϕ be the analytic solution of $F_w(\phi(t), t) = 0$, and put $f_{\mathrm{red}}(t) = F(\phi(t), t)$. Suppose, for an even $m = 2k \ge 4$, that

$$f_{\mathrm{red}}(r, y) = cr^m + c_1 r^{m+1} + \mathcal{J}_{m+2}(r, y) + O(s^{m+3}), \qquad c > 0, \tag{S.129}$$

where $\mathcal{J}_{m+2}$ is homogeneous of degree $m + 2$. Set $q = k + 1 = (m + 2)/2$. Then, for some fixed $b \in \mathbb{R}^h$,

$$\phi = br^q + O(s^{q+1}), \qquad \phi_r = qbr^{q-1} + O(s^q), \qquad \phi_y = O(s^q). \tag{S.130}$$

Moreover,

$$F(0, r, y) = f_{\mathrm{red}}(r, y) + \tfrac{1}{2} b^T H b \, r^{m+2} + O(s^{m+3}). \tag{S.131}$$

Thus transverse elimination changes the critical jet only by a radial term. In particular its restriction to $r = 1$ changes by a constant, without changing its active gradient. Neither positivity nor joint convexity of $\mathcal{J}_{m+2}$ is needed for this conclusion.

Proof. The implicit function theorem gives $\phi(0) = D\phi(0) = 0$. We first establish the uniform estimate

$$\|D^2\phi(t)[v,v]\|^2 \le CD^2 f_{\mathrm{red}}(t)[v,v] \qquad (\|v\| = 1). \tag{S.132}$$

Shrink the neighborhood so that $A(t) = F_{ww}(\phi(t), t)$ is uniformly positive definite. For $z = (D\phi(t)v, v)$, differentiation of $F_w(\phi(t), t) = 0$ and of f_{red} gives

$$z^T D^2 F(\phi(t), t)z = \partial_v^2 f_{\mathrm{red}}(t), \qquad D_w\{z^T D^2 F(w, t)z\}_{w=\phi(t)} = -A(t)\phi_{vv}(t).$$

In the second identity z is held fixed: differentiating the stationary identity twice gives $D^2(F_w)[z, z] + A\phi_{vv} = 0$, and mixed derivatives commute. For any unit transverse vector e, the scalar function

$$g(\tau) = z^T D^2 F(\phi(t) + \tau e, t)z$$

is nonnegative on a fixed interval $[-\delta, \delta]$, independently of small t and unit v, e. Bounded fourth derivatives on a smaller compact neighborhood give $|g''| \le M$, with $M > 0$ uniform. A nonnegative C^2 function on that interval satisfies

$$|g'(0)| \le \sqrt{2Mg(0)} + 2g(0)/\delta. \tag{S.133}$$

Indeed, apply Taylor's upper bound at distance $\min\{\delta, |g'(0)|/M\}$ opposite the sign of $g'(0)$. The two cases give respectively $g(0) \ge |g'(0)|^2/(2M)$ or $g(0) \ge \delta|g'(0)|/2$. Here $g(0) = \partial_v^2 f_{\mathrm{red}} \to 0$ uniformly in v and $g'(0) = -e^T A\phi_{vv}$. Taking the supremum over e, and using the uniform bound for A^{-1}, proves (S.132).

Analytic differentiation of (S.129) yields $D^2 f_{\mathrm{red}} = O(s^{2k-2})$. Hence (S.132), polarization, and integration from 0 give $D^2\phi = O(s^{k-1})$ and $\phi = O(s^{k+1})$. Let Φ be its homogeneous degree-$(k+1)$ jet, possibly zero. For a purely active direction $v = (0, \xi)$, $\partial_v^2 f_{\mathrm{red}} = O(s^{2k})$, so $\phi_{vv} = O(s^k)$. The degree-$(k-1)$ polynomial $D_y^2\Phi$ must therefore vanish. Homogeneity gives

$$\Phi(r, y) = br^{k+1} + r^k Cy \tag{S.134}$$

for a constant linear map C. This does not yet exclude an active-linear critical center term.

Apply (S.132) with $v = \partial_r$ at $t = \varepsilon(r, y)$, divide by ε^{2k-2}, and let $\varepsilon \downarrow 0$. Uniformity of the estimate gives

$$\|\Phi_{rr}(r, y)\|^2 \le C_0 cm(m-1)r^{2k-2}$$

for every (r, y) in a fixed bounded box. Substituting (S.134) and dividing by $|r|^{k-2}$ for $r \ne 0$ yields

$$\|k(k+1)br + k(k-1)Cy\| \le C_2|r|.$$

Letting $r \to 0$ with y fixed proves $C = 0$, since $k \geq 2$. The differentiated analytic remainder gives (S.130). In particular the argument uses arbitrarily small radial-to-active ratios, rather than only a bounded angular region. Finally Taylor expansion in w about its stationary center gives

$$F(0,t) - f_{\mathrm{red}}(t) = \tfrac{1}{2}\phi(t)^T H \phi(t) + O\big(s\|\phi(t)\|^2 + \|\phi(t)\|^3\big).$$

Since $2q = m + 2$, this proves (S.131). ∎

Remark (Derivative information and scope). The center conclusion already follows from $f_{\mathrm{red}} = cr^m + O(s^{m+1})$: the active Hessian estimate then gives $\phi_{vv} = O(s^{k-1/2})$, still excluding the degree-$(k-1)$ active Hessian of Φ. For later use, with $\delta = w - \phi(t)$, Taylor's integral formula gives an analytic matrix $\mathcal{H}(\delta, t)$ with $\mathcal{H}(0,0) = H > 0$ and

$$F(w,t) = f_{\mathrm{red}}(t) + \tfrac{1}{2}\delta^T \mathcal{H}(\delta,t)\delta, \quad F_j - \partial_j f_{\mathrm{red}} = O\big(\|\phi_j\|\|\delta\| + \|\delta\|^2\big), \quad j = r, y.$$

These are static statements. Tracking $w - \phi(r,y)$ along an exact Burg orbit requires its own discrete argument.

Remark (Optional feasible-tube version, static only). The same center conclusion holds if open-neighborhood convexity is replaced by analyticity on a neighborhood and convexity on $\{r \geq 0,\ \|w\| \leq \kappa r,\ \|y\| < \delta_0\}$, with $\kappa > 0$ and the same jet assumptions. Here the fixed interval in the preceding proof is unavailable. For each fixed direction (r,y) with $r > 0$, however, $\phi(\varepsilon r, \varepsilon y) = O(\varepsilon^2) = o(\varepsilon r)$. The graph therefore lies inside the tube for sufficiently small ε, and (S.133) applied on an interval of radius comparable to $r_t = \varepsilon r$ gives

$$\|\phi_{vv}(t)\| \leq C\big(\sqrt{\partial_v^2 f_{\mathrm{red}}(t)} + \partial_v^2 f_{\mathrm{red}}(t)/r_t\big).$$

Constants are direction-independent, although how small ε must be can depend on the direction. The general-direction orders ε^{k-1} and ε^{2k-3} first imply $\phi = O(s^{k+1})$ by analyticity; the active-direction orders ε^k and ε^{2k-1} imply $D_y^2\Phi = 0$. In the radial test, division by ε^{k-1} makes the second term vanish when $k > 2$. For $k = 2$ it is another constant multiple of r. Thus the same radial bound eliminates C by first taking $\varepsilon \downarrow 0$ and then $r \downarrow 0$ with y fixed. This proves the static jet extension only. It neither guarantees feasibility of the unconstrained center on every anisotropic path nor replaces the convexity hypotheses of any dynamical entry argument.

S5.3 Coercivity and the separate entry versions

Taylor's formula about the stationary transverse center gives the exact analytic representation

$$F(w,r,y) = f_{\mathrm{red}}(r,y) + \tfrac{1}{2}\delta^T H(\delta,r,y)\delta, \qquad H(0) \succ 0. \tag{S.135}$$

The fixed transverse coordinate maps are absorbed in constants. Differentiating at fixed (w, r, y) gives

$$\begin{aligned} F_r &= f_{\mathrm{red},r} + O(\|\phi_r\|\|\delta\| + \|\delta\|^2), \\ F_y &= f_{\mathrm{red},y} + O(\|\phi_y\|\|\delta\| + \|\delta\|^2), \\ F_u &= (\ell/N) f_{\mathrm{red},r} + K\delta + O(s\|\delta\| + \|\delta\|^2). \end{aligned} \tag{S.136}$$

Here K acts on the embedded transverse vector. The last line uses $D\phi = O(s^2)$ from the static lemma. The exact radial relation is $a^T F_u = F_r$, not merely a cancellation of $K\delta$.

Positivity of $\mathcal{J}_p(0, y)$ and compactness of the active unit sphere imply $\mathcal{J}_p(r, y) \geq c_0\|y\|^p - C|r|^p$, by Young's inequality for its mixed monomials. Absorb the $c_1 r^{m+1}$ term, $|r|^p$ loss, and analytic remainder $O(s^{p+1})$ on a smaller neighborhood. Uniform positivity of H then yields

$$f_{\mathrm{red}} \geq c_0(|r|^m + \|y\|^p), \qquad F \geq c_0(|r|^m + \|y\|^p + \|\delta\|^2). \tag{S.137}$$

In particular x_* is an isolated local feasible minimizer.

For an actual step $X = x^+$ put $\omega_i = x_i X_i$. The exact equations are

$$\Delta x_i = -\eta\omega_i(g_i - \mu), \qquad \mu = \frac{\sum_i \omega_i g_i}{\sum_i \omega_i}. \tag{S.138}$$

On any compact source set of bounded gradients the multiplier is a weighted mean. The two forms

$$x_i/X_i = 1 + \eta x_i(g_i - \mu), \qquad X_i/x_i = 1 - \eta X_i(g_i - \mu)$$

give uniform upper bounds for both ratios, since $x_i, X_i \leq 1$. Thus the target variance is comparable to the source-square variance $D_F(x) = \min_t \sum_i x_i^2(g_i - t)^2$. Whenever the source–target segment lies in the smooth neighborhood, Euclidean descent and $\omega_i \leq 1$ give

$$F(x) - F(X) \geq \eta(1 - \eta L/2) \sum_i \omega_i(g_i - \mu)^2 \geq c_0 \eta D_F(x). \tag{S.139}$$

This immediately applies to the convergent-tail version at all sufficiently late indices.

For local attraction, choose a convex affine ball of radius ρ around x_* on which the hypotheses and admitted bound hold. The affine gradient vanishes at x_*. The exact step and bounded weights imply $\|X - x\| \leq C\eta\|g(x)\| \leq C\eta L\|x - x_*\|$, after subtracting a common gradient component. Choose $0 < \rho_0 < \rho$ so every source in the smaller ball has its target in the larger ball. By (S.137), the closed feasible annulus $\rho_0 \leq \|x - x_*\| \leq \rho$ has a strictly positive minimum objective value. A sufficiently close initial point has smaller value. Its source–target segment lies in the large convex ball, so (S.139) prevents its target from lying in the annulus. Induction traps the orbit inside the smaller ball.

The exact Burg three-point identity, convexity and the same local smoothness bound give, for every nearby fixed interior comparator ξ,

$$\eta[F(X) - F(\xi)] \leq D_h(\xi, x) - D_h(\xi, X) - (1 - \eta L) D_h(X, x). \tag{S.140}$$

All points and comparisons lie in the local convex ball; $\nabla^2 h = \mathrm{diag}(x_i^{-2}) \succeq I$ supplies the relative-smoothness bound. If the decreasing objective limit were positive, choose an interior ξ close enough to x_* with smaller value and telescope (S.140) to a contradiction. Thus $F \to 0$, and local coercivity gives $x \to x_*$. This proves local attraction without any global extension. Under the separately stated global convexity, smoothness and unique-minimizer assumptions, the same comparator argument works on the entire simplex; compactness then gives all-start entry. These are standard convex mirror arguments, consistent with the global log-barrier convergence precedent [7].

On the protected supported tail, convexity and weighted Cauchy–Schwarz imply

$$F(x) \le \langle g, x - x_* \rangle \le C\sqrt{D_F(x)}.$$

One may subtract any common scalar from g: each missing-coordinate ratio $(x_i - x_{*,i})/x_i$ is one, and the supported ratios are bounded. Therefore

$$F_k - F_{k+1} \ge c_0 \eta F_k^2, \qquad F_k = O(T_k^{-1}). \tag{S.141}$$

Combining with coercivity, before any capture assertion, gives

$$r = O(T^{-1/m}), \quad y = O(T^{-1/p}), \quad \delta = O(T^{-1/2}). \tag{S.142}$$

Because $a_i, u_i > 0$, feasibility implies $\|u\| \le Cr$ and hence $\|w\| \le Cr$. The stronger static-center estimates additionally give

$$\phi = O(T^{-1/2-1/p}), \quad w = O(T^{-1/2}), \quad \phi_r = O(T^{-1/2}), \quad \phi_y = O(T^{-1/2}). \tag{S.143}$$

Indeed $r^q = O(T^{-1/2-1/m})$, $s^{q+1} = O(T^{-1/2-1/p})$, $r^{q-1} = O(T^{-1/2})$, and $s^q = O(T^{-1/2})$. These bounds precede all lower normal estimates.

S5.4 Radial reciprocity and every lower normal bound

Let $\psi = b_A^T(1/x_A)$ and

$$Q = B^T(1/x) = 1/u - \psi \mathbf{1}_N. \tag{S.144}$$

Since $B^T\mathbf{1} = 0$, the exact reciprocal map gives

$$\Delta Q = \eta F_u, \qquad \Delta(a^T Q) = \eta F_r. \tag{S.145}$$

The second identity differentiates along $u \mapsto u + ta$ at fixed y, w. By the full reduced jet and (S.142)–(S.143),

$$|f_{\mathrm{red},r}| \le C(r^{m-1} + s^{m+1}) \le C[T^{-(m-1)/m} + T^{-(m+1)/p}], \qquad |F_r - f_{\mathrm{red},r}| = O(T^{-1}).$$

Because $(m+1)/p > (m-1)/m$, summing (S.145) gives the signed upper bound $a^T Q \le CT^{1/m}$; the logarithmic accumulation of T^{-1} is absorbed. Restore the bounded supported term $(\sum_i a_i)\psi$. Every a_i/u_i is positive, so

$$c_0 T^{-1/m} \le u_i, r \le CT^{-1/m} \quad \text{for every } i, \qquad \frac{\|w\|}{r} = O(T^{-1/2+1/m}) \longrightarrow 0. \tag{S.146}$$

No active capture, transverse tracking, or radial monotonicity was used.

S5.5 The actual multiplier before active capture

This intermediate step is necessary: the coarse quadratic residual $O(\delta^2)$ in F_y cannot yet be absorbed at $\|y\| \asymp r$. Set

$$b_0 = (m+1)/p = 1 - 1/p, \qquad \gamma = 2/m.$$

The coarse box and (S.136) give

$$F_y = O(T^{-b_0}), \qquad F_u = O(T^{-1/2}), \qquad f_{\mathrm{red},r} = O(T^{-1+1/m}). \tag{S.147}$$

For example, $s^q\|\delta\|$ and $\|\delta\|^2$ are $O(T^{-1})$, whereas the reduced active force is $O(s^{m+1}) = O(T^{-b_0})$.

The exact affine covector

$$C_{y,A} = G^{-1}E_A^T \operatorname{diag}(x_{*,i}^{-2}), \qquad C_{y,I} = 0$$

satisfies $C_yE = I$, $C_yB = 0$, and $C_{y,A}x_{*,A}^2 = 0$. Choose the gradient representative $g_A = C_{y,A}^T F_y$, $g_I = F_u$; this changes only a common component on the simplex and does not replace the original admitted L. The actual source–target multiplier is

$$\mu = \frac{(C_{y,A}\omega_A)^T F_y + \omega_I^T F_u}{\sum_i \omega_i}. \tag{S.148}$$

The supported denominator is bounded below. Moreover

$$\omega_A = x_A^2 + \operatorname{diag}(x_A)\Delta x_A, \qquad \|\Delta x_A\| \le C\eta(\|F_y\| + |\mu|), \qquad \|\omega_I\| \le Cr^2.$$

Consequently

$$|\mu| \le C[s\|F_y\| + \eta\|F_y\|^2 + \eta\|F_y\||\mu| + r^2\|F_u\|].$$

Absorb $C\eta\|F_y\||\mu|$ on the late tail. Since $s\|F_y\| = O(T^{-1})$, $2b_0 > 1$, and the higher normal terms in (S.136) are bounded by $C\|\delta\|$, this yields

$$|\mu| \le C_\eta[T^{-1} + r^2(\|\delta\| + |f_{\mathrm{red},r}|)] = O_\eta(T^{-1} + T^{-1/2-2/m}). \tag{S.149}$$

This is a bound for the actual multiplier, not a source-square replacement.

Applying the exact increments directly, including $C_y\Delta x$, gives

$$\|\Delta u\| \le C_\eta\eta T^{-1/2-2/m}, \qquad \|\Delta y\| \le C_\eta\eta[T^{-b_0} + T^{-1/2-2/m}]. \tag{S.150}$$

Both endpoints and their straight segment lie in the same coarse box up to fixed constants. Integrating $D\phi$ along that segment with (S.143) gives the exact center-motion estimate

$$\|\Delta\phi\| \le C_\eta\eta[T^{-1-2/m} + T^{-3/2+1/p}]. \tag{S.151}$$

There is no omitted discrete Taylor remainder in this mean-value estimate.

S5.6 The intermediate transverse bootstrap

For a literal normal the rational identity is exact:

$$U_i = \frac{u_i}{1+\eta u_i(F_{u_i}-\mu)}, \qquad \Delta u_i = -\eta u_i^2(F_{u_i}-\mu) + \frac{\eta^2 u_i^3(F_{u_i}-\mu)^2}{1+\eta u_i(F_{u_i}-\mu)}. \tag{S.152}$$

All denominators tend to one. Its target correction is bounded by

$$C_\eta \eta^2 r^3(\|\delta\| + |f_{\mathrm{red},r}| + |\mu|)^2 \le C_\eta \eta^2 r^3 T^{-1}. \tag{S.153}$$

Expand the diagonal mobility D_u^2 around $r^2 D_a^2$ and project with P. The identity $PD_a^2(\ell/N) = 0$ removes the reduced radial force exactly at $u = ra$. With $A_0 = (PD_a^2 K)|_{E_0}$,

$$\begin{aligned} \Delta w &= -\eta r^2 A_0 \delta + R_w, \\ \|R_w\| &\le C_\eta \eta[r^2 s\|\delta\| + r\|w\|\|\delta\| + r\|w\||f_{\mathrm{red},r}| + r^2|\mu|] + C_\eta \eta^2 r^3 T^{-1}. \end{aligned} \tag{S.154}$$

Terms such as $r^2\|\delta\|^2$ are absorbed by the displayed small-tail terms. In $r\|w\||f_{\mathrm{red},r}|$ write $w = \phi + \delta$. The relative coefficient $|f_{\mathrm{red},r}|/r = o(1)$, and the other coefficients s, $\|w\|/r$ and r^2 also tend to zero. The remaining pure forces satisfy

$$\begin{aligned} r\|\phi\||f_{\mathrm{red},r}| &= O(T^{-3/2-1/p}), \qquad & r^2 T^{-1} &= O(T^{-1-2/m}), \\ r^4|f_{\mathrm{red},r}| &= O(T^{-1-3/m}), \qquad & r^3 T^{-1} &= O(T^{-1-3/m}). \end{aligned} \tag{S.155}$$

For $v, w \in E_0$, one has

$$\langle v, A_0 w\rangle_{D_a^{-2}} = v^T K w.$$

The projection correction vanishes because $D_a^{-2}a = \ell$ and $\ell^T v = 0$. Since $\ker K = \mathrm{span}\{a\}$ and $a \notin E_0$, A_0 is strictly positive and self-adjoint in this fixed metric. Thus $I - \eta r^2 A_0$ contracts its norm by $1 - c_0\eta r^2$ for late sources.

Subtract the actual center increment (S.151) from (S.154). Absorb the small coefficients and use normal comparability to obtain

$$\|\delta^+\|_* \le (1 - c_0\eta T^{-2/m})\|\delta\|_* + C_\eta\eta[T^{-1-2/m} + T^{-3/2+1/p}]. \tag{S.156}$$

The fixed-clock comparison proves

$$\delta = O_\eta(T^{-\beta}), \qquad \beta = \min\{1, \tfrac{3}{2} - \tfrac{1}{m+2} - \tfrac{2}{m}\} = \begin{cases} 5/6, & m = 4, \\ 1, & m \ge 6 \text{ even}. \end{cases} \tag{S.157}$$

This is an intermediate lag, not the final one. Its consequence is

$$F_y - f_{\mathrm{red},y} = O_\eta(T^{-\beta-1/2} + T^{-2\beta}) = o(T^{-1-1/m}). \tag{S.158}$$

Indeed $\beta + 1/2 > 1 + 1/m$ and $2\beta > 1 + 1/m$; at $m = 4$ these are $4/3 > 5/4$ and $5/3 > 5/4$. The problematic quadratic active residual is now subordinate without presuming $y = O(r)$.

S5.7 Critical capture in the physical limiting metric

Define the supported dual coordinate

$$z = G^{-1}E_A^T(-1/x_A + 1/x_{*,A}). \tag{S.159}$$

At $(u, y) = (0, 0)$, Burg orthogonality gives $z_y = I$ and $z_u = 0$. The analytic inverse-function theorem therefore gives

$$\begin{aligned} z = y + O((\|u\| + \|y\|)^2), \qquad y = \chi(u, z) = z + O((\|u\| + \|z\|)^2), \\ \chi_z = I + O(\|u\| + \|z\|), \qquad \chi_u = O(\|u\| + \|z\|). \end{aligned} \tag{S.160}$$

Only supported reciprocals are evaluated at the boundary. Pairing the exact mirror equations with E gives

$$\Delta z = -\eta G^{-1}F_y. \tag{S.161}$$

Put $R = \|z\|_G$. The full reduced jet, Euler homogeneity and positivity of $\mathcal{J}_p(0, y)$ give, for a fixed sufficiently large K_0 and all sufficiently late sources,

$$z^T f_{\mathrm{red},y} \geq c_0 R^{m+2} \quad (R \geq K_0 r), \qquad \|f_{\mathrm{red},y}\| \leq C(R + r)^{m+1}. \tag{S.162}$$

For the first inequality, first make r/R small. The homogeneous gradient $\mathcal{J}_{p,y}(r, y)$ is then a small perturbation of $\mathcal{J}_{p,y}(0, y)$, whose pairing with y is $p\mathcal{J}_p(0, y) \geq c_0\|y\|^p$. Next make the source small: both the analytic remainder and the quadratic inverse-chart correction are relatively small. This uses no joint convexity of $\mathcal{J}_p$.

By (S.158) and normal comparability, $F_y = f_{\mathrm{red},y} + e_y$ with $\|e_y\| = o(r^{m+1})$. Outside $R < K_0 r$ this error is absorbed in (S.162); on the whole tail $\|F_y\| \leq C(R + r)^{m+1}$. In the squared G norm of (S.161), the positive quadratic-step term is absorbed outside the barrier because $\eta R^m \to 0$. Hence

$$R^+ \leq R - c_0\eta R^{m+1} \quad \text{if } R \geq K_0 r.$$

Inside the barrier the whole step is $O_\eta(\eta r^{m+1})$. Enlarging the forcing constant combines both regions into (S.128). Therefore

$$\|y\| + \|z\| = O(T^{-1/m}) = O(r). \tag{S.163}$$

Active capture is a consequence of the full jets and entry assumptions, not an input to the transverse bootstrap.

S5.8 Final center-plus-lag tracking

Now $s = O(r)$, $\phi = O(r^q)$, $\phi_r = O(r^{q-1})$, $\phi_y = O(r^q)$, and $\delta = o(r)$. The forces satisfy

$$F_y = O(r^{m+1} + r^q\|\delta\| + \|\delta\|^2), \qquad F_u = O(r^{m-1} + \|\delta\|).$$

Repeating the actual multiplier absorption, with $s = O(r)$, yields

$$|\mu| \le C_\eta(r^{m+1} + r^2\|\delta\|). \tag{S.164}$$

For example $r\|\delta\|^2 = o(r^2\|\delta\|)$, and the fixed-$\eta$ squared active-force term is also subordinate since $\delta = o(r)$. The exact step bounds are

$$\begin{aligned} \|\Delta u\| &\le C_\eta\eta(r^{m+1} + r^2\|\delta\|), \\ \|\Delta y\| &\le C_\eta\eta(r^{m+1} + r^2\|\delta\| + \|\delta\|^2). \end{aligned} \tag{S.165}$$

Along their source–target segment, $|(r, y)| = O(r)$ and $r^+/r \to 1$. Integrate $D\phi$ to obtain

$$\|\Delta\phi\| \le C_\eta\eta[r^{m+q} + r^{q+1}\|\delta\| + r^{q-1}\|\delta\|^2]. \tag{S.166}$$

The last two terms are $o(\eta r^2\|\delta\|)$ since $q \ge 3$ and $\delta = o(r)$.

Projecting the exact rational normal map as in (S.154), the pure force from center-dependent mobility is $O(\eta r^{m+q})$, the multiplier force is $O(\eta r^{m+3})$, and the rational target correction is $O_\eta(\eta^2 r^{2m+1})$. All are $O_\eta(\eta r^{m+3})$; every remaining coefficient of δ is $o(\eta r^2)$. Subtract (S.166) and contract in the fixed transverse norm:

$$\|\delta^+\|_* \le (1 - c_0\eta r^2)\|\delta\|_* + C_\eta\eta r^{m+3}.$$

The external-clock comparison, already justified by (S.146), proves

$$\delta = O_\eta(T^{-(m+1)/m}) = O_\eta(r^{m+1}). \tag{S.167}$$

The tracked object is the actual analytic center. No position asymptotic has been differentiated to obtain this lag.

S5.9 Two radial coefficients and physical finite length

At the final scale, the center contributes a relative normal-mobility correction $O(r^{q-1}) = O(r^2)$. The reduced radial derivative is

$$f_{\mathrm{red},r} = mcr^{m-1} + (m+1)c_1 r^m + O(r^{m+1}), \qquad r^2 D_a^2(\ell/N) f_{\mathrm{red},r} = (r^2/N) f_{\mathrm{red},r} a.$$

After mobility the transverse force and multiplier are $O_\eta(r^{m+3})$. The exact rational correction is $O_\eta(\eta^2 r^{2m+1})$, also within the same budget. Thus

$$\begin{aligned} \Delta u &= -\eta(\lambda r^{m+1} + \lambda_1 r^{m+2})a + O_\eta(\eta r^{m+3}), \\ \Delta r &= -\eta(\lambda r^{m+1} + \lambda_1 r^{m+2}) + O_\eta(\eta r^{m+3}). \end{aligned} \tag{S.168}$$

Every $a_i > 0$, so all literal normals and r eventually strictly decrease. Taylor expansion gives

$$(r^+)^{-m} - r^{-m} = \eta m\lambda + \eta m\lambda_1 r + O_\eta(\eta r^2).$$

Its quadratic Taylor term is $O(\eta^2 r^m)$ and is absorbed because $m \geq 4$ and η is fixed. Summing with (S.146) yields

$$r_k = C_r k^{-1/m}[1 + O(k^{-1/m})]. \tag{S.169}$$

If $c_1 = 0$, the sum of r^2 instead gives $r_k^{-m} = m\lambda\eta k + O(k^{1-2/m})$ and the sharper relative error $O(k^{-2/m})$. No general zero-c_1 improvement is claimed when $c_1 \neq 0$.

At (S.167), the supported exact step and (S.164) imply

$$\|\Delta x\| \leq C_\eta \eta r^{m+1} \leq C'(r - r^+). \tag{S.170}$$

Telescoping proves finite physical length and $\mathcal{L}_k = O(r_k)$ before angular selection. Finally (S.135), capture and the reduced jet give $F = cr^m[1 + o(1)]$, hence

$$kF(x_k) \longrightarrow \frac{c}{\eta m \lambda} = \frac{N}{\eta m^2}. \tag{S.171}$$

S5.10 Angular selection by a fixed analytic lift

At the final lag, (S.136) gives $F_y = f_{\mathrm{red},y} + O_\eta(r^{m+q+1})$. Put $v = z/r$ and $\alpha = \eta r^m$. Capture and the actual inverse chart imply $y/r = v + O(r)$, so

$$F_y = r^{m+1}\mathcal{J}_{p,y}(1, v) + O_\eta(r^{m+2}).$$

The exact quotient identity $\Delta v = (\Delta z - v\Delta r)/r^+$, (S.161) and (S.168) yield

$$\begin{aligned} \Delta v = -\alpha G^{-1}\nabla V(v) + O_\eta(\alpha r), \qquad V(v) = \mathcal{J}_p(1, v) - \tfrac{\lambda}{2} v^T G v, \\ \Delta r = -\lambda\alpha r + O_\eta(\alpha r^2). \end{aligned} \tag{S.172}$$

The c_1 term changes the radial error but not the limiting potential. Its rescaling drift λv and the true supported Burg metric G must both be retained. Moreover

$$\alpha_k = \frac{1 + O(k^{-1/m})}{m\lambda k} \longrightarrow 0, \qquad \sum_{k<n} \alpha_k \sim \frac{\log n}{m\lambda} \longrightarrow \infty. \tag{S.173}$$

If $c_1 = 0$, the relative error improves to $O(k^{-2/m})$. The proved radial recurrence also makes r decay exponentially in the angular clock.

We spell out the analytic-gradient argument; summable perturbations alone do not prove point convergence. All v_k lie in a fixed compact set. Taylor expansion, positive definiteness of G^{-1}, and Young's inequality give

$$V(v) - V(v^+) \geq c_0\alpha\|\nabla V(v)\|^2 - C\alpha r^2.$$

Also $r^2 - (r^+)^2 = 2\lambda\alpha r^2 + O_\eta(\alpha r^3)$ up to the absorbed squared-step term. For a sufficiently large fixed L_1, let $\mathcal{W}(r, v) = V(v) + L_1 r^2$ and $q_k = (r_k, v_k)$. Throughout the entire late bounded tail,

$$\begin{aligned} \mathcal{W}(q_k) - \mathcal{W}(q_{k+1}) \geq c_0\alpha_k[\|\nabla V(v_k)\|^2 + r_k^2] > 0, \\ \|\Delta q_k\| \leq C\alpha_k[\|\nabla V(v_k)\| + r_k], \qquad \|\nabla\mathcal{W}(q_k)\| \asymp \|\nabla V(v_k)\| + r_k. \end{aligned} \tag{S.174}$$

In particular the weaker radial remainder $O(\alpha r^2)$ is harmless for the fixed lift. Its energy decreases to a finite limit h, with positive gaps at every finite tail index, and $\sum \alpha_k \|\nabla \mathcal{W}(q_k)\|^2 < \infty$.

The gradient tends to zero along the full tail. Its bounded-set Lipschitz variation per step is at most $C\alpha_k \|\nabla \mathcal{W}(q_k)\|$. If its norm exceeded a fixed ϵ infinitely often, this bound would force disjoint intervals of fixed positive cumulative α on which the gradient remains above $\epsilon/2$. Their energy drops contradict the finite gradient-square sum. A final excursion that never ended likewise contradicts $\sum \alpha_k = \infty$. Thus the compact cluster set lies in $\{\mathcal{W} = h, \nabla \mathcal{W} = 0\}$.

The ordinary local analytic gradient inequality, uniformized by a finite cover of that compact cluster set, gives $\theta < 1$ and $C > 0$ such that

$$(\mathcal{W}(q_k) - h)^\theta \le C \|\nabla \mathcal{W}(q_k)\|$$

on the whole sufficiently late tail. Compactness implies that the distance to the cluster set tends to zero, so this uniformization needs no prior local retention assertion. For a suitable concave desingularizer $\varphi(s) = C' s^{1-\theta}$ and $s_k = \mathcal{W}(q_k) - h > 0$, (S.174) gives

$$\begin{aligned} \varphi(s_k) - \varphi(s_{k+1}) &\ge \varphi'(s_k)[\mathcal{W}(q_k) - \mathcal{W}(q_{k+1})] \\ &\ge c_0 \alpha_k [\|\nabla V(v_k)\| + r_k] \ge c_1' \|\Delta q_k\|. \end{aligned}$$

Telescoping proves finite length in (r, v) and $v_k \to v_*$ with $\nabla V(v_*) = 0$. This uses the standard analytic-gradient inequality [2–4], not a boundary-uniform Burg KL property. Critical points need not be isolated. The proof supplies descent on the entire actual tail rather than importing a large-step retention assertion.

S5.11 Literal speed and the sharp physical tail

The physical chart gives

$$x - x_* = Bu + Ey = r(Ba + Ev_*) + o(r) = rd_* + o(r).$$

The speed needs a separate proof. Apply the actual inverse map $y = \chi(u, z)$ between consecutive sources. Its derivatives satisfy $\chi_z = I + O(r)$ and $\chi_u = O(r)$, while both increments are $O_\eta(\eta r^{m+1})$. Consequently

$$\Delta y = \Delta z + O(r)(\|\Delta u\| + \|\Delta z\|) + O((\|\Delta u\| + \|\Delta z\|)^2).$$

The reconstruction remainder $O_\eta(\eta^2 r^{2m+2})$ is within $O_\eta(\eta r^{m+2})$. Thus

$$\Delta y = -\eta r^{m+1} G^{-1} \mathcal{J}_{p,y}(1, v) + O_\eta(\eta r^{m+2}). \tag{S.175}$$

Criticality gives $G^{-1} \mathcal{J}_{p,y}(1, v_*) = \lambda v_*$. Together with the literal normal step this proves

$$\Delta x = -\eta \lambda r^{m+1} d_* + o(\eta r^{m+1}). \tag{S.176}$$

Since $d_{*,I} = a \neq 0$ and $\eta\lambda C_r^{m+1} = C_r/m$, the two scaled vector limits in the main theorem follow. Finally

$$\frac{\|\Delta x_k\|}{r_k - r_{k+1}} \longrightarrow \|d_*\|.$$

Positive radial decrements telescope to r_k. A tail squeeze proves

$$\mathcal{L}_k \sim \|d_*\| r_k, \qquad \|x_k - x_*\| \sim \|d_*\| r_k, \qquad \mathcal{L}_k / \|x_k - x_*\| \longrightarrow 1.$$

This concludes Main Theorem 9. Individual supported coordinates need not be eventually monotone. Tail ratio one is a classical gradient-geometric conclusion [1, Corollary 6.5]; here its proof comes from actual singular discrete increments.

S5.12 Exact homogeneous specialization and the quartic Schur term

We prove Main Corollary 8.1, including its stronger explicit bias, and retain the useful direct tracking estimates of this subcase. Suppose exactly

$$F(y,u) = \tfrac{1}{2} u^T K u + G_m(u) + J_{m+2}(y,u) \tag{S.177}$$

on the simplex, where both homogeneous even-degree polynomials are globally convex, $G_m(a) > 0$ and $J_{m+2}(y,0) > 0$ away from zero. Even homogeneity and convexity make both nonnegative. If $F = 0$, then $Ku = 0$, hence $u = ta$ with $t \geq 0$; positivity of $G_m(a)$ forces $t = 0$, and then active positivity forces $y = 0$. The polynomial defines a convex analytic germ on an open affine neighborhood and has the unique constrained minimizer x_*. Its extension and the original ambient admitted $\eta L < 1$ therefore supply global entry. The retained homogeneous coercivity is $F \geq c_0(\|u\|^m + \|w\|^2 + \|y\|^{m+2})$: near the positive kernel ray G_m controls the normal mth power; away from it the quadratic form does so, and a fixed quadratic fraction controls w. The loss $C\|u\|^{m+2}$ from the mixed critical polynomial is absorbed. Together with the shared energy estimate this also recovers $u = O(T^{-1/m})$, $w = O(T^{-1/2})$ and $y = O(T^{-1/(m+2)})$ directly.

Let $g_m = \nabla G_m(a)$ and $w_* = -A_0^{-1} P D_a^2 g_m$. Since A_0 is positive in the D_a^{-2} metric,

$$PD_a^2(Kw_* + g_m) = 0, \qquad Kw_* + g_m = \lambda \ell, \qquad \lambda N = a^T g_m = m G_m(a). \tag{S.178}$$

Stationarity of the transverse center means that $K\phi + \nabla G_m(ra + \phi) + \partial_u J_{m+2}(y, ra + \phi)$ annihilates E_0 and hence belongs to $\operatorname{span}\{\ell\}$. The implicit equation first gives $\phi = O(s^{m-1})$, and its lowest term is $r^{m-1} w_*$. The next errors have orders s^{2m-3} and s^{m+1}. Since $2m - 3 \geq m + 1$, uniform analytic inversion gives

$$\phi(r,y) = r^{m-1} w_* + O(s^{m+1}). \tag{S.179}$$

In particular there is no degree-m center term. This also follows from the oddness of the center under $(r,y) \mapsto (-r,-y)$.

To identify the full reduced critical jet, substitute (S.179) into (S.177). The quadratic energy and first G_m correction at order $2m-2$ combine as

$$r^{2m-2}\left(\tfrac{1}{2}w_*^TKw_*+g_m^Tw_*\right)=-\tfrac{1}{2}r^{2m-2}w_*^TKw_*.$$

Here $\ell^Tw_*=0$ and (S.178) give $g_m^Tw_*=-w_*^TKw_*$. If $m=4$, then $2m-2=m+2=6$, so this term must be retained; for even $m\geq 6$ it is beyond critical degree. The remaining terms have degree at least $m+3$: for example G_m's quadratic center correction has degree $3m-4$, the J_{m+2} center correction has degree $2m$, and the error in (S.179) paired with the leading force has degree $2m$. It follows that

$$\begin{aligned} f_{\mathrm{red}}(r,y)&=G_m(a)r^m+\mathcal{J}_{m+2}(r,y)+O(s^{m+3}),\\ \mathcal{J}_{m+2}(r,y)&=J_{m+2}(y,ra)-\mathbf{1}_{\{m=4\}}\tfrac{1}{2}(w_*^TKw_*)r^6. \end{aligned} \tag{S.180}$$

Thus $c=G_m(a)$, $c_1=0$, and $\mathcal{J}_{m+2}(0,y)=J_{m+2}(y,0)>0$. All reduced-jet hypotheses have been proved, including the absence of mixed degree-$(m+1)$ terms. At $r=1$ the Schur term is a constant, so the angular gradient is precisely that of $J_{m+2}(v,a)-\lambda v^TGv/2$. Joint convexity of the reduced critical polynomial is unnecessary.

The general theorem now gives $y=O(r)$ and $w=\phi+O_\eta(r^{m+1})$. By (S.179),

$$w=r^{m-1}w_*+O_\eta(r^{m+1}). \tag{S.181}$$

Its zero-c_1 radial law and all physical constants are exactly those in the corollary. In particular the homogeneous bias is a stronger special conclusion, not the generic analytic center.

Retained direct coordinate-weighted estimates. For clarity the exact subcase also admits the following direct tracking route after normal comparability and active capture, both already supplied above. Here $F_y=O(r^{m+1})$ and $F_u=Kw+O(r^{m-1})$. The actual multiplier absorption gives

$$\mu=O_\eta(r^2\|w\|+r^{m+1}),$$

and the rational target error is bounded by $C\eta^2r^3(\|w\|+r^{m-1})^2$. Expand the normal mobility and G_m in the exact step to obtain

$$\begin{aligned} \Delta u&=-\eta r^2D_a^2Kw-\eta r^{m+1}D_a^2g_m+R_u,\\ \|R_u\|&\leq C_\eta\eta(r\|w\|^2+r^m\|w\|+r^4\|w\|+r^{m+3}). \end{aligned} \tag{S.182}$$

The four errors respectively come from variation of mobility against Kw, variation of G_m and its mobility, the multiplier, and the critical normal force. The target correction has orders $\eta^2r^3\|w\|^2$, $\eta^2r^{m+2}\|w\|$, and η^2r^{2m+1}, each absorbed in the displayed bound. Thus no uniform $O(\eta^2)$ error is introduced.

Projecting, $w/r\to 0$ allows the variable linear coefficient to be absorbed, giving

$$\Delta w=-\eta r^2(A_0w+r^{m-1}b_*)+R_w,\quad b_*=PD_a^2g_m,\quad \|R_w\|\leq\eta r^2(\epsilon_k\|w\|+Cr^{m+1}), \tag{S.183}$$

where $\epsilon_k = O(\|w\|/r + r^{m-2} + r^2) \to 0$. The first fixed-clock comparison proves

$$w = O(T^{-(m-1)/m}) = O(r^{m-1}). \tag{S.184}$$

At this scale $r\|w\|^2$ and $r^m\|w\|$ have order $r^{2m-1} \le O(r^{m+3})$, since $m \ge 4$. Hence, for the bounded quotient $W = w/r^{m-1}$,

$$\Delta W = -\eta r^2(A_0 W + b_*) + O_\eta(\eta r^4). \tag{S.185}$$

This follows from the exact changing-denominator identity: $\Delta r = O(\eta r^{m+1})$, so denominator variation costs $O_\eta(\eta r^m) = O_\eta(\eta r^4)$. A second stable comparison gives $W - w_* = O(T^{-2/m}) = O(r^2)$, reproducing (S.181). Substitution of (S.178) gives the corollary's normal and radial speeds. In this subcase also $F = r^m G_m(a) + O(r^{m+2})$, since the transverse quadratic and first homogeneous correction have order $2m - 2 \ge m + 2$.

The only shrinking products used here are ηR^m, ηr^2 and ηr^m in active capture, transverse tracking and angular descent. The general analytic theorem and this corollary both concern the original fixed step, without a slow-step, outer-facet, or superflat-error assumption.

S5.13 Sharpness and the necessity of the full mixed jet

Proposition S5.1 (The center order is attained by a convex polynomial)**.** For every $m = 2k \ge 4$, the order $q = k + 1$ in (S.130) is sharp, even for a polynomial convex on a genuine open neighborhood and with a positive pure-active critical jet.

Proof. For scalar w, y, choose $\beta \ne 0$, $d > 0$, and put

$$F(w, r, y) = \tfrac{1}{2}w^2 + r^{2k} + \beta w r^{k+1} + Cw^2 r^2 + dy^{2k+2}.$$

Set $a_0 = 2k(2k-1)$, $B_0 = \beta k(k+1)$ and choose $C \ge B_0^2/(2a_0)$, $C > 0$. The (w, r) Hessian entries are exactly

$$A = 1 + 2Cr^2, \quad B = \beta(k+1)r^k + 4Cwr, \quad D = a_0 r^{2k-2} + B_0 w r^{k-1} + 2Cw^2.$$

Young's inequality gives $D \ge (a_0/2)r^{2k-2} + Cw^2$ and $B^2 \le 2\beta^2(k+1)^2 r^{2k} + 32C^2 w^2 r^2$. Since $A \ge 1$,

$$AD - B^2 \ge [a_0/2 - 2\beta^2(k+1)^2 r^2]r^{2k-2} + [C - 32C^2 r^2]w^2 \ge 0$$

whenever $r^2 \le \min\{a_0/[8\beta^2(k+1)^2], 1/(64C)\}$. The separate active Hessian is nonnegative, proving convexity on an open strip. The exact center and reduced objective are

$$\phi = -\frac{\beta r^{k+1}}{1 + 2Cr^2}, \qquad f_{\text{red}} = r^{2k} + dy^{2k+2} - \frac{\beta^2 r^{2k+2}}{2(1 + 2Cr^2)}.$$

Thus the first center coefficient is nonzero and the critical correction is purely radial. For $m = 6$, $\beta = d = 1$, $C = 3$, convexity holds on $|r| < 1/\sqrt{192}$ and $\phi = -r^4/(1 + 6r^2) = -r^4 + O(r^6)$. In particular analyticity and convexity alone do not force the older r^{m-1} transverse-center scale. ∎

Proposition S5.2 (Pure-axis jets do not ensure critical capture)**.** For every even $m \geq 4$, there is a globally convex polynomial with a positive transverse Hessian and pure flat-axis jets r^m and y^{m+2} through degree $m+2$, for which exact Burg orbits in an invariant interior slice do not satisfy $y = O(r)$ at any admitted fixed step. The obstruction is an active-dependent mixed term of degree $m+1$.

Proof. Write $p = m+2$, $R^2 = w^2 + r^2 + y^2$ and set

$$B_m = \frac{(m+1)(3m+2)}{8}, \qquad C_m = \frac{B_m}{m+4}, \qquad h(y) = 1 + \frac{y}{4} + \frac{y^2}{8}, \tag{S.186}$$
$$F_m(w,r,y) = \tfrac{1}{2}w^2 + r^m h(y) + y^p + C_m R^{m+4}.$$

All powers are even where required, so this is an entire polynomial. Its raw and reduced flat slices agree because $w = 0$ is its unique transverse minimum. Its axis restrictions are $r^m + C_m r^{m+4}$ and $y^p + C_m y^{m+4}$, while its mixed jet contains $r^m y/4$.

We verify global convexity explicitly. The (r, y) Hessian of $r^m h(y) + y^p$ has nonnegative diagonal entries $m(m-1)r^{m-2}h(y)$ and $r^m/4 + p(p-1)y^m$. Its determinant is

$$mr^{2m-2}Q_m(y) + m(m-1)p(p-1)r^{m-2}h(y)y^m, \quad Q_m(y) = \frac{3m-4}{16} - \frac{(m+1)y}{16} - \frac{(m+1)y^2}{32}.$$

For $|y| \leq 1$, $Q_m(y) \geq (3m-11)/32 > 0$, so the base is convex on that entire strip. Outside $R < 1$, the only possibly negative Hessian contributions have operator norms bounded by

$$\|D^2(r^m y/4)\| \leq \frac{m(m+1)}{4} R^{m-1}, \qquad \|D^2(r^m y^2/8)\| \leq \frac{(m+1)(m+2)}{8} R^m.$$

These bounds follow by summing absolute Hessian entries. For $R \geq 1$ their sum is at most $B_m R^m$, whereas the least Hessian eigenvalue of $C_m R^{m+4}$ is $B_m R^{m+2}$. The convexifier therefore dominates there; for $R \leq 1$ the base is already convex. This proves global convexity. Since $h(y) = (y+1)^2/8 + 7/8 > 0$, the origin is the unique zero and minimizer.

Use the four-coordinate simplex chart

$$x = (\tfrac{1}{2} - r + y, \tfrac{1}{2} - r - y, r + w, r - w), \qquad x_* = (\tfrac{1}{2}, \tfrac{1}{2}, 0, 0).$$

The normal quadratic block is $K = \frac{1}{4}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, with positive kernel ray $(1,1)$, and the supported metric is $G = 8$. Choose any ambient gradient-Lipschitz bound L on the simplex and $\eta > 0$ with $\eta L < 1$. For completeness, every interior orbit converges to x_*: exact Burg descent makes $F_m(x_k)$ nonincreasing, and its mirror comparison, with $B(x) = -\sum_i \log x_i$, is

$$\eta[F_m(x^+) - F_m(\xi)] \leq D_B(\xi, x) - D_B(\xi, x^+) - (1 - \eta L) D_B(x^+, x)$$

for every fixed interior comparator ξ. A positive limiting value would be contradicted by choosing an interior ξ near x_* with smaller value and telescoping. Thus $F_m(x_k) \to 0$;

compactness and uniqueness imply $x_k \to x_*$. The comparison follows from the exact mirror identity and $D^2B \geq I$ on the simplex.

The slice $\mathcal{S} = \{w = 0,\ 0 < r < 1/2,\ |y| < 1/2 - r\}$ is invariant, since $F_w = 0$ at $w = 0$ gives equal normal gradients and targets. Suppose an orbit in $\mathcal{S}$ satisfied $y = O(r)$. On that hypothetical tail,

$$F_r = mr^{m-1}[1 + O(r)], \qquad F_y = \tfrac{1}{4}r^m + O(r^{m+1}). \tag{S.187}$$

Use the ambient representative $g = (F_y/2, -F_y/2, F_r/2, F_r/2)$ on the slice, and put $\omega_i = x_i x_i^+$. The actual multiplier is $\mu = \sum_i \omega_i g_i / \sum_i \omega_i$. The exact coordinate equation is $\Delta x_i = -\eta\omega_i(g_i - \mu)$. Bounded gradients give bounded source-target coordinate ratios; convergence protects the two supported coordinates. Consequently,

$$\omega_3 + \omega_4 = O(r^2), \qquad \omega_1 - \omega_2 = 4(\tfrac{1}{2} - r)y + O_\eta(|F_y| + |\mu|),$$

and the multiplier denominator stays bounded below. Substitution and absorption of $C\eta|F_y||\mu|$ on the late tail give

$$|\mu| \leq C_\eta\{r|F_y| + \eta|F_y|^2 + r^2|F_r|\} = O_\eta(r^{m+1}).$$

Thus the literal rational normal update gives

$$r^+ = \frac{r}{1 + \eta r(F_r/2 - \mu)}, \qquad \Delta r = -\frac{\eta m}{2}r^{m+1}[1 + O_\eta(r)], \qquad r_k^m \sim \frac{2}{\eta m^2 k}. \tag{S.188}$$

The last equivalent follows by differencing r^{-m}; it has been derived only under the hypothetical capture assumption. Finally the supported dual coordinate satisfies the exact identity

$$z = \tfrac{1}{8}(1/x_2 - 1/x_1), \qquad \Delta z = -\tfrac{\eta}{8}F_y.$$

By (S.187), eventually $\Delta z \leq -\eta r^m/64$. The harmonic divergence in (S.188) forces $z_k \to -\infty$, contrary to $x_k \to x_*$ and hence $z_k \to 0$. Therefore every initial point in $\mathcal{S}$ has $\limsup_k |y_k|/r_k = \infty$. No alternative asymptotic rate is asserted. ∎

The obstruction concerns an omitted mixed hypothesis, not the center lemma: its full reduced jet violates (S.129) at degree $m+1$. At the critical degree $m+2$, mixed terms are instead retained in $\mathcal{J}_{m+2}$ and in the angular potential; (S.131) says only that eliminating the positive transverse block cannot alter their active gradient.

Arbitrarily small obstructing jets. The preceding global polynomial can be made arbitrarily close, coefficientwise, to $w^2/2 + r^m + y^{m+2}$: multiply the coefficients 1/4, 1/8 and C_m by $\epsilon \in (0, 1]$. On $|y| \leq 1$, its determinant factor is

$$(m-1)\epsilon/4 - m\epsilon^2/16 - (m+1)\epsilon^2 y/16 - (m+1)\epsilon^2 y^2/32 \geq \epsilon(3m - 11)/32 > 0.$$

Outside the unit ball the convexifier dominates $\epsilon B_m R^m$ exactly as before. The leading active force on a hypothetically captured tail is now $\epsilon r^m/4$, and the same harmonic contradiction applies. Thus critical capture is not an open property among all convex analytic objectives having only the specified pure-axis orders.

Allowed critical mixing changes the angular profile. In the same physical four-coordinate chart, for $0 \le \beta < 2$ and $d > 0$ define the symmetric perspective pair

$$F_{\mathrm{sym}} = \tfrac{1}{2}w^2 + \frac{r^m}{2}\big[(1-\beta y)^{-(m-1)} + (1+\beta y)^{-(m-1)}\big] + dy^{m+2}. \tag{S.189}$$

For $\beta > 0$ it is analytic on the open neighborhood $|\beta y| < 1$ of the entire closed simplex. For $t = 1 - \beta y$, the first perspective term r^m/t^{m-1} has Hessian

$$\frac{m(m-1)r^{m-2}}{t^{m+1}} \begin{pmatrix} t^2 & \beta r t \\ \beta r t & \beta^2 r^2 \end{pmatrix} \succeq 0.$$

The other term has the opposite β sign and the same positivity. This proves genuine open-neighborhood convexity; at $\beta = 0$ the formula is polynomial. All terms are nonnegative and their common zero is the origin, so global entry applies separately. The center is zero and

$$f_{\mathrm{red}} = r^m + \mathcal{J}_{m+2}(r,y) + O(s^{m+4}),$$
$$\mathcal{J}_{m+2}(r,y) = dy^{m+2} + A_\beta r^m y^2, \qquad A_\beta = \tfrac{1}{2}m(m-1)\beta^2.$$

For $\beta > 0$ this critical polynomial is not jointly convex: the leading term of its Hessian determinant as $y/r \to 0$ at $r \neq 0$ is $-2A_\beta^2 m(m+1)r^{2m-2}y^2 < 0$. The analytic theorem nevertheless applies, because $\mathcal{J}_{m+2}(0,y) = dy^{m+2} > 0$ and the full objective is convex. With the true metric $G = 8$ and $\lambda = m/2$, the angular potential is

$$V_\beta(v) = dv^{m+2} + (A_\beta - 2m)v^2.$$

At $m = 4, d = 1, \beta = 0$, its critical set is $\{0,\ \pm(8/3)^{1/4}\}$. At $\beta = 3/2$, it is $V_\beta(v) = v^6 + (11/2)v^2$ and its sole critical point is zero, so every interior-start orbit has $y/r \to 0$. These objectives have identical pure radial and active restrictions but distinct angular critical sets. The first critical set does not assert that every orbit selects a nonzero point; $y = 0$ is invariant. This example also shows why the general theorem cannot require joint convexity of the reduced critical jet.

S5.14 Convexity forces the degree split

We now prove Main Theorem 10 and its positive-Hessian refinement, using the same coupled-jet geometry. The zero-tilt result and its homogeneous bias have been proved above. In the tilted branch write $f = f_{\mathrm{red}}$, $p = m+2$, $q = p/2$, and let $b \in \mathbb{R}^d$ denote the active tilt. Use $b_0 \in E_0$ for the transverse-center coefficient to avoid confusion.

Proposition S5.3 (Forced tilt and transverse invariance)**.** Under Main (M.68), the degree-$(m+1)$ term is $c_1 r^{m+1} + r^m b \cdot y$. The pure-active polynomial $H(y) = \mathcal{J}_p(0,y)$ is convex and strictly convex. The function $H(v) + b \cdot v$ has one minimizer v_b, with $v_b \neq 0$ if and only if $b \neq 0$. Moreover,

$$\phi = b_0 r^q + O(s^{q+1}), \quad \phi_r = q b_0 r^{q-1} + O(s^q), \quad \phi_y = O(s^q), \tag{S.190}$$

and

$$F(0,r,y) - f(r,y) = \tfrac{1}{2} b_0^T K_w b_0 r^{m+2} + O(s^{m+3}), \qquad K_w = F_{ww}(0) \succ 0. \tag{S.191}$$

Thus c, c_1, b, H are identical on the raw slice and after reduction.

Proof. On a small fixed product convex box, the stationary transverse minimizer is interior and is the minimizer over the transverse box. Convex-combination comparison proves convexity of f. Its leading active Hessian is

$$f_{yy}(\varepsilon r, \varepsilon y) = \varepsilon^{m-1} (\mathcal{P}_{m+1})_{yy}(r,y) + O(\varepsilon^m).$$

Since $m-1$ is odd and convexity holds for both signs of small ε, $(\mathcal{P}_{m+1})_{yy} = 0$. Polynomial homogeneity gives the asserted degree split. Rescaling $f(0,y)$ proves convexity of H.

If equality in the convexity inequality for H held on a nontrivial segment, its polynomial restriction to that line would be affine on an interval and hence on the whole line. Its degree-p coefficient in the nonzero line direction d is $H(d) > 0$, a contradiction. Thus H is strictly convex. Positivity and homogeneity make $H + b \cdot v$ coercive, so it has one minimizer. Its gradient vanishes there; $\nabla H(0) = 0$ proves the last assertion.

The derivative-information remark after Lemma S12 uses only $f = cr^m + O(s^{m+1})$, so it applies here. More explicitly, its uniform inequality

$$\|D^2\phi(t)[v,v]\|^2 \le C D^2 f(t)[v,v], \qquad \|v\| = 1,$$

first forces order at least q for ϕ. The pure-active estimate excludes any active-quadratic term from the degree-q center, leaving $b_0 r^q + r^{q-1} C y$. The radial test at arbitrarily small $r/\|y\|$ eliminates C. This proves (S.190); stationary square completion proves (S.191). The same nonnegative-Hessian derivative estimate applied to $F_w(0,r,y)$ proves the center bound from the raw slice, without assuming equality of raw and reduced jets in advance. Equivalently, repeat the proof of Lemma S12 with the raw transverse force: its first possible degree-q term is radial, and eliminating it changes only the indicated radial square term.

The anisotropic conclusion is stronger than one on bounded y/r. For weights $\mathrm{wt}(r) = 1$, $\mathrm{wt}(y) = \theta = m/(m+1)$, every ordinary degree-$(q+1)$ center monomial has weight at least $\theta(q+1) > q$, with gap $(m-2)/[2(m+1)]$. The square-completion correction therefore begins at weight $2q = m+2$, strictly above the tilted active weight $m + \theta$. ∎

At $(r,0)$ the radial Schur complement gives the necessary condition

$$(\mathcal{J}_p)_{yy}(1,0) \succeq \frac{m}{c(m-1)} b b^T. \tag{S.192}$$

Indeed, $f_{rr} \sim cm(m-1)r^{m-2}$, $f_{ry} \sim m r^{m-1} b$, and $f_{yy} \sim r^m (\mathcal{J}_p)_{yy}(1,0)$. Divide the nonnegative complement by r^m and let positive r tend to zero. Thus the uncompensated expression $cr^m + r^m b \cdot y + H(y)$ is not convex for $b \neq 0$.

For any convex positive homogeneous H and any b, the model

$$F_0(w,r,y) = \tfrac{1}{2} w^T K_w w + \frac{c r^m}{(1 - d_0 \cdot y)^{m-1}} + H(y), \qquad d_0 = \frac{b}{c(m-1)}, \tag{S.193}$$

is convex and analytic on a small region with positive denominator. The middle term is the perspective of the convex scalar power cr^m; direct differentiation gives its positive-semidefinite rank-one radial/active Hessian. It attains equality in (S.192). The condition $b \neq 0$ is relatively open in these jet coefficients. Every zero-tilt germ can locally be approximated by nonzero-tilt convex analytic germs with the same c, H and Hessian: take a small convex combination with (S.193) using the same transverse Hessian, then use the raw-slice invariance just proved. This is a statement about this constrained local-jet class, not unrestricted genericity of convex analytic objectives.

S5.15 The bootstrap before tilted capture

This subsection extends the entry and preliminary tracking arguments in Sections S5.3–S5.4 and the subsequent transverse bootstrap. We verify every changed force order; the exact identities and scalar comparison proof are reused rather than duplicated.

Shrink the source box. The tilt is bounded by a small multiple of $|r|^m$. Positive H and Young inequalities give $\mathcal{J}_p(r,y) \geq c_H \|y\|^p - C|r|^p$. Absorb the c_1 term and analytic remainder, and use transverse positivity, to obtain

$$f \geq c_0(|r|^m + \|y\|^p), \qquad F \geq c_0(|r|^m + \|y\|^p + \|\delta\|^2). \tag{S.194}$$

For the actual target weights $\omega_i = x_i x_i^+$,

$$\Delta x_i = -\eta \omega_i (g_i - \mu), \qquad \mu = \frac{\sum_i \omega_i g_i}{\sum_i \omega_i}. \tag{S.195}$$

Source–target ratios are bounded on a bounded-gradient set and tend to one on a convergent tail. The smooth descent estimate is

$$F(x) - F(x^+) \geq \eta(1 - \eta L/2) \sum_i \omega_i (g_i - \mu)^2.$$

On a protected active tail, convexity and weighted Cauchy–Schwarz give $F \leq C\sqrt{D_F}$, where $D_F = \min_s \sum_i x_i^2 (g_i - s)^2$. The missing-coordinate comparator ratios equal one, while supported ratios are bounded. Thus for $T = 1 + \eta(k - k_0)$,

$$F = O(T^{-1}), \quad r = O(T^{-1/m}), \quad y = O(T^{-1/p}), \quad \delta = O(T^{-1/2}). \tag{S.196}$$

Positivity of the kernel ray gives $\|u\| \leq Cr$ and $\|w\| \leq Cr$. Combining (S.190) with (S.196) gives

$$\phi = O(T^{-1/2-1/p}), \qquad w = O(T^{-1/2}), \qquad \phi_r, \phi_y = O(T^{-1/2}).$$

The exact stationary Taylor representation yields

$$\begin{aligned} F_j &= f_j + O(\|\phi_j\|\|\delta\| + \|\delta\|^2), \quad j = r, y, \\ F_u &= (\ell/N) f_r + K\delta + O(s\|\delta\| + \|\delta\|^2). \end{aligned} \tag{S.197}$$

The tilt adds $mr^{m-1}b \cdot y$ to f_r and $r^m b$ to f_y. On the coarse box these are, respectively, $O(T^{-(m-1)/m})$ and $O(T^{-1})$, so the old pre-capture bounds remain

$$|f_r| \le C[T^{-(m-1)/m} + T^{-(m+1)/p}], \qquad \|F_y\| = O(T^{-(m+1)/p}), \quad \|F_u\| = O(T^{-1/2}). \tag{S.198}$$

Normal lower capture before active capture. Set $\psi = b_A^T(1/x_A)$ and $Q = B^T(1/x) = 1/u - \psi \mathbf{1}_N$. Exactly,

$$\Delta Q = \eta F_u, \qquad \Delta(a^T Q) = \eta F_r.$$

The second identity annihilates the transverse derivative. By (S.197), $F_r - f_r = O(T^{-1})$. Summation with (S.198) yields $a^T Q \le C T^{1/m}$. Restoring bounded ψ and using positivity of each a_i/u_i proves

$$u_i \ge cT^{-1/m}, \qquad u_i, r \asymp T^{-1/m}, \qquad w/r = O(T^{-1/2+1/m}) \longrightarrow 0. \tag{S.199}$$

No active ratio, radial monotonicity or final transverse lag was assumed.

Actual multiplier and preliminary lag. Use the affine representative $g_I = F_u$, $g_A = C_{y,A}^T F_y$, where $C_{y,A} = G^{-1} E_A^T \operatorname{diag}(x_{*,A}^{-2})$ and $C_{y,A} x_{*,A}^2 = 0$. The exact weights in (S.195), followed by absorption of the small $\eta\|F_y\||\mu|$ term, give

$$|\mu| \le C_\eta[T^{-1} + r^2(\|\delta\| + |f_r|)]. \tag{S.200}$$

Indeed the unsimplified bound is $C[s\|F_y\| + \eta\|F_y\|^2 + \eta\|F_y\||\mu| + r^2\|F_u\|]$, and (S.198) bounds its first two terms by $C_\eta T^{-1}$. Consequently

$$\|\Delta u\| \le C_\eta \eta T^{-1/2-2/m}, \qquad \|\Delta y\| \le C_\eta \eta [T^{-(m+1)/p} + T^{-1/2-2/m}].$$

Integrate $D\phi$ on the actual source–target segment, whose endpoints and segment satisfy the same coarse bounds. This gives the exact mean-value estimate

$$\|\Delta\phi\| \le C_\eta \eta [T^{-1-2/m} + T^{-3/2+1/p}]. \tag{S.201}$$

There is no discarded Taylor remainder here.

The literal rational normal map is $u_i^+ = u_i/[1 + \eta u_i(F_{u_i} - \mu)]$. Its quadratic correction has size $O_\eta(\eta^2 r^3 T^{-1})$. Expanding its source mobility about $r^2 D_a^2$, projecting by P and using $P D_a^2 \ell = 0$ gives

$$\begin{aligned} \Delta w &= -\eta r^2 A_0 \delta + R_w, \qquad A_0 = (P D_a^2 K)|_{E_0}, \\ \|R_w\| &\le C_\eta \eta [r^2 s\|\delta\| + r\|w\|\|\delta\| + r\|w\||f_r| + r^2|\mu|] + C_\eta \eta^2 r^3 T^{-1}. \end{aligned} \tag{S.202}$$

For $v, w \in E_0$, $\langle v, A_0 w\rangle_{D_a^{-2}} = v^T K w$. Hence A_0 is positive self-adjoint in this metric. All variable δ coefficients are small relative to r^2, by (S.199), $s \to 0$ and $f_r/r \to 0$. Split $w = \phi + \delta$ in the remaining mixed force and use the center estimate. Subtract (S.201) to obtain

$$\|\delta^+\|_* \le (1 - c_0\eta T^{-2/m})\|\delta\|_* + C_\eta\eta[T^{-1-2/m} + T^{-3/2+1/p}]. \tag{S.203}$$

The fixed-clock supersolution from Section S5.1 proves $\delta = O_\eta(T^{-\beta})$, with β in Main (M.70). The strict inequalities $\beta + 1/2 > 1 + 1/m$ and $2\beta > 1 + 1/m$ imply

$$F_y - f_y = o(T^{-1-1/m}) = o(r^{m+1}). \tag{S.204}$$

This derivation has used neither the zero-tilt capture nor its stronger post-capture lag.

Entry versions. For a sufficiently close interior start, (S.194) isolates the local minimizer. On a larger convex neighborhood, (S.195) bounds the target displacement by a constant times the source distance from x_*, since the affine gradient vanishes there. A smaller source ball has its targets in this neighborhood. Choose a local F-sublevel below the positive minimum on the intervening feasible annulus; descent prevents escape. The exact Burg three-point inequality against a fixed nearby interior comparator ξ is

$$\eta[F(x^+) - F(\xi)] \le D_h(\xi, x) - D_h(\xi, x^+) - (1 - \eta L)D_h(x^+, x).$$

If the decreasing objective limit were positive, choose ξ nearer x_* with smaller value and telescope to a contradiction. Thus $F \to 0$, and coercivity gives $x \to x_*$. Under the separate global convexity, smoothness and unique-minimizer assumptions, the same inequality works throughout the simplex; compactness gives convergence from every interior start. These are precisely the three entry alternatives in the main statement, with the original admitted L and fixed η.

S5.16 Tilted capture and literal normal asymptotics

The exact supported dual coordinate and its analytic inverse satisfy

$$\begin{gathered}\Delta z = -\eta G^{-1} F_y, \qquad z = y + O((\|u\| + \|y\|)^2), \\ y = \chi(u, z) = z + O((\|u\| + \|z\|)^2), \qquad \chi_z = I + O(\|u\| + \|z\|), \quad \chi_u = O(\|u\| + \|z\|).\end{gathered} \tag{S.205}$$

The vanishing linear normal term follows from the chosen affine chart. Set $R = \|z\|_G$. For $R \ge K\rho$, with K sufficiently large, Euler's identity and positivity of H give

$$z^T f_y \ge c_0 R^p, \qquad \|f_y\| \le C(R^{p-1} + r^m). \tag{S.206}$$

To verify uniformity before capture, note that $r/R \to 0$ on this region. The inverse chart changes y from z by a relative vanishing error. The tilt costs at most CRr^m

and is absorbed since $\rho^{p-1} = r^m$. Every degree-p mixed derivative has a vanishing r/R factor relative to H_y, while the analytic remainder is also smaller. These observations prove (S.206) on a small tail neighborhood.

Square the exact dual update in the G norm. The positive discrete term is $\eta^2 F_y^T G^{-1} F_y$; outside the ball its ratio to the coercive term is $O(\eta R^{p-2}) \to 0$. The residual (S.204) is smaller than r^m and is absorbed after increasing K. Inside $R \le K\rho$, the entire step is $O_\eta(\eta r^m)$. Combining both regions gives

$$R^+ \le R - c_0 \eta R^{m+1} + C_\eta \eta T^{-1}. \tag{S.207}$$

On a sufficiently small fixed interval the scalar map on the right is increasing. A large multiple $AT^{-1/(m+1)}$ is a supersolution: its derivative has exponent $1 + 1/(m+1) > 1$, whereas its power term and forcing have exponent one. Starting late and increasing A to dominate the initial value proves

$$z, y = O(T^{-1/(m+1)}) = O(\rho). \tag{S.208}$$

Now (S.190) yields

$$\phi = O(r^q), \quad \phi_r = O(r^{q-1}), \quad \phi_y = O(r^{\theta q}).$$

Here $\theta(q+1) > q$ controls the center remainder. Since $m\beta > q$, the preliminary lag gives $w = O(r^q)$. Expanding the reduced forces yields

$$\begin{aligned} f_r &= mcr^{m-1} + O(r^{m-1+\theta}), \qquad F_y = O(r^m), \\ F_u &= (\ell/N)mcr^{m-1} + O(r^{m-1+\theta}) + O(\delta). \end{aligned} \tag{S.209}$$

For the first estimate the tilt contributes $O(r^{m-1+\theta})$, while $(\mathcal{J}_p)_r = O(\rho^{m+1}) = O(r^m)$. The actual multiplier formula gives $\mu = O_\eta(r^{m+\theta})$; in particular $r^2\delta$ is subordinate since $m\beta + 2 > m + \theta$.

Insert these estimates and $u = ra + O(r^q)$ into the literal rational normal map. Its leading mobility is $r^2 D_a^2$, and $D_a^2 \ell = a$. The possible relative error exponents are at least θ, $m\beta - m + 1$, $q - 1$, $1 + \theta$, and m. Their safe minimum is κ of Main (M.70). Consequently

$$\Delta u = -\eta\lambda r^{m+1} a + O_\eta(\eta r^{m+1+\kappa}), \qquad \Delta r = -\eta\lambda r^{m+1} + O_\eta(\eta r^{m+1+\kappa}). \tag{S.210}$$

Every literal normal and r are eventually strictly decreasing. Power-reciprocal expansion gives

$$(r^+)^{-m} - r^{-m} = \eta m\lambda + O_\eta(\eta r^\kappa).$$

Since $r \asymp k^{-1/m}$ and $0 < \kappa/m < 1$, summation gives an error $O(k^{1-\kappa/m})$. Inversion yields the main radial equivalent with its relative power error. All nonradial reduced terms are $O(r^{m+\theta})$, the $c_1 r^{m+1}$ term is smaller, and $\delta^2 = o(r^m)$. Thus $kF(x_k) \to c/(\eta m\lambda) = N/(\eta m^2)$.

S5.17 The moving potential and unique attraction

The fractional substitution is analytical only; it changes neither the algorithm nor its metric. Define

$$\Psi(t,v)=\frac{f(t^{m+1},t^m v)-ct^{m(m+1)}}{t^{m(m+2)}}. \tag{S.211}$$

It extends real analytically through $t=0$ on every bounded v set. Indeed the tilt and pure-active H have exactly denominator degree $m(m+2)$. The term $c_1 r^{m+1}$ has one more degree; a degree-p monomial with j radial factors has j more; every ordinary degree at least $p+1$ has at least m more. The convergent analytic Taylor series therefore leaves no negative powers. In particular

$$\Psi(0,v)=H(v)+b\cdot v, \qquad \nabla_v\Psi(t,v)=r^{-m}f_y(r,\rho v). \tag{S.212}$$

Set $h=\eta r^{m-\theta}$ and $v=z/\rho$. On the captured region, (S.205) and (S.210) imply

$$\Delta v=-hG^{-1}\nabla_v\Psi(t,v)+O_\eta(h\rho). \tag{S.213}$$

Here all errors have the displayed quantitative strength. The dual-to-physical shift $y-\rho v=O(\rho^2)$ is multiplied by $f_{yy}=O(r^{m-\theta})$, giving force error $O(r^{m+\theta})$. The transverse residual obeys

$$\|F_y-f_y\|\le C(\|\phi_y\|\|\delta\|+\|\delta\|^2)=O(r^{\theta q+m\beta}+r^{2m\beta})=o(r^{m+\theta}). \tag{S.214}$$

Both gaps are strictly positive for every even $m\ge 4$. Finally division by ρ^+ contributes $O(h\rho)$ because $\Delta\rho/\rho=O(\eta r^m)=O(h\rho)$. These checks retain the true Burg matrix G.

The clock satisfies $h\asymp k^{-m/(m+1)}$, so $h\to 0$ and $\sum h=\infty$. The profile is bounded. Uniformly on its bounded set, (S.213) is

$$\Delta v=-hG^{-1}[\nabla H(v)+b+o(1)].$$

For $v\neq v_b$, strict convexity gives

$$(v-v_b)\cdot[\nabla H(v)-\nabla H(v_b)]>0.$$

Its minimum on each compact annulus about v_b is positive. Expanding the squared G-distance, its decrement is therefore a negative multiple of h on that annulus, up to $o(h)$. The increments are $O(h)$. Divergence of $\sum h$ forces entry into each smaller ball; sufficiently late outward crossings cannot skip a separating annulus, and its distance decrement prevents escape. Hence $v\to v_b$. The chart error divided by ρ vanishes, so $y/\rho\to v_b$ as well. Since $b\neq 0$, $v_b\neq 0$, and $r=o(\rho)$, the physical affine chart gives $x-x_*=\rho E v_b+o(\rho)$.

S5.18 Finite physical length at every forced profile

Summing $\|H_y(y)\| + \|b\|r^m$ would fail: the two leading forces have nonsummable harmonic magnitude and cancel. Nor may one replace Ψ by $\Psi(0,\cdot)$, because its first $O(t)$ correction is generally nonsummable in the fast clock. Instead keep (S.213) and the complete moving potential.

The already proved radial monotonicity makes t and ρ eventually decreasing. Thus

$$\sum |\Delta t| < \infty, \qquad \sum h\rho^2 = \sum \eta r^{m+\theta} < \infty. \tag{S.215}$$

On the compact captured (t,v) set, Ψ and its first two derivatives are bounded. Taylor expansion at fixed t, then the mean-value bound for changing t, gives

$$\Delta\Psi \le -c_0 h\|\nabla_v\Psi\|^2 + Ch\rho\|\nabla_v\Psi\| + Ch^2(\|\nabla_v\Psi\|^2 + \rho^2) + C|\Delta t|.$$

Cauchy–Young absorbs the mixed term, and $h \to 0$ absorbs the quadratic gradient term. Therefore

$$\Delta\Psi \le -c_1 h\|\nabla_v\Psi\|^2 + C_\eta h\rho^2 + C|\Delta t|. \tag{S.216}$$

No sign is assumed for the variation of Ψ in t. Its lower bound on this compact set and (S.215) imply

$$\sum h\|\nabla_v\Psi\|^2 < \infty.$$

Cauchy–Schwarz now yields

$$\begin{aligned}\sum \rho\|\Delta v\| &\le C\sum h\rho\|\nabla_v\Psi\| + C_\eta\sum h\rho^2 \\ &\le C\Big(\sum h\|\nabla_v\Psi\|^2\Big)^{1/2}\Big(\sum h\rho^2\Big)^{1/2} + C_\eta\sum h\rho^2 < \infty.\end{aligned} \tag{S.217}$$

Since $z = \rho v$, boundedness of v and variation of ρ give finite variation of z. Every u_i is eventually decreasing, so u has finite variation. The analytic inverse chart is uniformly Lipschitz on a small convex tail neighborhood; hence $y = \chi(u,z)$ and the physical x have finite Euclidean length. This argument needs no positive forced Hessian and no KL exponent at v_b. It does not assert finite unweighted length of v or a sharp physical tail ratio by itself.

S5.19 The positive-Hessian sharp refinement

Assume $A_b = \nabla^2 H(v_b) \succ 0$. Apply the analytic implicit-function theorem to $\nabla_v\Psi(t,v) = 0$, obtaining analytic $V(t)$ with $V(0) = v_b$ and $\bar y(r) = \rho V(t)$. In exact supported dual coordinates take

$$\bar z(u) = z(u,\bar y(r)), \qquad e = z - \bar z(u).$$

The comparison point uses the current complete u. Replacing it by $\rho V(t)$ would lose chart terms at the lag order. Since $z_y = I + O(\rho)$, $z_u = O(\rho)$ and

$$\bar{y}'(r) = \theta r^{\theta-1} v_b + o(r^{\theta-1}),$$

integration over the actual source–target u segment, using (S.210), gives

$$\Delta\bar{z} = -\eta\theta\lambda r^{m+\theta} v_b + o(\eta r^{m+\theta}). \tag{S.218}$$

This differentiates the explicit static center, not an orbit equivalent. The direct z_u contribution is smaller by a positive power of r.

The profile convergence already proved implies, uniformly on the active segment between $\bar{y}$ and y,

$$f_{yy} = r^{m-\theta}[A_b + o(1)], \qquad y - \bar{y} = [I + o(1)]e.$$

Combine the exact dual update, (S.214) and (S.218) to obtain

$$e^+ = (I - h[G^{-1}A_b + o(1)])e + h\rho^2[\theta\lambda v_b + o(1)]. \tag{S.219}$$

In the G metric the leading coefficient contracts by $1 - c_0 h$ on a late tail. Since the relative decay of ρ^2 is $O(\eta r^m) = o(h)$, a sufficiently large multiple of ρ^2 is a supersolution; hence $e = O(\rho^2)$. Dividing by ρ_+^2 gives the bounded recurrence for $l = e/\rho^2$,

$$\Delta l = h[-G^{-1}A_b l + \theta\lambda v_b + o(1)].$$

Linear annular contraction and $\sum h = \infty$ prove

$$l \longrightarrow \theta\lambda A_b^{-1} G v_b.$$

The physical/dual multiplier is $I + o(1)$, so this proves Main (M.77).

Substitute the lag into the exact dual update. The literal force gives

$$\Delta z = -\eta\theta\lambda r^{m+\theta} v_b + o(\eta r^{m+\theta}).$$

The inverse-chart derivatives in (S.205) make every reconstruction error smaller. The normal physical step is also smaller because $m + 1 > m + \theta$. This proves Main (M.78). Finally

$$\rho - \rho^+ = \eta\theta\lambda r^{m+\theta}[1 + o(1)].$$

Compare the step norm with this positive decrement. The relative error is uniformly vanishing on each sufficiently late tail; telescoping proves Main (M.79). For one active coordinate, $H(v) = a_0 v^p$ with $a_0 > 0$ and $v_b \neq 0$, so its Hessian at v_b is automatically positive.

S5.20 Classification of the polynomial obstruction

For the entire globally convex polynomial in Proposition S5.2, the fixed chart is

$$x = (1/2 - r + y,\ 1/2 - r - y,\ r + w,\ r - w).$$

Thus $a = (1, 1)$, $N = 2$, $G = 8$, $c = 1$, $b = 1/4$, $H(v) = v^{m+2}$ and $\lambda = m/2$. The forced profile is

$$v_b = -[1/(4(m + 2))]^{1/(m+1)},$$

with positive Hessian. The earlier proof supplies global convexity and a unique minimizer, so global entry applies to every interior start, not only the invariant slice used to refute $y = O(r)$. Hence

$$r_k \sim [2/(\eta m^2 k)]^{1/m}, \qquad y_k \sim v_b[2/(\eta m^2 k)]^{1/(m+1)},$$
$$\mathcal{L}_k \sim \sqrt{2}|v_b|[2/(\eta m^2 k)]^{1/(m+1)}.$$

These are applications of the tilted theorem, not extra hypotheses defining its scope.

Relationship to the retained branches. The reflection-symmetric class in Main Section 9 excludes the odd active tilt. Dominant balance, strict-convexity attraction and stable tracking are classical mechanisms; the assertions here concern their exact fixed-step use under the stated full mixed-jet geometry.

S5.21 Sharp endpoints at a corank-one forced Hessian

This subsection proves Main Theorem 12. Assume every hypothesis of the nonzero-tilt theorem, and additionally

$$\dim \ker \nabla^2 H(v_b) = 1.$$

The complete stable active block may have arbitrary dimension. We prove the pointwise estimate

$$Y = y/\rho, \qquad \Delta Y = o(h\rho), \qquad h = \eta r^{m-\theta}. \tag{S.220}$$

Since $h\rho \sim 1/(m\lambda k)$, this is equivalent to $k\Delta Y \to 0$. It yields the literal step first and then the sharp physical tail ratio; no full-vector static-lag equivalent is asserted. The finite energy budgets of Section S5.18 alone do not imply (S.220).

The stronger physical recurrence. Use the full moving potential Ψ from (S.211), now in the physical coordinate Y. It is analytic through $t = 0$, convex in Y for small $t \geq 0$, and

$$\Psi(0, Y) = H(Y) + b \cdot Y, \qquad f_y(r, \rho Y) = r^m \nabla_Y \Psi(t, Y).$$

Profile convergence gives $Y \to v_b$, while (S.214) gives $F_y - f_y = o(r^{m+\theta})$. Hence $F_y = o(r^m)$ as well. The exact dual step and the inverse chart give

$$\Delta y = -\eta G^{-1} f_y(r, y) + o(\eta r^{m+\theta}). \tag{S.221}$$

Indeed the coefficient error is $O(\rho)\Delta z = o(\eta r^{m+\theta})$; the normal term is $O(\eta\rho r^{m+1})$; and the quadratic Taylor error is $O((\|\Delta u\| + \|\Delta z\|)^2) = o(\eta r^{m+\theta})$ for fixed η. This use of the already selected profile is what improves the earlier $O(h\rho)$ error to a little-o error.

Put $\alpha_0 = \theta\lambda$. The literal normal increment proves

$$\begin{aligned} &\Delta\rho = -\alpha_0 h\rho^2[1 + o(1)], \qquad \Delta t = -\frac{\lambda}{m+1} h\rho t[1 + o(1)], \\ &h \to 0, \qquad \sum h = \infty, \qquad \sum h\rho = \infty, \qquad h\rho \sim \frac{1}{m\lambda k}. \end{aligned} \tag{S.222}$$

Divide (S.221) by ρ^+ and use $Y \to v_b$. The denominator correction is $O(h^2\rho\|\nabla\Psi\|) = o(h\rho)$. Thus

$$\Delta Y = -hG^{-1}\nabla_Y \Psi(t, Y) + \alpha_0 h\rho Y + o(h\rho). \tag{S.223}$$

All remainders here are evaluated on the actual sequence unless uniformity is stated. An arbitrary $O(h\rho)$ perturbation would not suffice.

Metric whitening and analytic stable reduction. Make the fixed linear change $V = G^{1/2}Y$, followed by a fixed orthogonal rotation. Write the resulting coordinates as (X, ξ), where ξ is the one-dimensional kernel direction of the limiting Hessian. Let $\Phi(t, X, \xi)$ be the transformed potential and (X_*, ξ_*) its limiting profile. No translation is made, so the dilation term retains its form:

$$\Delta X = -h\Phi_X + \alpha_0 h\rho X + o(h\rho), \qquad \Delta\xi = -h\Phi_\xi + \alpha_0 h\rho\xi + o(h\rho). \tag{S.224}$$

At the limit, $\nabla^2\Phi = \mathrm{diag}(A, 0)$ with $A \succ 0$. The analytic implicit-function theorem supplies $X = C(t, \xi)$ with $\Phi_X(t, C(t, \xi), \xi) = 0$ and $C_\xi(0, \xi_*) = 0$. Put

$$e = X - C(t, \xi), \qquad \psi(t, \xi) = \Phi(t, C(t, \xi), \xi), \qquad g = \psi_\xi. \tag{S.225}$$

For each small $t \geq 0$, minimizing over a fixed convex X box whose minimizer remains interior proves convexity of ψ on a fixed small ξ interval.

Strict convexity also holds; it must not be confused with Hessian positivity. Put $p = m + 2$. The pth directional derivative of $\Phi(0, \cdot)$ is $p!H(Ld) > 0$ for every nonzero direction d, where L is the invertible whitening map. It is uniformly positive for unit directions, independent of the base point. Analytic continuity preserves positivity on a sufficiently small compact box for small t. Convexity then precludes any nontrivial affine segment, giving strict convexity of Φ. For two minimizing graph points with distinct ξ,

apply strict convexity at their convex combination and then minimize in X; this proves strict convexity of ψ. Consequently

$$g(t,\cdot) \text{ is strictly increasing}, \qquad g_\xi \geq 0, \qquad g(0,\xi_*) = g_\xi(0,\xi_*) = 0, \tag{S.226}$$

and $g(0,\cdot)$ is not identically zero. Mean-value formulas on the actual transverse segment give

$$\Phi_X = [A + o(1)]e, \qquad \Phi_\xi = g(t,\xi) + B_k e, \qquad \|B_k\| \to 0. \tag{S.227}$$

The first relation uses the integral of Φ_{XX}, so contraction is Euclidean after whitening.

Entry into a stable force cone. Subtract the actual center increment $C(t^+,\xi^+) - C(t,\xi)$ from the first equation in (S.224). Its t contribution is $O(h\rho t)$ and its ξ coefficient tends to zero. Absorb the small linear terms using (S.227). There exist $c_0, C_0 > 0$ and $\varepsilon_k \to 0$ such that

$$\begin{aligned} \|e^+\| &\leq (1 - c_0 h)\|e\| + C_0 h\rho + \varepsilon_k h|g|, \\ |\Delta\xi| &\leq C_0 h(\rho + |g| + \varepsilon_k\|e\|). \end{aligned} \tag{S.228}$$

Since g_t is bounded and $g_\xi \to 0$ uniformly on actual source–target segments, the clock and these bounds imply

$$|\Delta g| \leq \varepsilon_k' h(\rho + |g| + \|e\|), \qquad \varepsilon_k' \to 0. \tag{S.229}$$

There is a K independent of $\varepsilon > 0$ such that for each fixed ε, eventually

$$\|e\| \leq K\rho + \varepsilon|g|. \tag{S.230}$$

Here are both entry and invariance. Take $K \geq 4C_0/c_0$ and $R = K\rho + \varepsilon|g|$. For these fixed constants, (S.222) and (S.229) imply

$$|\Delta R| \leq \delta_k h(R + \|e\|), \qquad \delta_k \to 0.$$

Start late enough that $\varepsilon_k \leq c_0\varepsilon/4$. The forcing in the first line of (S.228) is at most $(c_0/4)hR$. If $\|e\| \leq R$, then $\|e^+\| \leq (1 - 3c_0 h/4)R$, whereas $R^+ \geq (1 - 2\delta_k h)R$. Thus the cone is invariant late enough. Outside it, $W = \|e\| - R$ satisfies $W^+ \leq W - (c_0/2)h\|e\|$. If entry never occurred, $\|e\| \geq K\rho$ and $\sum h\rho = \infty$ would force $W < 0$, a contradiction.

Use one fixed ε in (S.230). Since $B_k \to 0$ and $\xi \to \xi_*$, the scalar equation becomes

$$\xi^+ = \xi - hg(t,\xi) + \alpha_0 h\rho\xi_* + R_k, \qquad |R_k| \leq \delta_k h(\rho + |g(t,\xi)|), \quad \delta_k \to 0. \tag{S.231}$$

Analytic force-level barriers. Fix $\varepsilon > 0$. For each small positive t, define the unique roots

$$g(t,\xi_-(t)) = t^m(\alpha_0\xi_* - \varepsilon), \qquad g(t,\xi_+(t)) = t^m(\alpha_0\xi_* + \varepsilon). \tag{S.232}$$

They exist in a fixed small interval about ξ_* because the limiting strictly increasing force has opposite signs at its endpoints. Both roots converge to ξ_*, with $\xi_- < \xi_+$.

We use the classical one-variable convergent Puiseux theorem only at this point. On a restricted analytic box the bounded root graphs in (S.232) are semianalytic, hence their uniquely defined functions are globally subanalytic. Each admits a convergent Puiseux expansion at zero [11, Proposition 1.8.4]. Their finite limits imply

$$t\xi'_-(t) \to 0, \qquad t\xi'_+(t) \to 0. \tag{S.233}$$

Alternatively, Weierstrass preparation at the finite-order zero in ξ, followed by convergent Newton–Puiseux, gives the same result. The discrete orbit is not asserted to be subanalytic. Because $t^+/t \to 1$, (S.222) and (S.233) give

$$\xi_\pm(t^+) - \xi_\pm(t) = o(h\rho). \tag{S.234}$$

For a fixed positive upper bound L_g on g_ξ on the box, the force gap implies

$$\xi_+(t) - \xi_-(t) \geq 2\varepsilon\rho/L_g. \tag{S.235}$$

The comparison maps

$$T_k^\pm(\xi) = \xi - hg(t,\xi) + \alpha_0 h\rho\xi_* \pm \delta_k h(\rho + |g(t,\xi)|) \tag{S.236}$$

are increasing on the fixed interval for sufficiently late k. Indeed g is increasing and L_g-Lipschitz; a difference bound gives slope at least $1 - hL_g(1+\delta_k) > 0$. This holds also at $g = 0$, without differentiating its absolute value. At the upper and lower force boundaries,

$$\begin{aligned} T_k^+(\xi_+(t)) - \xi_+(t) &= -\varepsilon h\rho + o(h\rho), \\ T_k^-(\xi_-(t)) - \xi_-(t) &= +\varepsilon h\rho + o(h\rho). \end{aligned} \tag{S.237}$$

Together with (S.234), monotonicity proves forward invariance of the moving interval $[\xi_-(t_k), \xi_+(t_k)]$.

Entry is also necessary to the argument. Above the upper boundary, the worst drift allowed by (S.231), as a function of $s = g(t,\xi)$, is

$$-s + \alpha_0\rho\xi_* + \delta_k(\rho + |s|).$$

For $\delta_k < 1$ this decreases in s, so it is at most its boundary value $-\varepsilon\rho + o(\rho)$. Allowing the boundary motion, the distance above decreases by at least $(\varepsilon/2)h\rho$. Divergence of $\sum h\rho$ rules out staying above forever; the same argument works below. A first crossing from above cannot skip the whole interval: monotonicity gives $\xi^+ \geq T_k^-(\xi_+(t)) = \xi_+(t) + O(h\rho)$, whereas (S.235) gives a gap of order ρ and $h \to 0$. The crossing from below uses T_k^+. Thus entry occurs and remains permanent. Since ε is arbitrary,

$$\frac{g(t_k,\xi_k)}{\rho_k} \longrightarrow \alpha_0\xi_*. \tag{S.238}$$

This includes $\xi_* = 0$; no nonzero kernel component is needed.

Literal step and tail. The cancellation gives $g = O(\rho)$, and the cone gives $e = O(\rho)$. Equation (S.231) then yields $\Delta\xi = o(h\rho)$. The actual center increment is $o(h\rho)$ because C_t is bounded, $\Delta t = O(h\rho t)$, and $C_\xi \to 0$. Hence

$$\Delta e = -h[A + o(1)]e + \alpha_0 h\rho X_* + o(h\rho).$$

Divide by ρ^+, whose relative change is $O(h\rho) = o(h)$. The bounded sequence $l = e/\rho$ obeys

$$\Delta l = h[-Al + \alpha_0 X_* + o(1)].$$

For $l_* = \alpha_0 A^{-1} X_*$ this gives

$$\|l^+ - l_*\| \le (1 - c_1 h)\|l - l_*\| + h\, o(1).$$

Divergence of $\sum h$ and late fixed-radius trapping prove $l \to l_*$. Return to the exact first equation in (S.224): its leading $h\rho$ terms cancel, so $\Delta X = o(h\rho)$. The fixed whitening transform and the scalar increment prove (S.220).

Finally $y = \rho Y$ and $Y \to v_b$ give

$$\Delta y = v_b \Delta\rho + o(h\rho^2) = -\eta\theta\lambda r^{m+\theta} v_b + o(\eta r^{m+\theta}).$$

The normal increment is smaller because $m + 1 > m + \theta$. The fixed affine physical chart proves the literal step. Its norm is

$$\|\Delta x_j\| = \|E v_b\|(\rho_j - \rho_{j+1})[1 + o(1)].$$

For each fixed tolerance the relative error is uniformly small on a sufficiently late tail. Summing and telescoping proves $\mathcal{L}_k \sim \|E v_b\|\rho_k \sim \|x_k - x_*\|$. This argument proves endpoints without a universal vector lag of order ρ^2.

Classical input and remaining scope. The Puiseux lemma, scalar order barriers and stable tracking are classical ingredients. A direct invocation of standard KL descent does not supply these endpoints: the usual fixed relative-error bound between gradient and step is not automatic in singular physical Burg coordinates [10, Section 2.3 and Theorem 2.9]. Even finite length alone gives neither a literal-step equivalent nor the tail/distance ratio. Continuous analytic gradient secant results [1] also do not identify a discrete nonautonomous weighted trajectory with a continuous gradient curve. The separate exact force cancellation above is essential.

For a forced-Hessian kernel of dimension at least two, scalar ordering has no direct vector replacement. Sharp pointwise speed and sharp tail equivalence remain open here; this is not a counterexample to either conclusion. The positive-Hessian lag theorem retains its stronger separate hypothesis. A universal full-vector static lag is not asserted for the corank-one theorem.

S6 Mixed polynomial rate types

The polynomial class, coefficient criterion and conclusions are stated in Main Theorem 13. Convexity is a separate hypothesis, not a consequence of coefficient positivity.

S6.1 Global entry and exact monotone dual coordinates

The two positive pure-axis terms and coefficient nonnegativity imply $F(y,u) > 0$ away from $(0,0)$ on D. Hence $x_* = (1/2, 1/2, 0)$ is the unique minimizer. The interior-comparator calculation in Main Lemma 2, which uses convexity only on the simplex, gives

$$\eta[f(x_{k+1}) - f(z)] \le D_h(z, x_k) - D_h(z, x_{k+1}) - (1 - \eta L) D_h(x_{k+1}, x_k) \tag{S.239}$$

for interior z, and $f(x_k) = O(\log k/k)$. Compactness and uniqueness therefore imply

$$x_k \to x_*, \qquad x_{1,k}, x_{2,k} \to \tfrac{1}{2}. \tag{S.240}$$

No active Hessian lower bound is needed. The entry estimate is used before sharp rates or finite length.

Lemma S6.1 (Exact reciprocal chart and physical variation)**.** *Define*

$$R = \frac{1}{x_1} - \frac{1}{x_2}, \qquad r = -\frac{R}{8} = \frac{y}{4x_1x_2}, \qquad Q = \frac{1}{u} - \frac{1}{2x_1} - \frac{1}{2x_2}. \tag{S.241}$$

Then

$$r^+ = r - \frac{\eta}{8} F_y(y,u), \qquad Q^+ = Q + \eta F_u(y,u). \tag{S.242}$$

The sign of r, hence of y, is preserved. If $r \neq 0$, its absolute value strictly decreases. The variable Q strictly increases and tends to infinity. On a late tail, $w = 1/Q > 0$ and the inverse chart is analytic, with

$$u = w + O(w^2), \qquad y = r[1 + O(w + r^2)]. \tag{S.243}$$

There is a constant K such that

$$\sum_{j \ge k} \|x_{j+1} - x_j\| \le K(w_k + |r_k|) \tag{S.244}$$

on that tail.

Proof. Pair (M.84) with $E = (1, -1, 0)$ and $B = (-1/2, -1/2, 1)$. Their total masses vanish and their objective derivatives are F_y and F_u, giving (S.242). These are coupled identities: the mixed monomials remain in both derivatives.

For $y \neq 0$, coefficient positivity gives $F_y/y > 0$. Along the feasible segment from $(0,u)$ to (y,u), the admitted ambient Lipschitz bound gives $F_{yy} \leq \|E\|^2 L = 2L$. Since $F_y(0,u) = 0$,

$$\frac{r^+}{r} = 1 - \frac{\eta}{2} x_1 x_2 \frac{F_y}{y} \geq 1 - \eta L x_1 x_2 \geq 1 - \frac{\eta L}{4} > 0. \tag{S.245}$$

Thus no nonzero orbit reaches or crosses $r = 0$, and $|r|$ strictly decreases. The branch $r = 0$ is exactly invariant. Also $F_u > 0$ for $u > 0$ because of the positive pure normal term. Hence Q strictly increases; (S.240) implies $Q \to \infty$.

Solving the quadratic equation for y in (S.241) gives

$$\begin{aligned} y &= \frac{2r(1-u)^2}{1 + \sqrt{1 + 16r^2(1-u)^2}}, \\ Q &= \frac{1}{u} - A_0(u,r), \qquad A_0(u,r) = \frac{1 + \sqrt{1 + 16r^2(1-u)^2}}{1-u}. \end{aligned} \tag{S.246}$$

The analytic map $w = u/[1 - uA_0(u,r)]$ has derivative one in u at the origin. Analytic inversion yields (S.243). The reconstructed physical point is Lipschitz in (w,r) on a fixed small chart. Both w and $|r|$ decrease to zero, while r has fixed sign. The total variation of these dual variables from time k is therefore at most $w_k + |r_k|$, which proves (S.244). Adding the finite initial segment proves finite physical length before any sharp rate or monotonicity of u is asserted. ∎

S6.2 The weighted support consequence of convexity

Lemma S6.2 (Necessary support inequality)**.** *Every supported monomial in* (M.82) *satisfies*

$$\frac{2a}{p} + \frac{b}{m} \geq 1. \tag{S.247}$$

Proof. For sufficiently small feasible (y,u), both $(2y,0)$ and $(0,2u)$ are feasible. Convexity and (M.83) give

$$0 \leq F(y,u) \leq \tfrac{1}{2} F(2y,0) + \tfrac{1}{2} F(0,2u) \leq K(|y|^p + u^m). \tag{S.248}$$

Set $y = t^{1/p}$, $u = t^{1/m}$ for small $t > 0$. Each nonnegative monomial is at most $2Kt$, which is impossible as $t \downarrow 0$ if its exponent in t is less than one. ∎

The converse fails. For $F = y^4 + u^4 + 7y^2u^2$, all supported monomials have weighted degree one for $p = m = 4$, but at $y = u = t > 0$ the Hessian is $t^2 \left(\begin{smallmatrix} 26 & 28 \\ 28 & 26 \end{smallmatrix}\right)$, with determinant $-108t^4$. These points are in the simplex interior for small t, so even local simplex convexity fails. This example isolates the role of the separately verified convexity hypothesis.

S6.3 Normal capture with a power remainder

The normal equation in (S.242) contains the full F_u. The following proof makes that mixed contribution summable without any fixed lift cancelling it.

Proposition S6.3 (Normal asymptotics before active classification)**.** *The normal profile, objective coefficient, and literal speed in* (M.86) *hold before using any of the active equivalents in Theorem 13.*

Proof. Write $q_0 = p-2$ and $\rho_k = |r_k|$. On a late tail, $y/r \to 1$. If $\rho_k > 0$, the pure active term in (S.242) gives

$$\rho_{k+1} \le \rho_k - c_0 \rho_k^{q_0+1}.$$

After increasing the starting time, $c_0 \rho_k^{q_0} < 1$. Taking inverse powers and using $(1-t)^{-q_0} \ge 1 + q_0 t$ yields a positive lower bound on $\rho_{k+1}^{-q_0} - \rho_k^{-q_0}$. Thus

$$\rho_k = O(k^{-1/(p-2)}), \qquad y_k = O(k^{-1/(p-2)}). \tag{S.249}$$

The invariant zero branch satisfies these bounds directly.

Independently, $Q_{k+1} - Q_k \ge \eta m a_m u_k^{m-1}$ and $u_k Q_k \to 1$. For the integer $m \ge 1$,

$$Q_{k+1}^m - Q_k^m \ge m Q_k^{m-1}(Q_{k+1} - Q_k) \ge \eta m^2 a_m (Q_k u_k)^{m-1}.$$

This has a positive lower bound, also when $m = 1$. Consequently

$$w_k, u_k = O(k^{-1/m}). \tag{S.250}$$

Only upper estimates have been proved so far; no lower normal capture has been used.

For every mixed monomial with $a, b \ge 1$, (S.249), (S.250), and Lemma S6.2 imply

$$\begin{aligned} |y_k|^{2a} u_k^b &= O(k^{-1-\delta_{ab}}), \\ \delta_{ab} &= \frac{2a}{p-2} + \frac{b}{m} - 1 \ge \frac{4a}{p(p-2)} > 0. \end{aligned} \tag{S.251}$$

Higher pure normal terms have positive margins $b/m - 1$ as well. Since $w/u = 1 + O(u)$, the exact reciprocal equation gives

$$w_{k+1} = \frac{w_k}{1 + \eta w_k F_u(y_k, u_k)}. \tag{S.252}$$

For the leading pure normal term, $w_k u_k^{m-1} = w_k^m [1 + O(u_k)]$. For each mixed derivative term, $w_k |y_k|^{2a} u_k^{b-1} = O(|y_k|^{2a} u_k^b)$. It follows that, with $c = \eta m a_m$,

$$\log w_{k+1} - \log w_k = -c w_k^m + \epsilon_k, \qquad |\epsilon_k| \le K k^{-1-\delta} \tag{S.253}$$

for some $0 < \delta < \min(1, 1/m)$, decreased to cover the finite monomial gaps. The leading transfer error is $O(k^{-1-1/m})$, and the logarithm error is $O(k^{-2})$ because $\eta w_k F_u = O(k^{-1})$. In particular, all mixed forces are retained in a controlled error before dividing by any normal lower scale.

Set $E_k = \sum_{j \ge k} \epsilon_j$ and $z_k = w_k \exp(E_k)$. Then $E_k = O(k^{-\delta})$ and exactly

$$\log z_{k+1} - \log z_k = -c w_k^m.$$

Therefore

$$\begin{aligned} z_{k+1}^{-m} - z_k^{-m} &= z_k^{-m}\{\exp(mcw_k^m) - 1\} \\ &= mc\exp(-mE_k) + O(w_k^m) \\ &= mc + O(k^{-\delta}) + O(k^{-1}). \end{aligned} \tag{S.254}$$

Summation, followed by inversion and $u = w + O(w^2)$, proves

$$u_k = (mc\,k)^{-1/m}[1 + O(k^{-\delta})] = Ck^{-1/m}[1 + O(k^{-\delta})],$$

decreasing δ if needed. This establishes the lower scale, rather than presupposing it.

Every pure active monomial is $O(k^{-p/(p-2)})$ or smaller. Together with (S.251) and the higher pure normal margins, this yields

$$F(y_k, u_k) = a_m u_k^m + o(k^{-1}), \qquad kF(y_k, u_k) \longrightarrow a_m C^m = \frac{1}{\eta m^2}.$$

It remains to prove the literal normal speed; a sequence equivalent alone would not justify differentiating it. The newly proved lower scale permits division of the monomial estimates by u_k^m, giving

$$F_u(y_k, u_k) = ma_m u_k^{m-1}[1 + O(k^{-\delta})]. \tag{S.255}$$

Take the ambient representative $g = (F_y/2, -F_y/2, F_u)$. With the actual source-target products $\omega_i = x_i X_i$ and $S = \sum_i \omega_i$, mass conservation in (M.84) gives

$$\mu = \frac{(\omega_1 - \omega_2)F_y/2 + \omega_3 F_u}{S}. \tag{S.256}$$

Here $S \to 1/2$, and the inverse chart and sign preservation give $\omega_1 - \omega_2 = O(\rho_k)$ and $\omega_3 = O(u_k^2)$. To see the first assertion directly, writing the next active displacement as Y and the next normal as U gives $\omega_1 - \omega_2 = (1-u)Y + (1-U)y$, while $|Y| = O(\rho_{k+1}) = O(\rho_k)$. By (S.251) and the pure active bound, $\rho_k|F_y| = O(k^{-1-\delta'})$ for some $\delta' > 0$. Since (S.255) has order $k^{-(m-1)/m}$, it follows that $\mu = o(F_u)$.

The exact physical normal equation is

$$U - u = -\eta u U(F_u - \mu). \tag{S.257}$$

The proved profile gives $U/u \to 1$. Multiplying (S.257) by $k^{1+1/m}$ now yields the limit

$$-\eta m a_m C^{m+1} = -C/m.$$

Its negativity proves eventual strict decrease of the literal u as well. ∎

S6.4 The exact active monomial recurrence

For a nonzero active start, $\rho_k = |r_k| > 0$ at every time. The exact first equation in (S.242), the inverse chart, and Proposition S6.3 imply, for some $\delta > 0$,

$$\frac{\rho_{k+1}}{\rho_k} = 1 - \sum_{\substack{a\geq 1,\ b\geq 0\\ c_{ab}>0}} A_{ab} k^{-b/m} \rho_k^{2a-2}(1 + e_{ab,k}), \qquad |e_{ab,k}| \leq K k^{-\delta}. \tag{S.258}$$

Indeed the relative factors are powers of $y_k/r_k = 1 + O(u_k + r_k^2)$ and of $u_k/(Ck^{-1/m}) = 1 + O(k^{-\delta})$. The bounds (S.249) and (S.250) make each error a positive power error, and the finite support permits a common exponent. The exact ratio is positive by (S.245); on a sufficiently late tail all factors $1 + e_{ab,k}$ are positive.

Proposition S6.4 (The faster-than-harmonic quadratic clock)**.** *If $\nu < m$, then* (M.88) *holds.*

Proof. The term $(a, b) = (1, \nu)$ in (S.258) alone yields $\rho_{k+1}/\rho_k \leq 1 - c_1 k^{-\nu/m}$. Since $0 < \nu/m < 1$, summing logarithms gives $\rho_k \leq K \exp(-c_2 k^{1-\nu/m})$. Every nonlinear active term is now negligible compared with $k^{-\nu/m}$, and every higher quadratic power is lower order. The square in the logarithm expansion is also lower order, because $\nu \geq 1$. Thus

$$\log \rho_{k+1} - \log \rho_k = -A_{1\nu} k^{-\nu/m}[1 + o(1)].$$

Summing the power clock proves $-\log \rho_k \sim A_{1\nu} k^{1-\nu/m}/(1 - \nu/m)$. Since $y_k/r_k \to 1$, the same equivalent holds for $|y_k|$. Higher quadratic clocks may themselves have divergent sums. This proof therefore asserts the logarithmic equivalent only; removing the leading exponential does not in general leave a fixed positive prefactor. ∎

Lemma S6.5 (A power-summable harmonic reduction)**.** *Suppose $\nu \geq m$ and use* (M.89)*. Then $v_k = k^\alpha \rho_k$ is bounded and, for some $\delta > 0$,*

$$\log v_{k+1} - \log v_k = \frac{\alpha - \kappa - P(v_k)}{k} + e_k, \qquad |e_k| \leq K k^{-1-\delta}. \tag{S.259}$$

Proof. Choose a monomial attaining α, and write $s = 2a - 2 > 0$ and $t = b/m < 1$. Its damping term gives, on a late tail,

$$\rho_{k+1} \leq \rho_k(1 - c_3 k^{-t} \rho_k^s).$$

Inverting as before yields $\rho_{k+1}^{-s} - \rho_k^{-s} \geq s c_3 k^{-t}$. Therefore $\rho_k = O(k^{-(1-t)/s}) = O(k^{-\alpha})$, proving boundedness before considering any limit of v_k.

For every nonlinear active monomial, the definition of α implies $b/m + \alpha(2a - 2) \geq 1$: this is immediate when $b \geq m$, and follows from the maximum when $b < m$. Equality is exactly membership in I. All noncritical nonlinear terms have a strictly positive gap above one. Quadratic terms have $b \geq m$; the harmonic one contributes κ/k, and all others

have a positive gap. On the attained bounded interval for v, the damping sum in (S.258) is therefore

$$\frac{\kappa + P(v_k)}{k} + O(k^{-1-\delta}).$$

The finite support makes the minimum nonzero gap positive. The logarithm expansion and $\alpha \log(1 + 1/k)$ each have error $O(k^{-2})$. Decreasing δ combines these errors with the coefficient errors and gives (S.259). ∎

S6.5 The three harmonic alternatives

We now complete the proof of Theorem 13. Let $d = \alpha - \kappa$ in (S.259). The function P is continuous, vanishes at zero, is strictly increasing on $(0, \infty)$, and tends to infinity.

Positive balance: $\kappa < \alpha$. There is exactly one $V > 0$ with $P(V) = d$. First establish a positive lower bound for v_k. Choose $h > 0$ so small that $P(h) < d/2$. On a sufficiently late tail, whenever $v_k \le h$, (S.259) has positive increment at least $d/(4k)$. An orbit remaining below h forever would therefore have $\log v_k \to +\infty$, a contradiction. Once it reaches h, any crossing from above has relative size $1 + O(1/k)$ by boundedness and (S.259); for large k it cannot land below $h/2$. While below h, it increases. Thus $v_k \ge h/2$ eventually.

On the resulting positive compact interval, the drift $d - P(v)$ is uniformly positive below $V - \varepsilon$ and uniformly negative above $V + \varepsilon$, whenever the indicated interval is nonempty. Harmonic divergence forces entry into every neighborhood of V: otherwise the logarithm would drift without bound with the appropriate sign. The same strict drift, together with one-step changes tending to zero, traps the sequence eventually in any slightly enlarged neighborhood. Letting its radius decrease gives $v_k \to V$. This proves (M.90).

Dominant harmonic stiffness: $\kappa > \alpha$. Subtract $\alpha \log(1 + 1/k)$ from (S.259). Since $P \ge 0$,

$$\log \rho_{k+1} - \log \rho_k \le -\frac{\kappa}{k} + O(k^{-1-\delta}) + O(k^{-2}).$$

Summation yields $\rho_k \le K k^{-\kappa}$. With this improved bound, every nonlinear term in (S.258) has exponent at least

$$\frac{b}{m} + \kappa(2a - 2) \ge 1 + (\kappa - \alpha)(2a - 2) > 1.$$

All higher quadratic terms and the coefficient error in the harmonic term are summable as well. The logarithm's square is $O(k^{-2})$. Hence

$$\sum_k |\log \rho_{k+1} - \log \rho_k + \kappa \log(1 + 1/k)| < \infty.$$

Thus $\log \rho_k + \kappa \log k$ converges to a finite real number. Its exponential is the strictly positive orbit-dependent D_0 in (M.91).

Critical harmonic stiffness: $\kappa = \alpha$. Set $E_k = \sum_{j\ge k} e_j$ and $z_k = v_k \exp(E_k)$. Since $E_k = O(k^{-\delta})$, (S.259) becomes exactly

$$\log z_{k+1} - \log z_k = -\frac{P(v_k)}{k}. \tag{S.260}$$

Thus z_k decreases to a nonnegative limit. A positive limit would keep $P(v_k)$ bounded away from zero and contradict harmonic divergence in (S.260); consequently $z_k, v_k \to 0$. By (M.92), $P(v) = A_* v^s[1 + o(1)]$ as $v \downarrow 0$. Therefore

$$\begin{aligned} z_{k+1}^{-s} - z_k^{-s} &= z_k^{-s}\{\exp(sP(v_k)/k) - 1\} \\ &= \frac{sA_*}{k}[1 + o(1)] + O(k^{-2}). \end{aligned} \tag{S.261}$$

The final bound is valid because $z_k/v_k \to 1$ and $z_k^{-s}P(v_k)^2 = O(v_k^s)$ is bounded. Summation gives $z_k^{-s}/\log k \to sA_*$, proving (M.93). All critical ties are retained in P; the smallest critical active power controls the final logarithm.

In all three alternatives, $y_k/r_k \to 1$ and the fixed sign transfer the result from ρ_k to y_k. The exactly invariant zero branch has already been handled. There are no nonzero exceptional separators in this reflection-symmetric class. This completes the proof of Theorem 13.

S6.6 Globally convex families realizing the alternatives

The examples below verify convexity of the whole objective independently. They also show that the active rate is not determined by the two pure-axis degrees.

Proposition S6.6 (A convex generating family)**.** *Let $s \ge 1$, $d \ge 2$ be integers, let q, m be positive even integers with $m < qd$, and let $A > 0$. Then*

$$F(y,u) = \frac{A}{2sd}(y^{2s} + u^q)^d + \frac{u^m}{m} \tag{S.262}$$

is globally convex on $\mathbb{R}^2$ and satisfies Theorem 13 with $p = 2sd$, $a_m = 1/m$, and $C = (m\eta)^{-1/m}$. Its genuine mixed terms admit no constant normal lift eliminating all active dependence.

Proof. The function $y^{2s} + u^q$ is convex and nonnegative on $\mathbb{R}^2$, and $t \mapsto t^d$ is convex and nondecreasing on $[0,\infty)$. Convex composition and addition prove global convexity, including after the affine substitution in (M.81). The binomial expansion has nonnegative coefficients, its face degree is $2sd$, and $m < qd$ makes u^m/m the leading normal term. Thus

$$u_k \sim (m\eta k)^{-1/m}, \qquad kF(y_k, u_k) \longrightarrow \frac{1}{\eta m^2}.$$

Although the active minimum sheet is the affine sheet $y = 0$, F_u contains genuine mixed terms. If $F_u - TF_y$ were independent of y for a constant T, parity would force $T = 0$: F_u is even in y, while the nonzero F_y is odd. With $T = 0$ its mixed active dependence remains. This is exactly the failure of the fixed affine-null cancellation. ∎

Corollary S6.7 (Quadratic active stiffness and its coefficient threshold)**.** *In* (S.262)*, let* $s = 1$ *and* $\nu = q(d-1)$*. For every* $y_0 \neq 0$*:*

1. *If* $\nu > m$*, then*
$$y_k \sim \operatorname{sgn}(y_0) \left[\frac{\eta A(2d-2)}{8}\, k\right]^{-1/(2d-2)}. \tag{S.263}$$
2. *If* $\nu < m$*, then*
$$-\log|y_k| \sim \frac{\eta A C^{\nu}}{8(1-\nu/m)} k^{1-\nu/m}. \tag{S.264}$$
3. *If* $\nu = m$*, the threshold is* $A_c = 4q$*. For* $0 < A < 4q$*,*
$$\frac{|y_k|}{u_k^{q/2}} \longrightarrow \sqrt{(4q/A)^{1/(d-1)} - 1}. \tag{S.265}$$
At $A = 4q$*,*
$$\frac{|y_k|\sqrt{\log k}}{u_k^{q/2}} \longrightarrow 1. \tag{S.266}$$
For $A > 4q$*, some orbit-dependent* $D_0 > 0$ *satisfies*
$$y_k \sim \operatorname{sgn}(y_0) D_0 k^{-A/(8m)}. \tag{S.267}$$

Proof. Differentiation gives $F_y = Ay(y^2 + u^q)^{d-1}$ and $F_{yy}(0,u) = Au^{\nu}$. The smallest quadratic stiffness power is therefore ν, with coefficient in (S.258) equal to $\eta A C^{\nu}/8$. If $\nu < m$, Proposition S6.4 immediately gives (S.264).

For $\nu > m$, index the nonlinear active monomials by $j = 2, \ldots, d$. Their candidate exponents are
$$\frac{1 - q(d-j)/m}{2j-2}$$
when the numerator is positive. As a function of j this ratio is strictly increasing when $m < q(d-1)$, so the pure active monomial $j = d$ uniquely maximizes it. There is no harmonic quadratic term. Hence $\alpha = 1/(2d-2)$ and the amplitude equation is $(\eta A/8)V^{2d-2} = \alpha$, proving (S.263).

When $m = q(d-1)$, all nonlinear active monomials tie at $\alpha = q/(2m) = 1/(2d-2)$, and $\kappa = A/(8m)$. Writing $R = V/C^{q/2}$, their combined polynomial satisfies
$$\kappa + P(V) = \kappa(1+R^2)^{d-1}.$$
Thus the subcritical equation is $\kappa(1+R^2)^{d-1} = \alpha$, which gives (S.265) and the exact threshold $A = 4q$. At equality, the smallest critical exponent is $s = 2$, and $A_* = \kappa(d-1)C^{-q}$, so $2A_* = C^{-q}$. The logarithmic conclusion of Theorem 13 gives (S.266). Above threshold, the same theorem gives the exponent $\kappa = A/(8m)$ and a positive prefactor, proving (S.267). ∎

Example S6.8 (Stretched exponential with an unchanged normal law)**.** The globally convex polynomial

$$F(y,u)=\frac{(y^2+u^4)^2}{4}+\frac{u^6}{6}$$

has $u_k\sim(6\eta k)^{-1/6}$ and $kF_k\to 1/(36\eta)$, while every nonzero active start satisfies

$$-\log|y_k|\sim\frac{3\eta(6\eta)^{-2/3}}{8}\,k^{1/3}.$$

Here the vanishing active Hessian on $y=0$ still accumulates a faster-than-harmonic clock.

Example S6.9 (Fixed degrees with a coefficient-dependent active power)**.** For

$$F_A(y,u)=\frac{A}{4}(y^2+u^4)^2+\frac{u^4}{4},\qquad A>0,$$

the degrees $p=m=4$, the leading normal coefficient, and the leading normal and objective constants are the same for all A. The active logarithm occurs at $A=16$, and the supercritical active power is $A/32$. This threshold comes from mixed stiffness competing with nonlinear damping. It arises from competing mixed stiffness and nonlinear damping, rather than an affine forcing term.

Corollary S6.10 (Changed active power without quadratic stiffness)**.** *In* (S.262)*, suppose* $s\ge 2$ *and put* $m_c=q(d-1/s)$*. There is no quadratic active monomial. The maximizing exponent in* (M.89) *is*

$$\alpha=\begin{cases}(2sd-2)^{-1}, & m<m_c,\\ [1-q(d-1)/m]/(2s-2), & m>m_c.\end{cases}\tag{S.268}$$

In the first case only the pure active term attains the maximum; in the second only the smallest active power does. If $m=m_c$*, all active monomials tie. In every case the positive amplitude is the unique solution of* $P(V)=\alpha$*.*

Proof. The active monomials are indexed by $j=1,\dots,d$ with active degree $2sj$ and normal degree $q(d-j)$. The relevant ratios are

$$f(j)=\frac{1-q(d-j)/m}{2sj-2}$$

for positive numerators. Regarding j as real, the sign of $f'(j)$ is the sign of m_c-m. Thus the largest index wins below m_c, the smallest index wins above it, and all tie at equality. Above m_c the numerator for $j=1$ is positive, since $m_c>q(d-1)$. No active degree is two, so $\kappa=0<\alpha$, and the amplitude conclusion follows from Theorem 13. ∎

Example S6.11 (A nonlinear mixed exponent)**.** Take $s=2,d=2,q=6,m=10,A=1$. Then

$$F(y,u)=\frac{(y^4+u^6)^2}{8}+\frac{u^{10}}{10}$$

is globally convex, with $p = 8$ and $F_{yy}(0, u) = 0$ for every u. Nevertheless its active exponent is $1/5$, rather than the pure-face value $1/6$. With $C = (10\eta)^{-1/10}$, its critical coefficient is $\eta C^6/8$, so

$$y_k \sim \operatorname{sgn}(y_0)\sqrt{\frac{8}{5\eta C^6}}\, k^{-1/5}, \qquad u_k \sim Ck^{-1/10}, \qquad kF_k \longrightarrow \frac{1}{100\eta}.$$

The changed exponent is caused by a nonlinear mixed force, without hidden quadratic active stiffness.

References

[1] K. Kurdyka, T. Mostowski, and A. Parusiński. Proof of the gradient conjecture of R. Thom. *Annals of Mathematics*, 152(3):763–792, 2000. Primary preprint. Relevant locations: Propositions 4.2 and 6.2, Sections 5–7 and 9.

[2] K. Kurdyka. On gradients of functions definable in o-minimal structures. *Annales de l'Institut Fourier*, 48(3):769–783, 1998. doi:10.5802/aif.1638.

[3] J. Bolte, A. Daniilidis, A. Lewis, and M. Shiota. Clarke subgradients of stratifiable functions. *SIAM Journal on Optimization*, 18(2):556–572, 2007. Primary preprint. Relevant locations: Theorem 11, Corollary 12, Remark 9.

[4] J. Bolte, S. Sabach, and M. Teboulle. Proximal alternating linearized minimization for nonconvex and nonsmooth problems. *Mathematical Programming*, 146:459–494, 2014. Author manuscript. Relevant locations in this version: Theorem 5.1 and Lemma 3.6.

[5] H. Chen, J. Fan, and A. M.-C. So. A Unified Framework for Iterate Convergence of Bregman Proximal Methods. arXiv:2608.05536v1, 2026. Relevant locations: Section 3, Theorem 1, Example 2, Proposition 4, and Section 6.

[6] M. C. Mukkamala, J. Fadili, and P. Ochs. Global convergence of model function based Bregman proximal minimization algorithms. *Journal of Global Optimization*, 83:753–781, 2022. doi:10.1007/s10898-021-01114-y. Relevant locations: Theorems 17–19 and their interior-cluster or interior-limit hypotheses.

[7] A. De Marchi, Y. Malitsky, and A. B. Taylor. Mirror descent algorithms with logarithmic barriers. arXiv:2608.22834v1, 2026. Relevant locations: Section 2, Theorem 1 (proximal mirror descent), Theorem 2 and equation (20) (exact mirror descent), and Theorem 3 (fixed-horizon tightness).

[8] C. T. C. Wall. *Singular Points of Plane Curves*. Cambridge University Press, 2004. Chapter 2: Puiseux theorem. Relevant input: convergent Puiseux solutions and finite local branch structure.

[9] F. Alvarez, J. Bolte, and O. Brahic, Hessian Riemannian gradient flows in convex programming, primary author manuscript. Author PDF.

[10] H. Attouch, J. Bolte, and B. F. Svaiter, Convergence of descent methods for semi-algebraic and tame problems: proximal algorithms, forward–backward splitting, and regularized Gauss–Seidel methods, *Mathematical Programming* **137** (2013), 91–129. Primary author manuscript.

[11] G. Valette. *On subanalytic geometry.* Author-hosted version dated 31 July 2025. Author notes. Relevant input: Proposition 1.8.4, p. 38, one-variable convergent Puiseux expansion.

[12] E. Pauwels. On the nature of Bregman functions. *Operations Research Letters*, 2024, article 107183. doi:10.1016/j.orl.2024.107183. Primary preprint, arXiv:2302.02689v2. Relevant locations: Theorem 1 and Section 4.3, unbounded kernels.

[13] E. Bierstone and P. D. Milman. Semianalytic and subanalytic sets. *Publications Mathématiques de l'IHÉS* **67** (1988), 5–42. doi:10.1007/BF02699126. Relevant input: Theorem 6.4, compact subanalytic Łojasiewicz inequality.